



Le réseau
de transport
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Multi-Projects Agreement for Onshore Converter
Station and Offshore Substation

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Sommaire

1	Introduction	6
1.1	Project description	6
1.2	Scope of validity	6
1.3	Definitions and Abbreviations	7
1.4	Principle of similarity	7
1.5	Incorporation of Employer's requirements	7
2	Sites, operating conditions	7
2.1	Sites	7
2.2	Operating conditions	8
2.2.1	Onshore Converter Station	8
2.2.2	Offshore Converter Station	10
3	Interfaces	10
3.1	Interface to OWFs and the Offshore AC grid	10
3.1.1	Control and communication interface to the OWFs	10
3.1.2	Protection Interface to the OWFs	11
3.1.3	Emergency power control OWFs	11
3.2	Interface to the Cable Contractor	11
3.3	Interface to the Employer	11
3.3.1	Interfaces with Employer Remote Control Centres	11
3.3.2	Interfaces with Employer Onshore Substation	11
3.3.3	Converter transformers monitoring system	12
3.3.4	DTS/DAS system (applicable to both Onshore Converter Station and Offshore Substation)	12
3.3.5	Metering system (applicable only to the Offshore Substation)	12
3.4	Auxiliary power onshore	13
4	Quality	13
5	HSE & sustainable design	13
5.1	HSE specifications	13
5.2	Sustainable design	14
6	Project management	14
7	Key performance indicators	14
7.1	Transmission capacity	14
7.2	Reliability, Availability, Maintainability and Safety (RAMS)	15
7.2.1	General requirements	15
7.2.2	System performance requirements and capabilities	15
7.3	Transmission losses	18
7.3.1	Definition and determination of transmission station losses	18
7.3.2	Losses statcom case determination	18

7.3.3	Contractual assessment of transmission station losses	19
7.3.4	Transmission Station Losses calculation during design stage	19
7.3.5	Transmission Station Losses verification using measurements during FAT	20
7.3.6	Transmission Station Losses – real time continuous measurement – non contractual	20
8	System basis of design	21
8.1	Employer AC system description	21
8.1.1	AC system voltage and frequency	21
8.1.2	AC short circuit levels	21
8.1.3	Background harmonics	22
8.1.4	Harmonic impedances	22
8.2	AC system performance requirement	22
8.2.1	Voltage and frequency ranges	22
8.2.2	Power quality and harmonics	23
8.2.3	Negative sequence of AC voltage	26
8.2.4	Earthing requirement	26
8.2.5	Lightning protection requirements	27
8.3	HVDC system rating	27
8.3.1	Reactive power capability	27
8.3.2	DC transmission voltage and current range	29
8.4	Operating modes	30
8.4.1	Operating mode OM1: bipolar with DMR	31
8.4.2	Operating mode OM2: symmetrical bipolar	31
8.4.3	Operating mode OM3: asymmetrical monopolar	32
8.4.4	Operating mode OM4: STATCOM onshore	32
8.4.5	Operating mode OM5: Round power test for commissioning	32
8.4.6	Requirements related to start-up, shutdown and restart of the HVDC Link	33
8.5	HVDC system grounding	34
8.6	HVDC System dynamic behaviors	35
8.6.1	Dynamic response definitions	35
8.6.2	Active power control	36
8.6.3	Onshore AC voltage & reactive power control	37
8.6.4	Offshore converter station control	44
8.6.5	Frequency control	45
8.6.6	Stability under dynamic conditions	45
8.6.7	Advanced functionalities	53
8.6.8	System statuses and sequences	55
8.6.9	Loss of inter-station telecommunication	59
8.7	Electromagnetic Compatibility (EMC)	59
8.7.1	Radio Frequency Interference (RFI)	59
8.7.2	Electromagnetic emissions and electrostatic fields limits (EMF)	59
8.7.3	Emission and immunity inside and near the converter station	60
8.8	Audible noise level	60
8.8.1	During construction	60

8.8.2	During operation	60
8.9	Station layout and HV switchgear	62
8.9.1	Onshore station	63
8.9.2	Offshore station.....	65
8.10	High voltage equipment	67
8.10.1	General requirements	67
8.10.2	Insulation coordination	68
8.10.3	Maximum allowable overvoltages and currents on HVDC cable terminations	70
8.10.4	Converter transformers.....	71
8.10.5	Auxiliary supply transformer (AST).....	73
8.10.6	Gas Insulated Switchgear	74
8.10.7	AC/DC equipment.....	75
8.10.8	Resistor, Reactor and Capacitor (RLC).....	76
8.10.9	HV and MV cable inside converter station	78
8.10.10	Converters	78
8.10.11	Replica	80
8.10.12	Dynamic Breaking System (DBS)	80
8.11	General Control & Protection System	81
8.11.1	General Requirements	81
8.11.2	Control Requirements	81
8.11.3	Data Network Infrastructure and Software	88
8.11.4	Protection Requirements	88
8.11.5	Testing Requirements (FST/FAT)	91
8.12	Auxiliary systems	92
8.12.1	Auxiliary power systems.....	92
8.12.2	Auxiliary support systems (onshore & offshore).....	96
8.12.3	Fire protection (onshore & offshore)	96
8.13	IT and information security (Cyber security).....	97
8.14	Communication technology	98
8.14.1	Core concepts	98
8.14.2	Main telecom requirements.....	98
8.15	Electrical system studies and models.....	99
8.15.1	Electrical system studies	99
8.15.2	Models.....	101
8.16	Onshore civil works	102
8.17	Operation, preparation of operation and maintenance	102
8.17.1	Operation and maintenance documentation.....	102
8.17.2	Asset Management	105
8.17.3	Spare parts.....	106
8.17.4	Briefing and training.....	106
8.18	Commissioning	107
8.18.1	Responsibilities and Communication	107
8.18.2	Testing of the Converter System	109

8.18.3	Commissioning of the Converter System	109
8.18.4	Trial Operation	110
8.18.5	Acceptance	112
9	Multiterminal readiness variant	112
10	Standards, regulations and special requirements	112
11	Appendix I: Reference document and standards	114
12	Appendix II: List of attachments	117

1 Introduction

1.1 Project description

[C11-GENE-010001]

The 525 kV – 2000 MW projects - Fécamp Grand Large 1 (HA1), Fécamp Grand Large 2 (HA2) and Golfe du Lion Centre (GLC) and Bretagne Nord Ouest (BNO) - in the framework of PPE3 (Plan Pluriannuel de l'Energie 3) consists of bipolar HVDC topology for each project with a Guaranteed Active Power Link Capacity of 2000 MW and a rated DC voltage of ± 525 kV using Voltage Source Converter for each HVDC Link.

[C11-GENE-010002]

Each HVDC link – grid connection system – consists of an Offshore Substation, an Onshore Converter Station and a HVDC Cable Link (HVDC cables and Dedicated Metallic Return (DMR)) between the two stations. Figure 1 shows schematically the HVDC Link and the corresponding system boundaries.

The length of HVDC Cable Link is described in (BOD_02_Interface_Basis_of_design).

The Inter Array Cable (IAC) voltage is 132 kV.

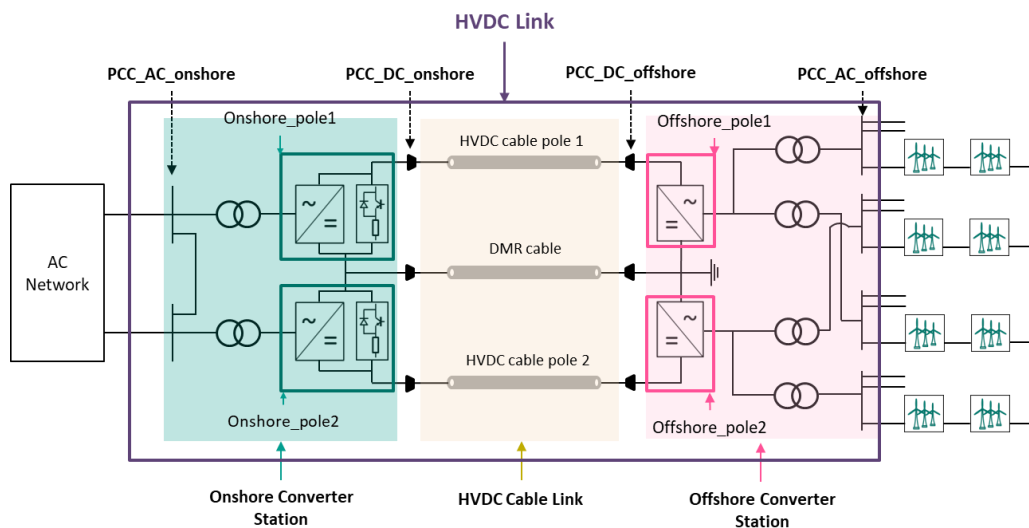


Figure 1– Schematic setup of HVDC Link and illustration of system boundaries. The position of grounding is only for illustration purpose.

All the requirements in this document are applicable for each HVDC Link individually, except if stated otherwise.

The main electrical system parameters are summarized in the Table 1.

Rated active power transmission capacity at Pcc_dc_offshore	2000 MW
Rated DC voltage	± 525 kV
Nominal AC grid voltage offshore	132 kV
Nominal AC grid voltage onshore	400 kV
Nominal frequency (onshore & offshore)	50 Hz

Table 1: Main electrical system parameters for 525 kV DC – 2000 MW projects

1.2 Scope of validity

This document describes the Contractor's scope of supply and services.

The scope of supply and services of the Contractor includes at least the following points:

- All components, systems and works required to establish a low-maintenance, durable, functional, operationally safe and highly available HVDC Link.

- Trial operation of the entire HVDC Link after final installation and commissioning
- Project management
- Ensuring that all components and (sub)systems, which are required for the specific functions and operation of the HVDC Link are taken into account.
- Proactive contribution to the design collaboration with the Employer and Cable contractor, anticipate possible requirements beforehand as far as possible.

1.3 Definitions and Abbreviations

[\[C11-GENE-010004\]](#)

Definitions are in (Def) .

1.4 Principle of similarity

[\[B21ai-GENE-110566\]-N3](#)

All Converter Stations of same technology provided by the Contractor to Employer, including in other projects, shall be as much identical as possible. Equipment shall be interchangeable when technically possible. At least the high voltage equipment used in each HVDC Link shall be interchangeable. Any differences between the systems shall be justified by the Contractor and agreed with the Employer.

1.5 Incorporation of Employer's requirements

[\[CBOD-GENE-100029\]](#)

Throughout this document, the Employer's minimum requirements with respect to planning, design, fabrication, delivery, assembly, installation, testing and commissioning and performance of the Onshore and Offshore Substations are described in a functional manner and with limited detail to accelerate the tender phase. However, detailed Employer's Requirements (ERs) shall be met by Contractor.

2 Sites, operating conditions

2.1 Sites

[\[B21ai-SITE-110002\]-N3](#)

The site, network, cable characteristics listed in this section shall be used in the design of all equipment for the Onshore and Offshore Converter Stations unless specified otherwise elsewhere in the Technical Specification (in this specific case, the Contractor shall ask the Employer which requirement shall apply).

The performances of the HVDC Link shall be guaranteed for the characteristics and conditions listed in this section (regardless of how long these ambient conditions have prevailed). As such, the HVDC Link shall be stable under steady state, dynamic and transient or abnormal conditions for all combinations of impedances and voltages of the connected AC networks within the boundary described in section 8.1 and 8.2. In addition, the HVDC Link shall be self-protecting under all operational conditions.

[\[B21ai-SITE-110003\]-N3](#)

For the network surrounding the converter stations:

Onshore Converter Station

Any equipment to consider nearby are provided in (Call_off).

Any specific information relating to the link between Onshore Converter Station and Employer Existing AC Substation are provided in (Call_off).

Offshore Converter Station

OWF (Offshore Wind Farm) will be connected to the Offshore Converter Station.

2.2 Operating conditions

[B21ai-SITE-110012]-N3

The ambient and environmental conditions stated in this chapter shall be used for the design of all equipment and buildings/platform of Converter Stations (unless applicable standards or regulations consider more restrictive environmental conditions which shall be then applied).

If specified otherwise elsewhere in the set of technical specifications, the Contractor shall ask the Employer for decision on the chosen environmental criteria.

The performance of the HVDC Link specified in this document shall be guaranteed under any conditions described in this section.

More details relating to site and environmental conditions will be provided in (Call_off).

[B21ai-SITE-110482]

Performances of each Converter Station shall be guaranteed under the conditions listed below (section 2.2.1 and 2.2.2), with and without considering the use of redundant cooling capacity for temperature.

Redundant cooling capacity shall be used and activated automatically during Converter Station operation without disruption of service, whenever the cooling capability is not sufficient to prevent damage on valve equipment or to avoid a reduction of power capacity (for example in case of temperature higher than here required).

Contractor shall provide the curve with active power limitation as a function of the relevant parameters used by Contractor's protection for each Converter Station. In any case, when the temperature exceeds the upper threshold, the Guaranteed Active Power Link Capacity shall be gradually reduced, decreasing smoothly from 100% to 0% over at least a 2°C range. This reduction shall trigger a warning and runback shall be triggered only by the before mentioned Contractor's protections.

2.2.1 Onshore Converter Station

[CBOD-SITE-100001]

Performances of each Onshore Converter Station shall be guaranteed under the conditions listed in Table 2.

	HA1, HA2	GLC	BNO	Unit
Altitude above sea level	<1000	<1000	<1000	m
Maximum air temperature*	40°C / 42°C for without / with redundancy	42°C / 44°C for without / with redundancy	Call-Off	°C
Minimum air temperature	-25	-25	-25	°C
Maximum daily rainfall	Call-Off	Call-Off	Call-Off	mm/day
Maximum rainfall per annum	Call-Off	Call-Off	Call-Off	mm/year
Wind Pressure**	HPV	HPV	HPV	
Maximum Wind Velocity***	Call-Off	Call-Off	Call-Off	m/s
Snow/Ice loading****	Yes	Yes	Yes	
Solar radiation	Call-Off	Call-Off	Call-Off	W/m ²
Ground strike point density (Nsg as per IEC 62858)	Call-Off	Call-Off	Call-Off	Strokes/km ² /year
Keraunic level	Call-Off	Call-Off	Call-Off	

Table 2 – Environmental Data

All outside electrical equipment shall be mechanically designed in accordance with (CCG_P) and (Arrete_17_2001).

*: The specified maximum air temperature shall be considered for cooling and HVA/C systems. These values shall be confirmed in (Call_off). For converter transformers, reactors, capacitors, resistors, surge arresters, insulators, bushings, circuit breakers, disconnecter switches, instrument transformers, GIS, the maximum ambient air temperature between the specified values and the values indicated in the latest corresponding IEC standard shall be applied. In any case, at least 40°C shall be considered. For low voltage apparatus of main HV equipment, a maximum air temperature of 70°C shall be considered.

**: For secondary gantries (such as bus bars), HPV level is required as indicated in (CCG_P) section 2 and (B22a). For HV equipment, HPV also applies. For primary gantries, local conditions apply.

***: This applies to HV equipment and not to gantries

****: For all gantries, loading due to frost is to be considered and shall be in accordance with (CCG_P) section 2.

[B21ai-SITE-110013]-N3

All buildings (and other non-electrical structures or equipment such as transformer bay walls, structures of outdoor cooling units, fences, etc.) are to be compliant with the latest versions of Eurocodes (and considering national amendments) and French standards. Additional data is provided in Table 3.

Contractor shall take into consideration requirements of section “structures requirements” and “building requirement” of (B22b_i).

The seismic level is detailed in Table 3.

	HA1, HA2	GLC	BNO
Snow load (NF EN 1991 1 3)	A1	A2	Call Off
Seismic Level (Decret 2010 1255)	1	4 (could be lowered to 3 at Call Off)	Call Off
Wind Velocity (NF EN 1991 1)	2	3	Call Off

Table 3 – Additional Environmental Data (Eurocodes)

[B21ai-SITE-110488]

Following requirements regarding seismicity shall be considered:

The Onshore Converter Station design shall take into account the seismic level of the location (Table 3). The Onshore Converter Station shall be considered as an installation of category of IV of (Arrete_22102010_Seismic) (this category includes installations whose protection is essential for the needs of civil security and national defence as well or the maintenance of public order, as such it includes public energy distribution centres) and the acceleration considered for the design shall comply with (Arrete_22102010_Seismic).

As such and in accordance with article R565-5 of “Code de l'Environnement”, the equipment and systems located within the Onshore Converter Stations’ buildings must be capable of remaining in service and delivering its performances in case of earthquake.

The buildings of the Onshore Converter Station, and all equipment of the Onshore Converter Station installed in buildings shall be designed in accordance with (but not limited to):

- (Arrete_22102010_Seismic) related to the classification and the earthquake-proof construction rules that are applicable to the buildings of the category “normal risk”.
- “Arrêté du 19 juillet 2011” modifying the “arrêté du 22 octobre 2010” related to the classification and the earthquake-proof construction rules that are applicable to the buildings of the category “normal risk”.
- Any other superseding applicable «Arrêté» and regulatory requirements.
- EUROCODE 8.
- The seismic tests, the testing procedures, the choice of severity for components and apparatus of the Onshore Converter Station shall comply with IEC standards (IEC60068_3_3), (IEC62271_207), (IEC62271_300) and other relevant standards if applicable.

For outdoor equipment

- Structure and support of equipment shall comply with EUROCODE 8

- The seismic tests, the testing procedures, the choice of severity for components and apparatus of the Onshore Converter Station shall comply with IEC standards (IEC60068_3_3), (IEC62271_207), (IEC62271_300) and other relevant standards if applicable.
- For reactors, capacitors and resistors please consider requirement [\[B22aiii-GENE-440181\]](#) made in (B22a_iii) specification (consideration of IEC standard 62271-300 for filter capacitors and reactors).
- For converter transformers, the requirements indicated in (B22a_i) shall apply.

The fulfilment of the seismic requirements for the systems and sub-systems shall be proved at least through calculations at engineering stage. Regarding equipment, the fulfilment of the seismic requirements shall be demonstrated by tests or by suitable test certification.

[\[B21ai-HSDB-110622\]-I](#)

The Employer will provide the following studies at (Call_off):

- A hydrological study
- A geotechnical study
- An acoustic study
- A topographic survey
- An environmental survey
- An UXO survey

2.2.2 Offshore Converter Station

[\[CBOD-SITE-100028\]](#)

Performances of each Offshore Converter Station shall be guaranteed under the air temperature conditions listed in Table 4.

	HA1, HA2 Offshore Converter Station		GLC Offshore Converter Station		BNO Offshore Converter Station	
Maximum air temperature*	Without*	With*	Without*	With*	Without/with*	°C
	+32	+34	+35	+37	Call-off	°C
Minimum air temperature	-10		-10		Call-off	°C

Table 4 – Offshore Converter Station ambient temperature. *Without/with redundancy

Performances of each Offshore Converter Station shall be guaranteed up to sea water temperature specified in (BOD_01). For the record, air cooling is the preferred solution and sea water cooling is considered as a deviation, see (ERS_6_5).

3 Interfaces

[\[CBOD-INTE-100030\]](#)

The interfaces described in (BOD_02_Interface_Basis_of_design) and (SOW03) shall be applied. In this section, only control and protection (C&P) interfaces are detailed. C&P interfaces are further detailed in specification (B22a_v) and represented on specification (B22av6).

3.1 Interface to OWFs and the Offshore AC grid

3.1.1 Control and communication interface to the OWFs

[\[CBOD-INTE-100031\]](#)

The Contractor shall interface the control and protection (C&P) system with the OWF Controllers of the Producers in order to provide signals from the Offshore Substation (e.g. circuit breaker position, current and voltage measurements, power limitations...). As part of these signals the Contractor shall provide to the OWF Controllers of the Producers measurements of the onshore grid frequency.

[\[CBOD-INTE-100032\]](#)

The communication to the OWFs shall be implemented and monitored. Type of interfaces (hardwired, IEC 61850, IEC 104...) shall be agreed in the Engineering phase. A signal exchange list shall be agreed between the Employer, the Contractor and the OWF Operator. For all interfaces, the location of the OWF controller (onshore and/or offshore) shall be taken into account in the design.

Additional information is in (B22a_v).

3.1.2 Protection Interface to the OWFs

[\[CBOD-INTE-100033\]](#)

Each IAC cable feeder will be protected by redundant non-unit protections. The protection devices settings shall be agreed between the Producers, the Contractor and the Employer. The Contractor is responsible for testing, implementation and verification of the settings. Selectivity principle shall be enforced for each individual feeder.

3.1.3 Emergency power control OWFs

[\[CBOD-INTE-100034\]](#)

In the event of possible restricted grid situations, e.g. power reduction of OWF, emergency shutdown ..., the Contractor shall implement an Emergency Power Control (EPC). The EPC signals shall be processed by the converter control, synchronized with the DBS, if required, and forwarded to/coordinated with the OWF Controller.

The EPC signals (quantity, type, order ...) shall be coordinated with the Employer during the Engineering Phase.

3.2 Interface to the Cable Contractor

[\[CBOD-INTE-100035\]](#)

The interface with cable contractor is described in (BOD_02_Interface_Basis_of_design) and (SOW03).

3.3 Interface to the Employer

3.3.1 Interfaces with Employer Remote Control Centres

[\[CBOD-INTE-100036\]](#)

The C&P shall exchange data with different Employer Remote Control Centers (RCC) through the Remote Control Interface (RCI) by IEC 60870-5-104 protocol gateway. Employer requirements are detailed in section 3.2 and Appendix 31 of (B22a_v).

3.3.2 Interfaces with Employer Onshore Substation

Tranche Raccordement (TRacc)

[\[CBOD-INTE-100037\]-I](#)

The TRacc is a Bay Unit which is supplied by Employer and installed in the Employer Existing AC Substation, in a relay building (BR) near AC apparatus of the HVDC Link feeder. There will be as many TRacc as HVDC Link feeders. It ensures the control and protection of Employer AC apparatus.

[\[CBOD-INTE-100038\]](#)

An interface is required for the exchanges of information between the control system of the Employer Existing AC Substation (TRacc) and the Onshore Converter Station control system provided by the Contractor. This interface will be called "Interface HVDC" (IF-HVDC) in this document.

Cubicle(s) associated with IF-HVDC interface shall be provided by the Contractor and shall be installed in the BR of the Employer Existing AC Substation, near to TRacc bay unit.

[\[CBOD-INTE-100039\]](#)

IF-HVDC shall include I/O modules to interface with the TRacc using hardwired signals. IF-HVDC is then communicating with the Contractor C&P system through optic fibres installed between the Employer Onshore Substation and the Onshore Converter Station.

[\[CBOD-INTE-100040\]](#)

The list of signals will be discussed during the design phase between the Employer and the Contractor but includes for example: feeder disconnect, earthing switch, circuit breaker positions, circuit breaker trip order, circuit breaker open/close command...

This list will not exceed 30 signals per HVDC Link feeder.

Employer requirements regarding TRacc are further detailed in section 8.1.1 of (B22a_v).

Integration of Employer protections

[\[CBOD-INTE-100041\]](#)

Protection equipment for the 400 kV AC link(s) between the Onshore Converter Station and the Employer Substation will be provided by the Employer in a cubicle (one cubicle per HVDC Link feeder). Contractor shall integrate this cubicle in the Onshore Converter Station and shall interface it with its control and protection system.

Contractor shall provide dedicated CT and VT circuits for these protections.

Contractor shall allocate space, access, power supply to install and connect this cubicle.

Employer requirements regarding integration of Employer protections are further detailed in section 8.1.2 of (B22a_v).

3.3.3 Converter transformers monitoring system

[\[CBOD-INTE-100042\]](#)

Employer will provide a transformer monitoring solution (for gas analysis...) for the onshore & offshore converter transformers. Further information on the transformer monitoring solution is detailed in section 5.6 of (B22a_i). Contractor shall integrate this solution, including its installation, its cabling, power supply and its communication with Employer network interfaces installed within the Onshore & Offshore Converter Station. It has to be noted that no communication with Contractor C&P system is required (transformer monitoring system will directly interface with Employer network equipment).

[\[CBOD-INTE-100043\]](#)

Employer will be in charge of testing and commissioning activities associated with this system. Contractor shall be requested to collaborate with Employer for testing and commissioning of transformer monitoring system.

3.3.4 DTS/DAS system (applicable to both Onshore Converter Station and Offshore Substation)

[\[CBOD-INTE-100261\]](#)

The interface with DTS/DAS system is described in (BOD_02_Interface_Basis_of_design) and (SOW03).

3.3.5 Metering system (applicable only to the Offshore Substation)

[\[CBOD-INTE-100044\]](#)

Employer will provide metering cubicles for OWF metering. Contractor shall integrate these cubicles in the Offshore Substation.

[\[CBOD-INTE-100045\]](#)

Contractor shall also provide: CT/VT, measurement circuits, power supply and data signal transmission cables between metering cubicles and Employer's communication gateways and between metering cubicles and OWF control system / communication gateways on the Offshore Substation. It has to be noted that no communication with Contractor C&P system is required (metering cubicles will directly interface with Employer network equipment and OWF control system / communication gateways on the Offshore Substation).

[\[CBOD-INTE-100046\]](#)

Employer will be in charge of testing and commissioning activities. Contractor shall be requested to collaborate with Employer for testing and commissioning of metering system.

3.4 Auxiliary power onshore

[\[CBOD-INTE-100047\]](#)

For the auxiliary power of the Onshore Converter Station, Source 2 (see description in section 8.12.1.2) – Option 2 shall be considered. Final choice will be confirmed by Employer in (Call_off).

- Option 1: A separate MV line running from the existing Employer AC Substation,
- Option 2: A separate MV line running from the nearest distribution substation (from DSO).

[\[CBOD-INTE-100048\]](#)**For option 1:**

The Contractor shall be responsible for doing all the modifications at the secondary side of the existing HV/MV transformer in Employer Existing AC Substation.

Employer will bear the cost of energy in this case.

[\[CBOD-INTE-100049\]](#)**For option 2:**

The Contractor shall be responsible for liaising with the local DSO (Distribution System Operator) and organizing this supply.

Contractor shall bear the cost or energy necessary for the Works in this case.

connection type shall be “coupure d'artère” or “double derivation” as described in the NFC 13-100 standard section 312.

[\[CBOD-INTE-100050\]](#)

The Contractor shall provide the protection scheme for the complete auxiliary system. The Contractor shall coordinate the protections in own scope of supply and with the Employer and/or DSO protection system.

4 Quality

[\[CBOD-QUA-100051\]](#)

Employer's requirements stated in (QUA_01), (QUA_02) and (QUA_03) shall be applied.

5 HSE & sustainable design

5.1 HSE specifications

[\[CBOD-HSE-100052\]](#)

Employer requirements in (HSE) shall be applied.

[\[CBOD-HSE-100053\]](#)

The safety requirements address all aspects of the project. Hereafter some specific examples of safety requirements.

- Electrical operations: In accordance with French legislation, (NFC18_510_1) shall be applied for all electrical operations, and its implementation in the Employer's rules.

- Fall prevention measures: In compliance with French legislation, collective measures shall be employed to prevent falls from height. The use of a secure working platform like Mobile Elevated Work Platform (MEWP) shall be required. It should be noted that, within the valve hall, the valve structure itself is prohibited from being used as a working platform.
- Chemical product compliance: Chemical products shall be aligned with the AMAC list. Contractor can propose alternatives, subject to approval by Employer. Material Safety Data Sheets (MSDS) of all products (used during installation, operation, maintenance or contained in equipment to which operators should be exposed) shall be provided.
 - o Chemical substances and preparations for the operation and the maintenance of fittings are not allowed if they are classified in:
 - Carcinogenic Category "1 or 2", "3".
 - Mutagenic Category "1 or 2", "3".
 - Toxic for reproduction Category "1 or 2", "3".
 - o The above categories refer to the directives 1999/45/EC (concerning preparations) and 67/548/EC (concerning substances) of the European parliament and of the council.
 - o The following categories are forbidden from formulation and from each substance or ingredient or any preparation used in the formulation: R40, R45 ...

5.2 Sustainable design

[\[CBOD-SUST-100054\]](#)

The "Ecodesign for Sustainable Products Regulation (ESPR)" proposition establishes a framework for the setting of ecodesign requirements for durable products and repeals the Directive 2009/125/EC.

In this context, the Employer's environmental objectives are the integration of the principles of eco-design and the circular economy for its network infrastructure.

The Employer's requirements detailed in (B21a_v) shall be applied.

6 Project management

[\[CBOD-PROJ-100055\]](#)

Employer's requirements stated in (PRO) shall be applied.

7 Key performance indicators

[\[C11-SYST-010012\]](#)

The system shall be designed to ensure that the system's performance do not decrease over the system's operational lifetime.

In case of bipolar project, all faults, if they are limited to one HVDC Converter Pole, shall only affect the faulty Converter Pole. The unaffected HVDC Converter Pole shall remain in operation.

7.1 Transmission capacity

[\[C11-SYST-010024\]](#)

The HVDC Link shall be designed to meet the following active power and losses requirements.

- The minimum Guaranteed Active Power Link Capacity is 2000 MW at the PCC-DC Offshore sending end. This power is the Rated Capacity of the HVDC Link.
- The minimum Guaranteed Active Power Pole Capacity is 1000 MW for each pole of offshore converter station (positive pole and negative pole) – half of the minimum Guaranteed Active Power Link Capacity. This power is the Rated Pole Capacity of the HVDC Pole Link.
- Furthermore, the HVDC Link shall allow power flow in every direction (at minimum to accomplish the energization of and to supply auxiliary power to the Offshore Converter Station including the energization of the Offshore AC Network and the OWFs from the Onshore Converter Station and during the commissioning by round power tests).

[\[C11-SYST-010164\]](#)

As there will be overplanting of OWF of up to 10% (200 MW), it is required that the Converter Station withstands this potential overload during a sufficient time to trip the relevant IAC feeder or to send an active power reduction to the OWF. As such, a time of 4 s can be used for the design and the final duration shall be discussed during execution phase with Employer and Producer.

7.2 Reliability, Availability, Maintainability and Safety (RAMS)

7.2.1 General requirements

[\[C11-SYST-010009\]](#)

The HVDC Converter Stations shall be designed for minimum life of 40 years including 2 years of T&I and commissioning at start of lifetime and 1 year of decommissioning at end of lifetime, except for:

- SCADA / HMI which shall be designed for minimum 10 years;
- Control and protection system which shall be designed for minimum 25 years;
- Auxiliary systems for HVDC system (heating, ventilation and air conditioning, cooling control ...) which shall be designed for minimum 25 years. For auxiliary systems related to Offshore Converter Platform, please refer to Employer's ERS.

7.2.2 System performance requirements and capabilities

7.2.2.1 Reliability and availability requirements

[\[C11-SYST-010013\]](#)

For bipolar HVDC transmission topology, each Converter Pole on Converter Station shall be able to operate independently of other and each Converter Pole shall be capable of being operated and maintained whether or not the other is in service.

[\[C11-SYST-010014\]](#)

Outage: see (CIGRE_TB590) edition 2014, Outage definition.

Converter Scheduled Outage: see (CIGRE_TB590) edition 2014, Scheduled Outage definition. The "suitable time" mentioned in (CIGRE_TB590) is fixed to 1 month. As such, if an outage cannot be postponed for more than one month, it shall be considered as a "Forced Outage".

Converter Forced Outage: see (CIGRE_TB590) edition 2014, Forced Outage definition.

[\[C11-SYST-010015\]](#)

Definition of Converter Scheduled Energy Unavailability (SEU): The Scheduled Energy Unavailability (SEU) has the meaning given in (CIG_TB590) edition 2014.

For the Onshore Converter Station, preventive maintenance activities shall be scheduled considering that the maintenance staff will be limited to 21 people working 8 hours per day during normal working hours within the interval 7:30 am to 5:00 pm (Central European Time) and with one-hour and thirty minutes break at mid-day, Monday to Friday.

This limit of 21 persons includes the Employer's maintenance personnel. For maintenance levels 3, 4 and 5 and for following systems: climate system and air conditioning, cooling system, diesel generator and fire detection systems. The personnel needed for these maintenance activities is not to be considered as part of the team of 21 people.

The Contractor may consider different working conditions (regarding working hours, working days, shifts, ...) in order to increase the availability, but in any case, the considered conditions shall be compliant with national Labour Code.

For the Offshore Converter Station, working conditions shall be compliant with Transport Code for non-seafaring personnel. In any case, the Contractor shall clearly describe the working conditions.

The HVDC Link shall not be put back into service between daily maintenance periods (i.e. overnight). Each outage shall consider a 4 hours duration for the establishment (on the first day to prepare the maintenance) and 4 hours duration for the re-establishment (on the last day to put the link back in service) of safe working conditions according to national regulations, which shall be considered as part of the Scheduled Outage and consequently as part of Converter Scheduled Energy Unavailability.

The calculation of SEU and all other preventive maintenance activities shall consider this size of crew and the mentioned working time constraints.

[C11-SYST-010016]

Definition of Converter Forced Energy Unavailability (FEU): The Forced Energy Unavailability (FEU) has the meaning given in (CIG_TB590) edition 2014.

Definition of Annual Converter Forced Outage (AFO):

- AFO_Pole corresponds to the sum of number of Converter Forced Outages related to each HVDC Pole Link independently (1 and 2) in a Monitoring Year on the Offshore Converter Station and the Onshore Converter Station. In this case, the loss of transmission capability is maximum of 50%.
- AFO_Bipole corresponds to the sum of number of Converter Forced Outages related to the whole HVDC Link in a Monitoring Year. AFO_Bipole is counted only in cases where the HVDC Link lost 100% of its transmission capacity as a direct or immediate consequence of a single event.

[CBOD-SYST-100002]

If the Offshore Converter Station is in Scheduled or Forced Outage, it shall be possible to operate the Onshore Converter Station in STATCOM mode.

[CBOD-SYST-100056]

Definition of Energy Unavailability (EU): see (CIGRE_TB590) edition 2014, Energy Unavailability definition.

Definition of Energy Availability (EA): see (CIGRE_TB590) edition 2014, Energy Availability definition.

[C11-SYST-010017]

The HVDC Link excluding HVDC Cable Link shall meet the availability and reliability guaranteed performances of Table 5.

EA	$\geq 98.5\%$
SEU Threshold	$SEU \leq 0.7\%$ per Monitoring Year during the MAD Monitoring Period except the year with preventive maintenance for neutral yards. In this year, $SEU \leq 0.82\%$
FEU Threshold	$FEU \leq 0.54\%$ per Monitoring Year during the Taking Over Monitoring Period
Annual Converter Forced Outage (AFO) Threshold	$AFO_Pole \leq 3$ during the Taking Over Monitoring Period $AFO_Bipole < 0.04$ (design value)

Table 5 – Availability and reliability performance requirements

Performances shall be measured as defined in (B22c).

[C11-SYST-010018]

The interval period between two consecutive Converter Scheduled Outages for Preventive Maintenance (as defined in (B22c)) shall be:

- Not less than two (2) Monitoring Years (biennial maintenance campaigns) for each HVDC Pole Link. During the first Monitoring Year, one of the two HVDC Pole Links shall be in Scheduled Outage for preventive maintenance.
- Not less than four (4) Monitoring Years for neutral yards. During Taking Over Monitoring Period, at least one preventive maintenance on neutral yards shall be performed. Neutral yards are

defined as the equipment on Converter Station where a maintenance activity on these equipment require the outage of the whole HVDC Link. Contractor shall list those equipment.

Each HVDC Pole Link shall have preventive maintenance at different time period than the other HVDC Pole Link, except the maintenance of neutral yards. When preventive maintenance activity requires an outage of both HVDC Pole Links, efforts shall be made to minimize the outage duration.

- The maximum allowable duration for preventive maintenance shall be 5 consecutive days for each HVDC Pole Link (including the duration for the Preventive Maintenance of neutral yards when applicable).
- The maximum allowable duration for preventive maintenance of neutral yards shall be 1 day.

In exceptional circumstances such period as defined by Contractor may be reduced/increased by the Employer for one specific Outage by a few months with no resulting impact on equipment performance and associated warranties.

The interval between two preventive maintenance visits (e.g. equipment inspection) performed while the Offshore Converter Station in in operation shall not be less than 3 months.

The interval between two preventive maintenance visits (e.g. equipment inspection) performed while the Onshore Converter Station in in operation shall not be less than 1 months.

[\[C11-SYST-010111\]](#)

In addition to the SEU requirements, the time necessary to perform preventive maintenance tasks while the Onshore Converter Station is in operation shall be limited to an amount of 1200 man-hours (for the Employer maintenance staff as described in [\[C11-SYST-010015\]](#)) per Monitoring Year on the whole lifetime. Values shall be measured and compliant with the target requirement during the Monitoring Period.

[\[C11-SYST-010108\]](#)

For each Scheduled/Forced Outage onshore, the Contractor must provide a written confirmation to de-energize or/and to energize the HVDC link. This confirmation shall be communicated by means of a written report following a defined template (to be defined between Contractor and Employer) as well as an email to the address given by the Employer.

No other communication method shall be accepted.

[\[C11-SYST-010159\]](#)

Total maintenance (with and without outage) workforce for one year for the Offshore Substation including all the preventive maintenance, the under-water survey, coating inspection shall be less than 3500 man-hours.

7.2.2.2 Reliability and availability requirements during commissioning

[\[C11-SYST-010162\]](#)

Definition of Commissioning Forced Outage:

A Commissioning Forced Outage is defined as an Outage in section 7.2.2.1 leading to an impossibility to inject energy at PCC-AC Onshore for Producer as planned by his production plan.

The amount of energy that the Producer could not inject at PCC-AC as planned by his production plan during the Commissioning Forced Outages between MAD and Taking Over shall not exceed 1440000 MWh.

7.2.2.3 Reliability and availability assessment criteria

[\[C11-SYST-010020\]](#)

The fulfilment of the requirements on reliability and availability shall be demonstrated through studies at engineering stage and in operation, including studies performed in similar HVDC projects.

7.3 Transmission losses

7.3.1 Definition and determination of transmission station losses

[B21ai-LOSS-110022]-N3

The **Transmission Station Losses** are defined as the value of losses between PCC-AC and PCC-DC for each Converter Station (inc. fixed and variable losses) reached on the different reactive power, active power stated hereafter and for ambient and operation conditions as specified in (IEC61803).

The Transmission Station Losses shall be assessed for each Converter Station independently.

The Contractor shall provide a table stating the Transmission Station Losses for each of the operational conditions mentioned below:

- Active power at PCC-AC Offshore from 0 MW to Guaranteed Active Power Link Capacity (intervals of power not higher than 10% of the Guaranteed Active Power Link Capacity)
- Reactive power at PCC-AC of Q_{min} , Q_{max} , and $Q=0$, with Q_{min} and Q_{max} corresponding to the reactive power requirement as mentioned in section 8.3.1.
- Same active and reactive powers shall be considered for two poles of HVDC link.
- For all of the operating points described in the previous bullet points, the Converter Station shall be in non-blocking state.
- Nominal DC voltage at PCC_DC_offshore is 525 kV
- Cable resistance in accordance with maximum resistance value as described in (BOD_02_Interface_Basis_of_design) for contract execution phase.
- In operating mode OM1 (8.4.1) and OM3 (8.4.3).

[B21ai-LOSS-110023-a]-N3

The determination of the Transmission Station Losses at Design and FAT stages shall be carried out considering the guidelines given in (IEC61803) and (IEC62751).

[B21ai-LOSS-110023-b]-N3

The losses of each equipment/system included in the scope of supply shall be taken into account in the determination of the Transmission Station Losses.

[B21ai-LOSS-110023-c]-N3

Losses determination shall be described in a detailed report and separated in different parts. Following shall be considered in Transmission Station Losses:

- All part in clause 5 of (IEC61803) and appendix A of (IEC62751) when applicable.
- Converter transformer losses shall be computed according to (B22a_i), section "Losses".
- If active harmonic damping is used, it is admitted that it should not be taken into consideration as part of Transmission Station Losses as long as such function is an adaptive control active when the harmonic content at time of operation exceeds the limits fixed in section 8.2.2.
- An unmanned Offshore Substation is to be considered for the calculation: all loads that are not required for unmanned operation shall be considered out of service. As such:
 - o Only the relevant Heating, Ventilation and Air Conditioning for operation (e.g.: ambient conditions control for some equipment or C&P rooms) shall be considered in operation.
 - o Cranes shall be considered out of service, except its monitoring.
 - o Only relevant lighting shall be considered in operation.
 - o All diesel generators shall be considered in standby mode, etc.

7.3.2 Losses statcom case determination

[B21ai-LOSS-110491]-N3

An additional power losses calculation shall be supplied with the Onshore Converter Station operating in STATCOM (with DC cable disconnected), considering:

- Reactive power of Q_{min} , Q_{max} and $Q=0$
- Nominal DC voltage

7.3.3 Contractual assessment of transmission station losses

[B21ai-LOSS-110590]-N3

The Contractor shall indicate the Transmission Station Losses for each Offshore and Onshore Converter Station for the following operating points, in operating mode OM1 (8.4.1):

Offshore converter station			Onshore converter station		
Losses	P (% of Guaranteed Active Power Link Capacity)	Q (pu)	Losses	P (% of Guaranteed Active Power Link Capacity)	Q (pu)
L_off_0	0	0	L_on_0	0	0
L_off_25	25	0	L_on_25	For contractual purpose, a resistance of $R = 1 \Omega$ shall be considered for each HVDC pole cable and the DMR.	0
L_off_50	50	0	L_on_50		0
L_off_75	75	0	L_on_75		0
L_off_100	100	0	L_on_100		0

Table 6 – Converter Station losses

For the operating point at 0 MW / 0 Mvar, Converter Stations are in a non-blocking state, i.e. in idling operating state as defined in (IEC62747).

For contractual purpose only, Transmission Station losses include exclusively:

- Valves
- Converter and DC reactors
- DC chopper if any
- Transformer
- Auxiliaries essential to maintain operation of the HVDC system (cooling, monitoring, ventilation, C&P, HVAC, aux power)
- Platform systems (water, HVAC, emergency lights, communication, safety, etc.)

The losses shall be calculated with the conditions listed in Table 7.

	Onshore	Offshore
Ambient temperature	20°C	20°C
Atmospheric pressure	101.3 kPa	101.3 kPa
Relative air humidity	< 60 % RH	< 60 % RH
AC frequency	50 Hz	50 Hz
AC system voltage at PCC_AC	400 kV	132 kV
Number of power transformer in operation	6 x single phase	4 x tri phase
Tap position on power transformer	rated tap	NA (no tap changer)
Reference temperature of transformer windings	85°C for ON cooling mode / 90°C for OD cooling mode	
Sea water temperature	10°C	10°C

Table 7 : Operation conditions for loss calculation

Total Converter Losses are defined as

$$L = 0.13 \times (L_{\text{off}_0} + L_{\text{on}_0}) + 0.39 \times (L_{\text{off}_{25}} + L_{\text{on}_{25}}) + 0.12 \times (L_{\text{off}_{50}} + L_{\text{on}_{50}}) + 0.16 \times (L_{\text{off}_{75}} + L_{\text{on}_{75}}) + 0.20 \times (L_{\text{off}_{100}} + L_{\text{on}_{100}})$$

The Total Converter Losses shall not exceed the **Guaranteed Maximal Total Converter Losses** (defined in (Exhibit_6)).

7.3.4 Transmission Station Losses calculation during design stage

[B21ai-LOSS-110026]-N3

During design stage, the Transmission Station Losses shall be re-evaluated via two different means:

- By the power losses calculation that will be given as the sum of the mathematical calculation of each equipment losses.
- By the Electro-Magnetic Transient Program calculation, coupled with a thermal model. Models shall be provided by Contractor to Employer for validation.

[\[B21ai-LOSS-110027\]-N3](#)

The power losses justification must be provided not more than one year after the Commencement Date (during Design Stage).

[\[B21ai-LOSS-110028\]-N1](#)

The EMT model for losses justification shall be consistent with the rest of the EMT simulations and model. Employer shall have the possibility to validate the given results.

7.3.5 Transmission Station Losses verification using measurements during FAT

[\[B21ai-LOSS-110025\]-N3](#)

Not more than three months after the end of the Factory Acceptance Tests (FAT): The Contractor shall re-evaluate the Transmission Station Losses calculation, using the measurements of equipment/system losses (in particular for the converter transformer as stated in (B22a_i) and the measured characteristics (in particular for the converter valves e.g.: resistance measurements) established during the FAT, and considering the requirements made in (B22a_i), (B22a_iv), (B22av_2) and (IEC62751). The measurement on the auxiliary systems on site shall also be included to establish to Transmission Station Losses.

The Contractor shall propose prior to this re-evaluation a clear and detailed methodology in order to include these measurements, for review and approval by the Employer.

For equipment/systems that have factory test conditions (both operational and environmental) different from the operating conditions stated in section 7.3.1 or (IEC62751) or (IEC61803), suitable methods to correct such conditions shall be used by the Contractor and detailed in its report. Any discrepancy between the test and the requested conditions shall be duly highlighted and justified for the Employer's acceptance.

The **Actual Station Losses** correspond to the values of Transmission Station Losses for both Onshore and Offshore Converter Stations, as evidenced during the FAT / in calculation according to relevant IEC standards (IEC61803) and (IEC62751) in accordance with the requirements stated in section 7.3.1.

Actual Total Converter Losses is the value of Total Converter Losses with the **Actual Station Losses**. Each Transmission Station Losses of Table 6 shall be defined.

7.3.6 Transmission Station Losses – real time continuous measurement – non contractual

[\[B21ai-LOSS-110501-a\]-N1](#)

The Transmission Station Losses shall be continuously assessed electrically via CT/VT measurement. Accuracy shall be less than 100%.

[\[B21ai-LOSS-110501-b\]-N1](#)

The valve losses shall be continuously assessed by estimating losses dissipated via the cooling system (as referred in section 4.3.2 "Thermal measurement of the valve section losses" of (CIGRE_TB844) based on inlet/outlet temperature and flow rate of the cooling).

[\[B21ai-LOSS-110501-c\]-N1](#)

Active and reactive power load of Auxiliary Power Supply equipment shall be continuously assessed.

[\[B21ai-LOSS-110501-d\]-N1](#)

It shall be possible to display and export trends in the HMI of the Converter Station and TFR, corresponding to electrical measurement of Transmission Station Losses, load of auxiliary power supply, coolant inlet temperature, coolant outlet temperature, coolant flow rate.

8 System basis of design

8.1 Employer AC system description

8.1.1 AC system voltage and frequency

[\[CBOD-ACGR-100003\]](#)

Nominal operating voltage is 400 kV onshore and 132 kV offshore.
Nominal frequency is 50 Hz onshore and offshore.

8.1.2 AC short circuit levels

[\[C11-ACGR-010150\]](#)

All the performances of the Technical Specification shall be guaranteed within the ranges of AC short circuit levels stated in this section.

8.1.2.1 Onshore

[\[C11-ACGR-010078\]](#)

Onshore AC short circuit level is stated in Table 8. For any value of short circuit level in between the indicated values below, the Onshore Converter Station shall be capable of staying connected and the Contractor shall list the limitations on the performances. These values will be confirmed at (Call_off).

Short Circuit Level (SCL) kA	HA1, HA2	GLC	BNO
Minimum	2.96	Call-off	Call-off
Maximum	40	Call-off	Call-off

Table 8 – Short circuit levels onshore

[\[C11-ACGR-010151\]](#)

If the HVDC Link must switch between operating control modes (grid forming mode/regular and vice versa) to operate efficiently at different short circuit levels, the transition between those operating modes shall be automated and smooth/bumpless.

8.1.2.2 Offshore

[\[C11-ACGR-010152\]](#)

One or several Offshore Wind Farms (from one or more than one Producers) with a total rating indicated in section 7.1 will be connected to the offshore AC network at PCC_AC_offshore. Their maximum short circuit current for the rating coming from the OWF shall be considered to be 1.4 times the nominal current of a wind turbine at 132kV. This value will be updated during execution phase by Producer, see (*BOD_02_Interface_Basis_of_design*). Maximal permanent (I_k) and peak (I_p) short circuit current as defined in (*IEC60909_0*) shall be determined by the Contractor in a study provided to the Employer.

[\[C11-ACGR-010153\]](#)

The equipment rated short circuit current for the Offshore Converter Station HV equipment shall be determined according to [\[C11-ACGR-010152\]](#) and with the contribution of HVDC Link.

8.1.3 Background harmonics

8.1.3.1 Onshore

[\[B21ai-GENE-110419\]](#)

The voltage harmonics background levels at the Employer's Existing AC Substation (excluding the connection from the Onshore Converter Station) is provided in (BOD_02_Interface_Basis_of_design)

8.1.3.2 Offshore

As the offshore network is only made up of the Converter Stations and Offshore Wind Farms (OWF) with their Inter Area Cables, there are no background harmonics. OWF are requested to be compliant with (Art_8_3_1) and (Arr_09_06_2020).

8.1.4 Harmonic impedances

8.1.4.1 Onshore

[\[B21ai-SITE-110010\]](#)

The AC network harmonic impedances at the Employer's Existing AC Substation (excluding the connection from the Onshore Converter Station) is provided in (BOD_02_Interface_Basis_of_design). The locus of the AC system impedances for varying operating conditions is defined to be within the envelope of these areas.

If relevant, Contractor shall consider the impact of the impedance between Onshore Converter Stations, if being in his scope of supply.

8.1.4.2 Offshore

[\[B21ai-SITE-110571\]](#)

The offshore AC network harmonic impedances depend both on:

- The Inter Area Cable data (including layout and electrical characteristics) and
- The model of the wind turbines and wind farm controller. As this data will only be available in an advanced stage of the execution, Contractor shall make use of its own experience with such projects (with OWF) for his design studies. Precise data will be provided according to (BOD_02_Interface_Basis_of_design) and if IP clauses with Producer allow it.

8.2 AC system performance requirement

8.2.1 Voltage and frequency ranges

[\[C11-ACGR-010076\]](#)

Converter Stations shall be capable of staying connected to the network and capable of operating at maximum active power capacities and reactive power as described in 7.1 and 8.3.1, within the ranges of the network voltage at the PCC-AC Onshore and PCC-ACC offshore specified in Table 9. Nominal operating voltage is 400 kV onshore and 132 kV offshore. 1 pu is 400 kV and 132 kV for onshore and offshore respectively.

	AC voltage range		Time period
	Value [kV]	Value [p.u]	mins
Onshore	340 – 420	0.85 – 1.05	Unlimited
	420 – 440	1.05 – 1.1	60
Offshore	112.2 – 118.8	0.85 – 0.9	60
	118.8 - 145.2	0.9 – 1.1	Unlimited

Table 9 : Onshore and offshore voltage ranges.

The time periods of the table above can occur an unlimited number of times. If voltage goes beyond the above-mentioned limits, the Converter Station shall not trip as long as the inherent capabilities of

equipment are not exceeded. In addition, fault ride through capabilities described in section 8.6.6 shall be fulfilled.

[C11-ACGR-010077]

Converter Stations shall be capable of staying connected to the network and capable of operating at maximum active and reactive power capacities, within the ranges of AC system frequency at the PCC-AC Onshore and PCC-AC Offshore specified by Table 10. Nominal frequency is 50 Hz onshore and offshore.

Frequency range	Minimum operating period
47.0 Hz - 47.5 Hz	60 seconds
47.5 Hz - 49.0 Hz	90 minutes
49.0 Hz – 51.0 Hz	Unlimited
51.0 Hz - 51.5 Hz	90 minutes
51.5 Hz – 52.0 Hz	15 min

Table 10 – Onshore and offshore AC system frequency variations

The time periods of the table above can occur an unlimited number of times.

The Converter Stations shall be capable of staying connected to the AC network for a maximum speed of frequency variation of 2.5 Hz/s, measured at any point in time as an average of the rate of change of the frequency for the previous 1 second.

[B21ai-SPGE-110468]

In normal operating conditions, the maximum allowable rms voltage step induced at PCC-AC by an action in the converter station shall not be greater than 3%. These actions include notably: operation on equipment, connection of the Converter Station, closing of circuit breakers, energisation of the HVDC Link.

8.2.2 Power quality and harmonics

[B21ai-HARM-110032]

For each Converter Station, the following requirements shall be fulfilled for any network conditions, defined in section 8.1, any topology at onshore AC substations or offshore AC substations and any Converter Stations operating conditions. The Contractor shall ensure that the total harmonic distortions (current and voltage) and the flicker at the PCC-AC stay within the limits defined below, so that it does not cause any impairment on the AC onshore and offshore grid.

NOTE: terms and definitions of this part are based on (IEC61000).

8.2.2.1 Compliance criteria

8.2.2.1.1 Flicker

[B21ai-HARM-110034]

The Onshore Converter Station shall not cause any deterioration in the existing flicker levels.

8.2.2.1.2 Current harmonics

8.2.2.1.2.1 HVDC cables

[C11-SYST-010103]

The equivalent harmonic disturbance current (refer to (CIGRE_TB811)), denoted I_{eq} , generated by the converter in the HVDC cables shall not exceed 400 mA.

The Contractor shall use the definition of I_{eq} in Equation 6-13 of (CIGRE_TB811), along with the following assumptions:

- The harmonic currents shall be from the worst-case operating point and include the effect of any cross-modulation of AC harmonics onto the DC system.
- The ITU-T psophometric system weighting system should be used for P_n (i.e. Appendix I of ITU-T of K.68).
- For frequencies above 6kHz, a psophometric weighting of 6kHz should be used.
- An H_n of 1 should be used. (Note: this may be updated after the results of the telecommunication infrastructure survey is undertaken, but before the contract is signed).
- The frequency range should be from 20Hz to 20,000Hz.
- I_{eq} should include the cable screen/sheath in addition to the main conductor currents.

In addition, the Contractor shall consider in their design the magnetic induction phenomenon on the HVDC cables and shall determine the electrical and magnetic interaction with existing infrastructure near the route of the DC cable. The requirement [C11-SYST-010043] of (B21a_iv) shall be satisfied.

8.2.2.1.2.2 Onshore

[B21ai-HARM-110036]-N3

The total harmonic currents at PCC-AC Onshore after connection of the Onshore Converter station shall not exceed the limits of Table 11.

	Rank	French limits (A rms)
Odd harmonics (individual)	3	19,702
	5 and 7	24,249
	9	9,093
	11 and 13	15,155
	>13	9,093
Even harmonics (individual)	2	9,093
	4	4,547
	>4	3,031
THD _i (%)	$\sqrt{\sum_{n=2}^{50} \left(\frac{I_n}{I_1}\right)^2}$	4.8

With I_1 nominal current value at 400kV.

Table 11 – Requirements for current harmonics in the onshore side

The total distortions at PCC-AC Onshore comprise of the background harmonics, the Onshore Converter Station emission, and any amplification of the background harmonics by the Onshore Converter Station impedance.

8.2.2.1.2.3 Offshore

[B21ai-HARM-110528]-N3

The Contractor shall make an allowance of time in order to modify the Offshore Converter Station control in coordination with the OWF in case of current harmonics issues declared by either the Employer, the OWF or the Contractor. See 8.2.2.1.4 for active control.

8.2.2.1.3 Voltage harmonics

[B21ai-HARM-110502]

Interharmonic level shall not exceed 0.2% from 50Hz until 5000Hz, in compliance with (61000_3_6).

[C11-SYST-010042]

The voltage Total Harmonic Distortion (THD) in the HVDC cables shall not exceed 2% of the nominal DC voltage. Harmonics up to 5 kHz and with a resolution of 1 Hz shall be considered.

8.2.2.1.3.1 Onshore

[B21ai-HARM-110037]

The total distortions at PCC-AC Onshore comprise of the background harmonics, the Onshore Converter Station emission, and any amplification of the background harmonics by the Onshore Converter Station impedance.

The total voltage harmonics level after the connection of the HVDC Link at PCC-AC Onshore shall respect following total limits $V_{lim,n}$:

- sum of the value indicated in 8.1.3.1 plus 0.2 point for ranks 2 to 50.
- For ranks above 50, the harmonic voltage level shall be less than 0.2% of the fundamental.

In case active damping is used on a given harmonic rank, shall the background harmonic level be higher than the one described in 8.1.3.1, the amplification factor shall remain constant or below the initial factor (i.e. the factor designed with the harmonic background level of 8.1.3.1).

The THD increase after connection of the HVDC Link shall be lower than the value indicated in 8.1.3.1 plus 0.3 point. The THD computed only for ranks 51 to 100 shall be less than 1%.

All the performances of the Onshore Converter Station described throughout the specification documents (including noise, RFI requirement, active and reactive power capability, etc.) shall be guaranteed for total voltage harmonics level below $V_{lim,n}$.

The Onshore Converter Station performances described in sections 7, 8.2, 0, 8.4 and 8.6 (e.g.: active and reactive power capability, AC voltage and frequency ranges for which the Onshore Converter Station shall stay connected, etc.) shall remain guaranteed for network voltage harmonics and THD levels below values indicated in Table 12.

As such, all equipment shall be rated to withstand these levels.

However, other requirements related to noise for instance may be relaxed for network voltage harmonics level above the total voltage harmonics limits $V_{lim,n}$.

[B21ai-SITE-110009]

Throughout the lifetime of the Converter Station, harmonic voltages at PCC-AC Offshore and at Employer Existing AC substations may increase until the values indicated in Table 12.

		Voltage (% of fundamental)
Odd harmonics (individual)	3	<4%
	5 & 7	<4%
	9	<2%
	11 & 13	<3%
	15 & 21	<1%
	17 & 19	<2%
	23 & 25	<1.5%
Even harmonics	2	<3%
	4	<2%
	6 to 24	<1%
THD (%)		<6%

Table 12 – Maximum AC network harmonic levels during the lifetime of the station

8.2.2.1.3.2 Offshore

[B21ai-HARM-110038]

Harmonic contribution of the Offshore Converter Station at PCC-AC offshore shall be below the planning level in table 2 of (IEC61000_3_6) divided by two.

8.2.2.1.4 Harmonic filtering and damping

[\[B21ai-COBA-110537\]](#)

Contractor shall implement an active harmonic filtering control, capable of filtering voltages and currents at each Converter Pole up to rank 20. This control will depend on the OWF and its design shall therefore be further discussed with Employer during execution stage.

[\[B21ai-COBA-110538\]](#)

In addition, active damping shall also be possible by modifying the damping of each Converter Pole control impedance for frequencies up to 2 kHz.

The Contractor shall refer to section 7.3 of (CIGRE_TB754) for the definitions of active filtering and damping. Active damping can be seen as a “virtual resistor” to as to provide additional network damping. The Converter Station shall create an opposing voltage to achieve the desired damping of the whole system through impedance adaptation.

This function shall mitigate distortion magnitudes at specific individual harmonics or over a broad harmonic frequency range.

It shall be possible to enable/disable such function by the operator.

[\[B21ai-COBA-110628\]](#)

Contractor shall implement a low frequency damping controller for the Offshore Converter Station, in the range of 1 to 40 Hz.

8.2.2.2 Acceptance

[\[B21ai-HARM-110039\]](#)

If for a particular harmonic rank, meeting the compliance criteria requires harmonics filters in the scope of the Converter Station or any other modification in the design, this shall be brought to the knowledge of the Employer for its review and acceptance. The Contractor shall detail which points or area within the harmonic impedances lead to such design change for further assessment by Employer.

8.2.3 Negative sequence of AC voltage

[\[B21ai-GENE-110418\]-N2](#)

Each Converter Station performance shall be guaranteed for a negative phase sequence component of (based on (IEC61000_4_30)):

- 1% of the AC voltage at the PCC-AC measured over a 10 min period.
- 2% of the AC voltage at the PCC-AC measured over a 3 sec period.

Each Converter Station shall be able to withstand a maximum negative phase sequence component of 2% of the AC voltage at the PCC-AC (either continuous or short-term basis) without accelerated aging of the equipment. It shall remain connected and stable in such network conditions.

8.2.4 Earthing requirement

8.2.4.1 Onshore earthing system

[\[B21ai-EART-110572\]](#)

Earthing grid of the Onshore Converter Station shall be connected to the earthing of the Employer Existing AC Substations, and existing transmission lines, as specified in (B22b_i). The DSO network (if any) shall also be considered.

[C11-ACGR-010080]

For the Onshore Converter Station, earthing of the primary of the converter transformer shall be as follows:

$$Z_z + 3Z_e > 75 \Omega$$

with Z_z being the zero-sequence impedance of the converter transformer and Z_e the earthing impedance at primary neutral – seen from the 400kV network.

The ratio of zero sequence and direct sequence impedance between 1 and 3 at PCC-AC can be considered instead of the equation above.

8.2.4.2 Offshore earthing system

[C11-ACGR-010154]

The Offshore Converter Station shall have a reference to earth on the AC grid side on each bus bar section. Converter System Contractor shall determine, design and install the offshore earthing solution (rigid or impedance earthing with a dedicated impedance or at the neutral of the converter transformer) in compliance with:

- Protection requirements as described in section 8.11.4.
- Performance requirement in this document
- Rating of equipment and cables in the Offshore Converter Station.

8.2.5 Lightning protection requirements

The Onshore Converter Stations shall be protected against lightning (including AC, DC yard, neutral yard, cable terminations) according to Employer's requirements mentioned in (B22b_i).

8.3 HVDC system rating

8.3.1 Reactive power capability

[C11-SYST-010048]

For the HVDC Link (all values are for PCC-AC)

- Onshore Converter Station shall be designed to meet the reactive power requirements presented in **Table 13** at PCC-AC Onshore. This requirement encompasses the provision of reactive power within the specified AC voltage range and the active power up to Guaranteed Active Power Pole Capacity/ Guaranteed Active Power Link Capacity. (1pu = 400kV for voltage, 1pu = 1000 Mvar per pole and 2000 Mvar per HVDC Link for reactive power).

AC voltage range		Time periode	Reactive power range per Converter Pole		Reactive power range per Converter Station	
Value [kV]	Value [p.u]	[min]	Value [Mvar]	Value [pu]	Value [Mvar]	Value [pu]
340 – 380	0.85 – 0.95	Unlimited	0 – 320	0 – 0.32	0 – 640	0 – 0.32
380 – 420	0.95 – 1.05	Unlimited	-350 – 320	-0.35 – 0.32	-700 – 640	-0.35 – 0.32
420 – 440	1.05 – 1.1	60	At least -350	At least -0.35	At least -700	At least -0.35

Table 13: AC voltage and reactive power ranges for onshore converter station.

Reactive power and voltage diagram is illustrated in the figure below.

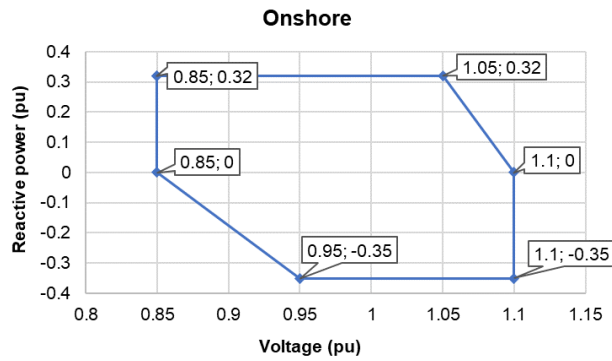


Figure 2: UQ diagram for Onshore Converter Station

The time periods mentioned in table above can occur an unlimited number of times. If voltage goes beyond the mentioned limits, the Converter Station shall not trip as long as the inherent capabilities of equipment are not exceeded. Moreover, maximum reactive power capability shall not suddenly drop when voltage crosses the voltage threshold mentioned above.

- Offshore Converter Station shall be designed to meet the following reactive power requirements at PCC-AC Offshore (1 pu = 132 kV for voltage and 1 pu = 1000 Mvar per pole for reactive power. This requirement encompasses the provision of reactive power within the specified AC voltage range and the active power up to Guaranteed Active Power Pole Capacity/ Guaranteed Active Power Link Capacity.

AC voltage range		Time period	Reactive power range per pole	
Value [kV]	Value [p.u]	[min]	Value [Mvar]	Value [pu]
138.6 – 145.2	1.05 – 1.1	Unlimited	-140 – 50	-0.14 – 0.05
125.4 – 138.6	0.95 – 1.05	Unlimited	-140 – 140	-0.14 – 0.14
118.8 – 125.4	0.9 – 0.935	Unlimited	-50 – 50	-0.05 – 0.05
112.2 – 118.8	0.85 – 0.9	60	-50 – 50	-0.05 – 0.05

Table 14: AC voltage and reactive power ranges for offshore converter station

Reactive power and voltage diagram is illustrated in the figure below for the voltage range [0.85 pu – 1.1 pu] at active power equals to rated capacity of the HVDC Link.

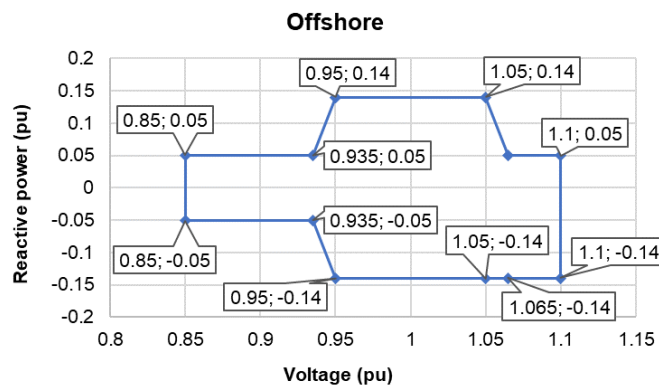


Figure 3: UQ diagram for Offshore Converter Station.

Active and reactive power capability is described in the figure below for the voltage range [0.95 pu – 1.05 pu].

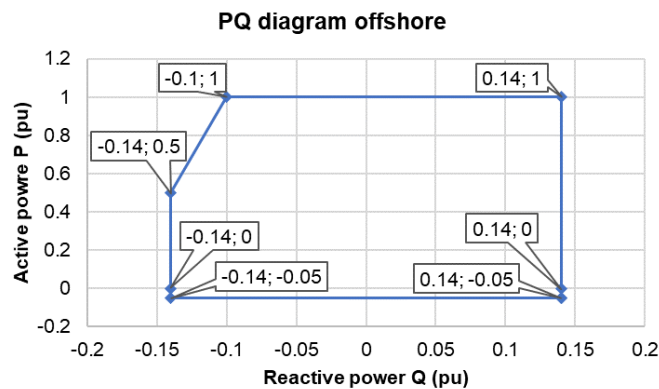


Figure 4: PQ diagram for Offshore Converter Station

The time periods mentioned in table above can occur an unlimited number of times. If voltage goes beyond the mentioned limits, the Converter Station shall not trip as long as the inherent capabilities of equipment are not exceeded. Moreover, maximum reactive power capability shall not suddenly drop when voltage crosses the voltage threshold mentioned above.

- In any case, the Converter Station shall only trip in case current ratings of equipment are exceeded and not based on a hard-set control threshold. As such, trip of Converter shall not be based on guaranteed PQU capability but on inherent limits.
- The Contractor shall therefore provide the maximal inherent PQU capability of the Converter Station, to be proved through a study.
- The reactive power capability limit of each Converter Pole shall be available at each electrical node corresponding to that Converter Pole. For example, the whole reactive power of Converter Pole shall be available at each MV side of converter transformer (IAC side of the transformer) if the other node is at 0 Mvar.

[C11-SYST-010049]

- It shall be possible to control the reactive power output of each Converter Pole and Converter Station independently from the other Converter pole and Converter Station.
- The time necessary to reach a new reactive power setpoint shall be in accordance with requirement [B21ai-VORE-110160] for the Onshore Converter Station and [B21ai-EMCO-110608] for the Offshore Converter Station, assuming that tap changer (if any) remains in its initial steady state tap position.
- It shall be possible to fully use the inherent reactive capacity of the Converter Stations across the complete active power operating range. The Converter System (Converter Stations) Contractor shall present the reactive power capacity in the range from zero active power to the Guaranteed Active Power Link Capacity/ Offshore Maximum Active Power in both directions, within the specified AC system voltages defined above.
- The Converter System (Converter Stations) Contractor shall submit the diagrams at the PCC-AC Onshore and PCC-AC Offshore [U,Q] to Employer for review and acceptance.
- For the Offshore Converter Station, the [U,Q] diagram shall be provided for 3 converter transformers or four converter transformers in operation.
- For voltage support during exceptional situations, refer to 8.6.6.
- An installation of additional reactive power compensation systems, e.g. capacitor banks and reactors, is not permitted to meet the requirements of the reactive power performance level.

8.3.2 DC transmission voltage and current range

[C11-SYST-010040]

The operational voltage and current range for DC power transmission shall be:

- Rated DC voltage for both cable and converter stations: $U_0 = \pm 525$ kV

- Maximum continuous DC voltage for both cable and converter stations (including harmonics, ripples and measurement tolerances): $U_m = 1.05U_0 = \pm 550$ kV
- Minimum DC voltage: to be defined by Contractor
- Nominal Rated Current for both HVDC Cable Link (including DMR cable) and Converter Stations (including harmonics, ripples and measurement tolerances) $I_m = 1970$ A

The DC voltage control shall keep the DC voltage at Offshore Converter Station as close as possible to its rated DC voltage. The maximum continuous DC voltage must not be exceeded at any time and location in the HVDC Link by the operation of converter system during operation.

[C11-SYST-010041]

A DC current limitation (Converter Station protection) shall be integrated to the protection system, intended to limit the current to a user-defined value as set by the Employer.

8.4 Operating modes

[C11-SYST-010050]

The HVDC Link shall be capable:

- Of exporting the active power fed into the offshore AC network at PCC-AC Offshore by the OWF.
- Of feeding the AC offshore network from the Onshore Converter Station to energise the IAC and/or the OWF.
- Of energising and starting the Offshore Converter Station from the Onshore Converter Station.

[CBOD-SYST-100004]

The operating modes of HVDC Link presented in Table 15 shall be fully supported by Contractor. For all operating modes, no permanent current flow through ground is allowed. The protection shall be designed and tested to ensure this requirement. The default operating mode is OM1.

Operating mode	Abbreviation
Bipolar with DMR	OM1
Symmetrical bipolar without DMR (rigid bipolar)	OM2
Asymmetrical monopolar	OM3
STATCOM onshore	OM4
Round power test (commissioning only)	OM5

Table 15: Operating modes for HVDC Link

[CBOD-SYST-100005]

Table 16 outlines the minimum requirements of transition capability from one operating mode to another one without power interruption.

*: no power interruption even in case of fault inside and outside the Converter Station.

From / To	OM1	OM2	OM3	OM4	OM5
OM1	/	Yes*	Yes*	Yes	/
OM2	Yes	/	/	Yes	/
OM3	Yes	/	/	Yes	/
OM4	/	/	/	/	/
OM5	/	/	/	/	/

Table 16: Transition between different operating modes for HVDC link

[CBOD-SYST-100006]

Given that the DMR and neutral section of the HVDC link are on the grounding, detecting faults in this area might be challenging. Contractor is required to propose efficient fault detection strategies to promptly identify any fault for DMR and neutral yards, thereby minimizing ground current.

[CBOD-SYST-100007]

Contractor shall detail the operating procedures to be implemented to ensure the transition from between different operating modes in the objective of maximizing transported energy by HVDC Pole Link and HVDC Link.

The transition between the different grid topologies shall not generate any disturbance or malfunction on the control system and in general in the HVDC Link operation.

8.4.1 Operating mode OM1: bipolar with DMR

[CBOD-SYST-100008]

OM1 is the default operating mode of the HVDC Link. The transmission of Guaranteed Active Power Link Capacity shall be possible. The currents on the HVDC Pole Link 1 and on the HVDC Pole Link 2 can be different and the difference in current circulates through the DMR.

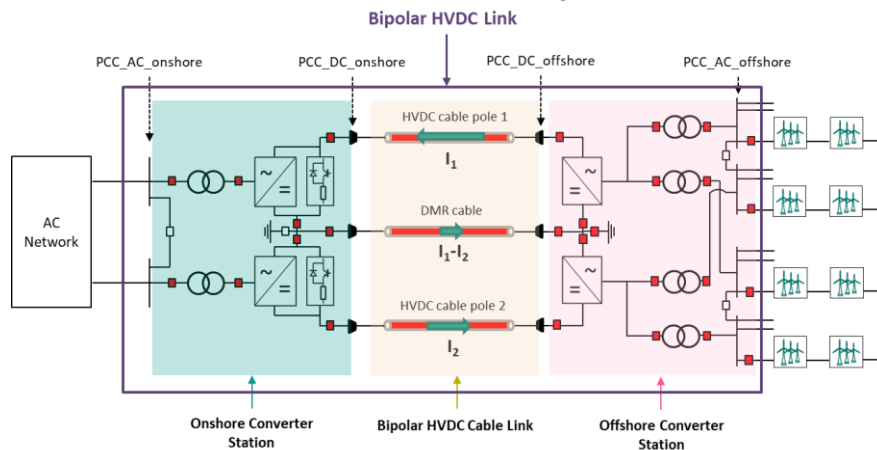


Figure 5: Operating mode 1 (OM1) schematic diagram of a HVDC Link.

8.4.2 Operating mode OM2: symmetrical bipolar

[CBOD-SYST-100009]

OM2 is an operating mode where the DMR is not available. In this mode, it is crucial to ensure a perfect balance of current on both Converter Poles to prevent any current circulation through the ground. The contractor shall propose an effective strategy for a transition from OM1 to OM2 without any power interruption, even in case of DMR fault detection. The proposal needs to be verified and validated by Employer. As an example, one possible approach to achieve current balance is by coupling offshore AC electrical nodes.

The transmission of minimum Guaranteed Active Power Link Capacity shall be possible and if necessary for operational reason, an active power reduction order shall be dispatched to the OWF.

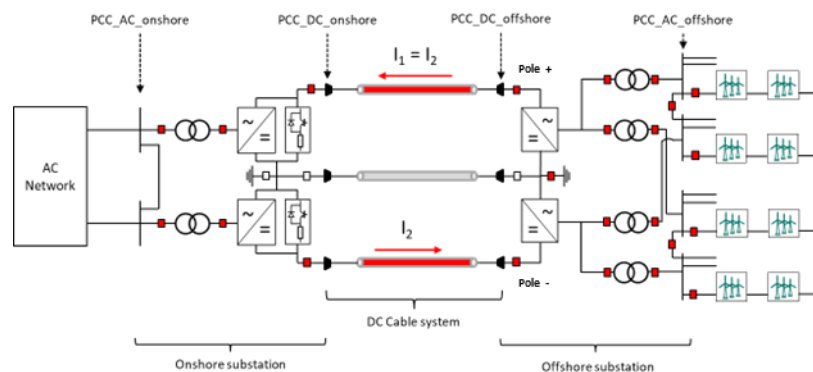


Figure 6: Operating mode 2 (OM2) schematic diagram of a HVDC Link.

8.4.3 Operating mode OM3: asymmetrical monopolar

[CBOD-SYST-100010]

The operating mode OM3, one of the HVDC Pole Link is not available (onshore Converter Pole, Offshore Converter Pole or HVDC pole cable), the HVDC Link shall be operated as an asymmetrical monopole with the remaining HVDC Pole Link (Figure 7). The transmission of Guaranteed Active Power Pole Capacity shall be possible (half of Guaranteed Active Power Link Capacity). It shall be possible to reconfigure the AC offshore substation to optimise the power generated by wind turbines. However, if the wind power production exceeds the Guaranteed Active Power Pole Capacity, it shall trip the relevant IAC feeder or to send an active power reduction to the OWF.

If possible, the Onshore Converter Pole of the out of service HVDC Pole Link shall be in STATCOM mode.

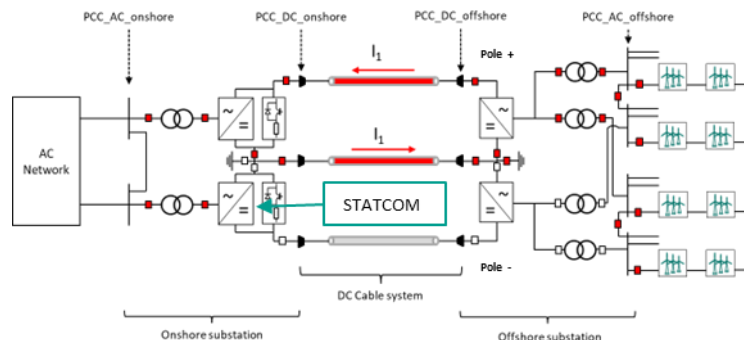


Figure 7: Operating mode (OM3) schematic diagram of a HVDC Link.

8.4.4 Operating mode OM4: STATCOM onshore

[C11-SYST-010052]

The Onshore Converter Station shall use its inherent reactive power capacity in STATCOM mode to control the AC voltage or reactive power with no active power transmission between the two Converter Stations. This requirement is applicable for both Onshore Converter Poles.

The STATCOM mode shall include U-mode and Q-mode.

[B21ai-COBA-110466]

It shall be possible to operate the Onshore Converter Station in STATCOM mode in case of works on the DC cable or on the Offshore Converter Station.

8.4.5 Operating mode OM5: Round power test for commissioning

[CBOD-SYST-100011]

To partially commission the HVDC link when the OWFs are unavailable, the operating mode OM5 is required. In this mode, power is continuously circulated between the Onshore and Offshore Converter Stations through both Converter Poles, HVDC cables and DMR. The power is transmitted from one Onshore Converter Pole to its corresponding Offshore Converter Pole and then returned to the Onshore Converter Station via other Converter Pole. The maximum power that can be sent shall be proposed by Contractor to Employer to not exceed the rating of DMR cable.

Figure 8 shows an example of schematic diagram of a HVDC link operated in OM5.

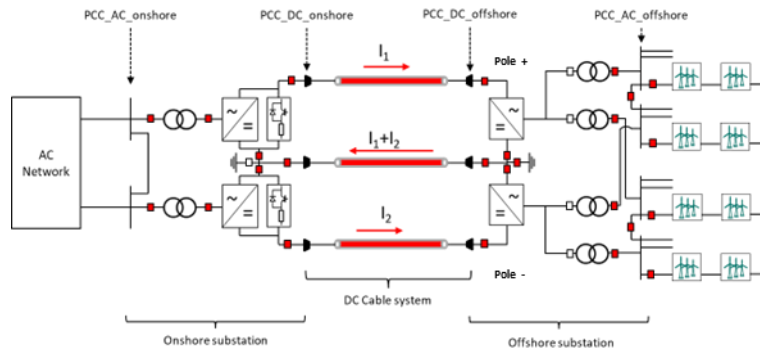


Figure 8: Operating mode 5 (OM5) schematic diagram of a HVDC Link.

8.4.6 Requirements related to start-up, shutdown and restart of the HVDC Link

8.4.6.1 Start-up requirement

[C11-SYST-010070]

It shall be possible to start the HVDC Link within a time shorter than 10 minutes, this time is defined between the following instants:

- From a state where all grounding switches of the Converter Stations are opened.
- To a state where Converter Stations are in (POW) status as described in section 8.6.8.

8.4.6.2 Shutdown requirement

[C11-SYST-010071]

When a shutdown instruction is initiated by the HVDC Link control (either automatically or manually from an HVDC operator), it shall be possible to access the valve hall in the Onshore Converter Station in less than 60 minutes from initiation of shutdown instruction (including discharge of the HVDC Link and time necessary for cool down period of valve hall as stated in [B22aiv-FUNC-450005] of (B22a_iv). The HVDC Link discharge time is defined between the following instants:

- From a state where Converter Stations are blocked
- To a state where pole earthing switches are closed.

Final value shall be communicated prior to Taking-Over by the Converter System Contractor to the Employer.

[CBOD-SYST-100012]

The shutdown instruction shall be applied either to the entire HVDC link or independently to each HVDC Pole Link. In case of a shutdown for a single pole, the required time is the same with the one mentioned in [C11-SYST-010071].

[B21ai-VORE-110593]

Emergency shutdown of either the HVDC Pole Link and the HVDC Link shall be possible either upon request of the Operator or in case of internal HVDC Link fault.

[C11-SYST-010072]

Additionally, the voltage threshold value to start the disconnection/earthing sequence shall be lower than 5% of 525 kV. The final value may be adjusted by Employer during design stage depending on the converter and cable studies.

[C11-SYST-010157]

The Contractor shall indicate if the Dynamic Braking System can be used to accelerate the discharge time of the DC Cable and the Converter Stations.

8.4.6.3 Restart requirement

[\[C11-SYST-010073\]](#)

Duration requirements for restart are described hereafter for different situations:

- Trip in an energised state of the entire Converter Station, of a single pole of Converter Station, HVDC Pole Link and HVDC Link: less than 30 minutes. Contractor shall optimize the duration depending on the equipment ratings and states (e.g.: charging state of valves, temperature of pre-insertion resistor, etc.). Contractor may highlight a critical charging state enabling a quick restart (less than 3 minutes).
- Trip at early stage of starting sequence where the Converter and HVDC Link have not yet been charged up to a critical point: as fast as possible and less than 3 minutes.

In all the cases described above, it is assumed that protections are acknowledged if previously triggered. Trips may be due to an external event (not due to Converter malfunction) or due to an obvious cause clearly identified and cleared before restart.

In any case, it shall be possible to launch three consecutive starting sequences with at most 30 minutes between each attempt.

The Converter System Contractor shall highlight to Employer limitations due to valve submodule discharging if any.

8.4.6.4 Cable discharge procedure

[\[C11-SYST-010074\]](#)

The Converter System Contractor shall provide a discharge system and protocol which ensures that after any shutdown (either initiated by e.g. an HVDC operator or by a trip) of the HVDC Link, the HVDC Link including both HVDC cables and DMR is safely discharged and can be safely energized afterwards (on the basis of its own converter design, and the rest of these technical specifications, in particular provisional HVDC Cable System design data also provided in (BOD_02_Interface_Basis_of_design)).

During project execution, Converter System contractor shall confirm the design of the discharge protocol (if any update is necessary from tender stage). Cable Contractor will confirm its compliance with this protocol.

[\[C11-SYST-010062\]](#)

The Converter System Contractor shall provide a means of safely discharging the HVDC cables without polarity reversal.

8.5 HVDC system grounding

[\[CBOD-SYST-100013\]](#)

The Converter Station Contractor shall determine, design and install the earthing solution (rigid or impedance earthing with a dedicated impedance) of the DMR. The earthing point shall be either on onshore converter station and offshore converter station but in the same time, only one grounding point of DMR shall be allowed.

[\[CBOD-SYST-100014\]](#)

The Converter Station Contractor shall justify the design of earthing point of DMR to be compliance with the insulation coordination as mentioned in 8.10.2.

[\[CBOD-SYST-100015\]](#)

If the proposed DMR earthing design shows a non-zero current crossing at the AC grid side during fault event, the Converter Station Contractor shall propose means to overcome this problem.

8.6 HVDC System dynamic behaviors

[CBOD-SYST-100016]

The system dynamic behavior shall be respected for the entire Converter Station and for each Converter Pole regardless the coupling mode of AC side. This requirement applies for both Onshore Converter Station and Offshore Converter Stations.

8.6.1 Dynamic response definitions

[B21ai-DYNR-110467]

The following definitions shall apply regarding performances:

- **Reference:** Signal on which a step is applied
- **Measure:** Physical measured value of the signal (e.g.: AC voltage, power at PCC-AC)
- **Overshoot:** Difference between maximal value reached and the reference value during a transition
- **Activation time:** Time difference between the change in value of the reference signal (step order) and the 10% value of the step
- **Rise time to 90%:** Time difference between the change in value of the reference signal (step order) and the time when the measure reaches 90% of the final reference value (with a possible overshoot)
- **Settling time to 95%:** Time difference between the change in value of the reference signal (step order) and the time when the measure remains in the [95%-105%] of the final reference value

The above-mentioned definitions are illustrated on Figure 9 and apply to a step.

- **Reference:** Signal on which a ramp is applied
- **Tracking error time:** Maximal difference between the time when the reference reaches a value and the time when the measure reaches the same value during ramping
- **Tracking error value:** Maximal difference between the values of reference and measure at a given time during ramping
- **Steady-state static error:** Difference between the final steady-state value of the measure and the reference

The above-mentioned definitions are illustrated on Figure 10 and apply for a ramped signal.

Definitions of Figure 9 and Figure 10 are linked: For example, assuming a first order system, settling time to 95% with a step as input is 3τ (τ defined as time constant to reach 63% of step value) and tracking error time is equal to τ for the same system with a ramp as input instead of a step. As such, for sake of clarity, tracking error time is used to define dynamics on ramped signals hereafter.

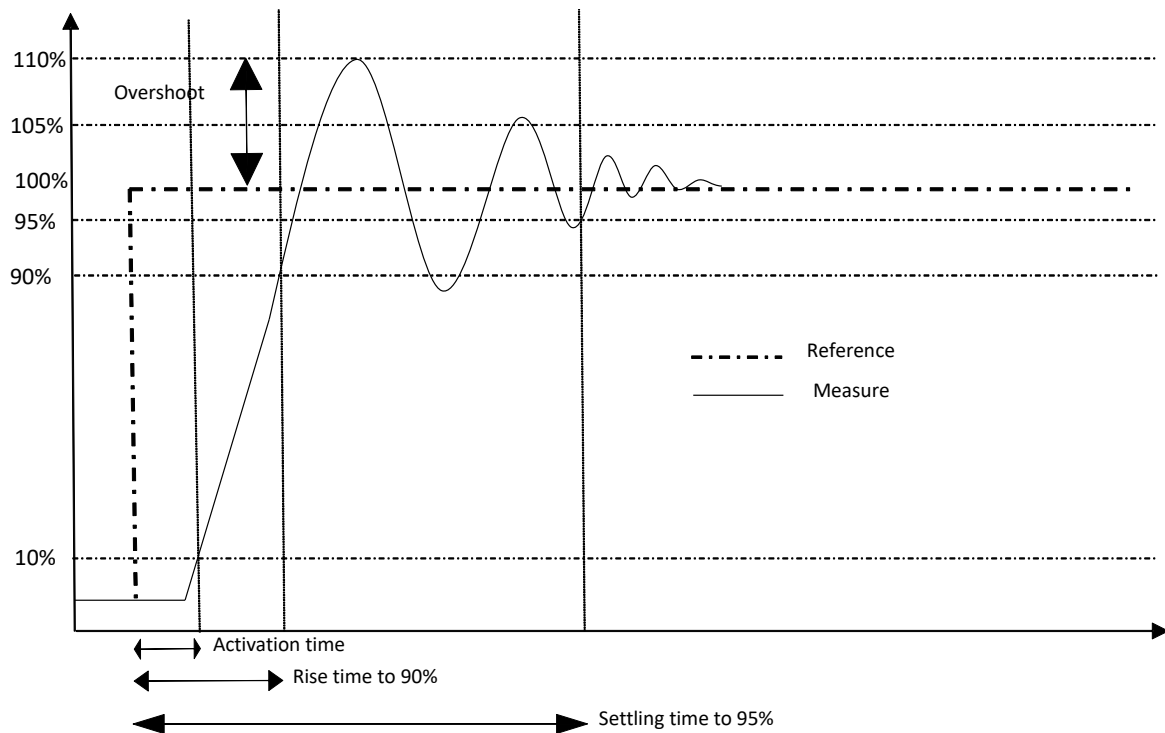


Figure 9 – Dynamic response definition – part 1

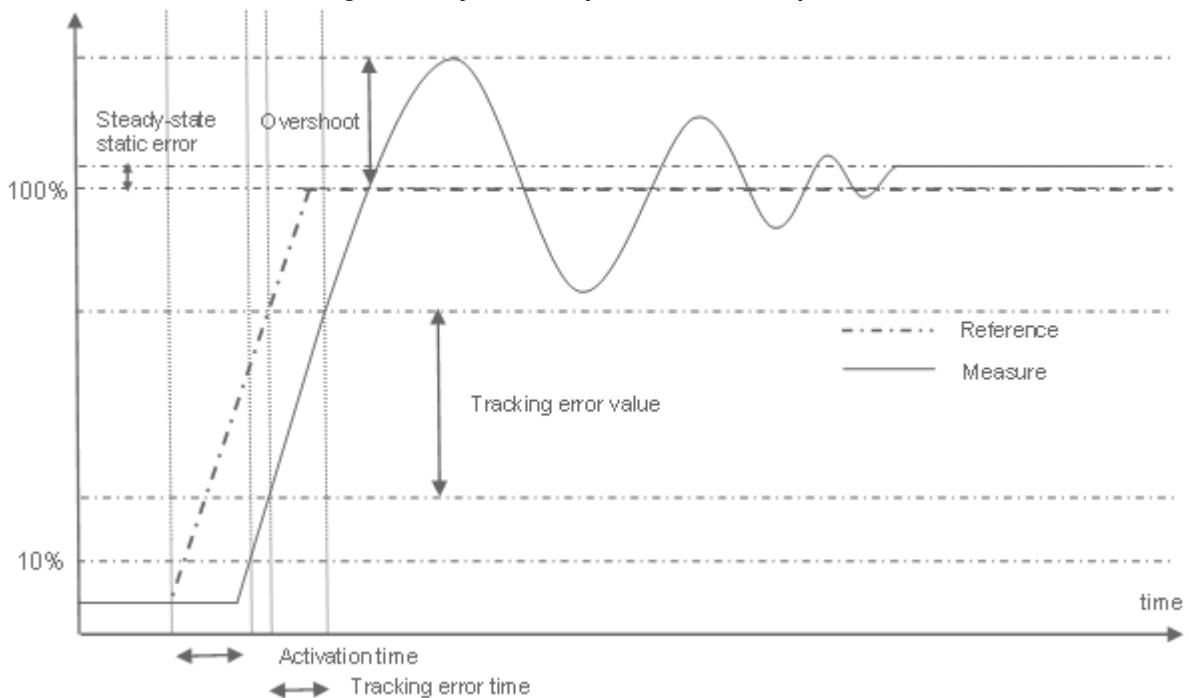


Figure 10 – Dynamic response definition – part 2

8.6.2 Active power control

[B21ai-APCO-110592]

The control system shall measure the active power flow at each PCC-AC Offshore Interface, PCC-AC-Offshore and at PCC-AC Onshore. The control system shall indicate the Maximal Active Power Capacity Available in real-time at each Converter Pole. These values may be different from the Guaranteed Active Power Pole Capacity because of following events (list non exhaustive):

- Loss of cooling capacity,
- Disconnection of converter transformer on the Offshore Converter Station, etc.

8.6.3 Onshore AC voltage & reactive power control

8.6.3.1 General

[\[B21ai-COBA-110181\]](#)

The Onshore Converter Station shall operate in one of the following selectable system operating control modes:

- Bipole Mode (BPM): This is a normal operating mode where there is an automatic management of reactive power/voltage of the HVDC Onshore Converter Station by fully controlling and coordinating the two Converter Poles. The operating configuration and sequences (even protective sequences) are automatically coordinated and synchronized.
- Single Pole Mode (SPM): This is a normal operating mode which allows an automatic management of reactive power/voltage of each Onshore Converter Pole independently of the status of the other Onshore Converter Pole (Onshore Pole 1 and Onshore Pole 2)

[\[B21ai-COBA-110082\]](#)

Every control function for reactive power/voltage management and for stability under dynamic conditions shall be available in these two modes of operation control.

[\[B21ai-COBA-110083\]](#)

The default operating control modes shall be BPM for reactive power/voltage. In this mode, the same values shall automatically be applied for both poles.

[\[B21ai-COBA-110085\]](#)

In case of bus splitting at the Onshore PCC-AC (two electrical nodes with each Pole on one node at Onshore Converter Station), operation control mode is not expected to automatically switch from BPM to SPM. This action will be performed by the operator.

[\[B21ai-COBA-110104\]-N3](#)

Transitions between control functions shall be stable, smooth, bump less and shall not cause any system status change or malfunction in HVDC Link operation. Such transitions shall be achieved with a change in output power of less than 1 MW and a change in reactive power of less than 1 MVar and shall not cause any electrical disturbances or unintended control or protection actions.

A simultaneous AC network event during a transition shall not lead to a trip of the Converter Stations in compliance with section 8.6.6.

In particular, transitions between BPM/SPM shall be done independently of the reactive power control (U-mode and Q-mode) in a smooth and bumpless way.

[\[B21ai-VORE-110131\]](#)

Two control modes shall be available at the Onshore Converter Station, namely AC Voltage Control (U-Mode) and Reactive Power Control (Q-Mode). The choice of the control mode in each Converter Pole is independent from the other.

8.6.3.2 AC voltage control (U-mode)

[\[B21ai-VORE-110132\]](#)

In U-Mode, the Onshore Converter Station shall be capable of contributing to voltage control at PCC-AC Onshore by provision of reactive power within the capability limits defined for the Onshore Converter Station, by means of an AC voltage target which can be set by the Operator.

[\[B21ai-VORE-110134\]](#)

The AC voltage controller shall be able to take its reference U_{target} locally from HMI (U_{refLOCAL}) or remotely from RCC – Remote Control Centres ($U_{\text{refREMOTE}}$).

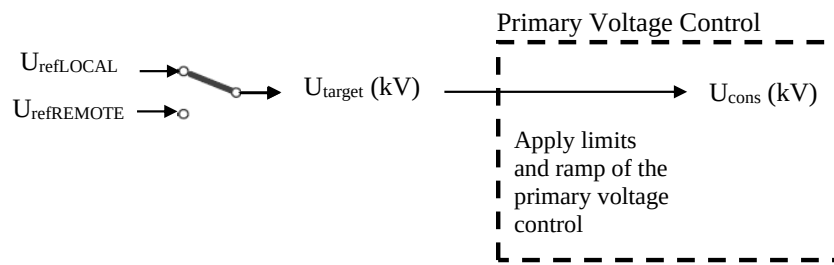


Figure 11 – Reference for AC primary voltage control

U_{cons} is the reference for primary AC voltage control after limitations and ramp.

[B21ai-VORE-110137]

For each Converter Pole, at least PCC-AC Onshore voltage measure (U_{PROD}), PCC-AC Onshore voltage setpoint (U_{cons}), lambda and ramp rate, as well as the margin to the capability limits of the Onshore Converter Station shall be available on the OWS and RCC.

8.6.3.2.1 Primary voltage control

[B21ai-VORE-110138]

Each Onshore Converter Pole shall be able to accept orders of new voltage setpoint U_{target} at least every second.

Voltage reference requirements:

The voltage setpoint U_{target} shall be settable to all values included in the range defined in Table 17 (understood as a minimum requirement)

Parameters	Associated minimum required range
U_{target}	[0 kV – 500 kV] with a 0.1kV minimum step

Table 17 – Voltage reference requirements

Employer shall be able to adjust on the HMI, independently for each Converter Pole, the range within which U_{target} can be set, by setting U_{target_min} and U_{target_max} values. U_{target_min} and U_{target_max} values can take any value in the range defined in the above table, and $U_{target_min} < U_{target_max}$.

Voltage reference ramp rate requirements:

- Voltage ramp rate applied to U_{target} for Onshore Converter Pole shall be settable on the HMI by the Operator to all values included in the range defined in Table 18 (understood as a minimum requirement). Voltage ramp rate value shall be configurable independently for each Converter Pole.
- It shall be possible to stop the ramp at any value between the old reference and the new order upon reception of a “stop ramping” order
 - o Manually from an Operator
 - o Automatically when the AC voltage exceed a settable overvoltage or undervoltage threshold.

Parameters	Associated minimum required range
Ramp rate (applied to U_{target}) U_{Ramp_Rate}	[1 kV/min – 500 kV/min] with 1 kV/min as minimum step

Table 18 – Voltage reference ramp rate requirements

[B21ai-VORE-110139]

In steady state and without limitations, the voltage control shall respect the following law:

$$U_{target} = U_{cons} = U_{PROD} + \lambda * Q_{PROD}$$

Where:

- U_{target} (kV) is the reference for primary voltage control

- U_{cons} (kV) is the reference for primary voltage control after limitations and ramp
- U_{PROD} (kV) is the voltage at the PCC-AC
- Q_{PROD} (Mvar) is the reactive power at the PCC-AC
- λ (kV/Mvar) is a settable parameter, defined as the change in voltage (in kV) that results from a change in reactive power equal to 1 Mvar.

Note that, U_{target} can be different from U_{cons} during ramping or when limitations are reached.

[B21ai-VORE-110140]

The measurements U_{cons} , U_{PROD} and Q_{PROD} shall be sent to the Employer's telecommunication interface each 5 seconds and shall be filtered to reject frequencies above 0.05 Hz.

[B21ai-VORE-110141]

This control will be performed without a dead-band on the voltage setpoint. Voltage variation insensitivity shall be less than 0.02% of the nominal voltage (400kV).

[B21ai-VORE-110142]

The parameter λ shall be settable by the operator on the HMI to all values defined in Table 19 (understood as a minimum requirement). λ value shall be configurable independently for each Onshore Converter Pole.

Parameters	Associated minimum required range
$1/\lambda$	[10 Mvar/kV – 333 Mvar/kV] with a 1 Mvar/kV minimum step

Table 19 – Parameter λ

Here after is an illustration of what is required, with $U_{\text{target}} = 400$ kV, and $U_{\text{PROD}} = 390$ kV.
(U_{network} is U_{PROD} and Q is Q_{PROD} in Figure 12).

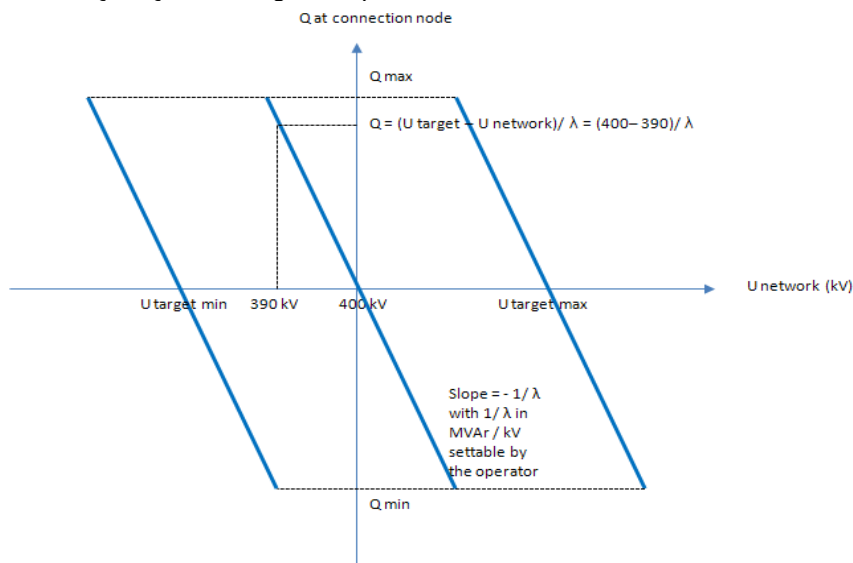


Figure 12 – Illustration of primary voltage control

[B21ai-VORE-110143]

AC voltage control dynamic requirements

The voltage control dynamics shall satisfy the two following requirements and shall be settable for each Onshore Converter Pole independently:

- For a step of U_{target} (with no ramp) in the normal operating range (voltage, reactive power):
 - o Settling time to 95% shall be settable within the fastest possible time (less than 1s) and 60 seconds (2 seconds by default).
 - o Steady-state static error shall be no greater than 0.2% of the nominal voltage.
 - o Overshoot shall be no greater than 10%.
- For a change of U_{PROD} , with U_{target} and λ considered as constant:

- Activation time less than 0.2 seconds.
- Rise time to 90% shall be settable within 0.1 and 10 seconds.
- Settling time to 95% shall be settable between 0.1 and 10 seconds.
- Steady-state static error shall be no greater than 0.2% of the nominal voltage.
- Overshoot shall be no greater than 10%.

The Contractor shall indicate the possible ranges of the parameters of the dynamic performances for a step of U_{target} or U_{PROD} so that they are consistent.

[B21ai-VORE-110145]

When the control of the Onshore Converter Pole is in Remote mode, (see (B22a_v), section “Operating and Control Modes”), U_{target} shall be equal to the signal $U_{\text{refREMOTE}}$ sent by RCC. In case of invalid value or contingency, the last valid value of $U_{\text{refREMOTE}}$ shall be used for setting U_{target} without time limit. U_{refLOCAL} shall be used if the control of the Converter Pole is in LOCAL mode.

[B21ai-VORE-110147]

After a start-up sequence, as soon as the Converter Pole is in STATCOM or POW status, default mode shall be U-mode.

When AC voltage control is enabled, initial U_{target} shall be initialized to the current measurement of U_{PROD} so that no reactive power is delivered or absorbed by the Onshore Converter Pole at the activation of U-mode.

When U-Mode is activated:

- when transiting the control of the Onshore Converter Pole from LOCAL to REMOTE, U_{target} shall be maintained to the last active setpoint value of U_{refLOCAL} as long as the first valid setpoint $U_{\text{refREMOTE}}$ has not been received from RCC.
- when transiting the control of the Onshore Converter Pole from REMOTE to LOCAL, U_{target} shall be maintained to the last active setpoint value of $U_{\text{refREMOTE}}$ as long as the first valid setpoint U_{refLOCAL} has not been validated from HMI.

8.6.3.2.2 Participation of the onshore converter station into Employer secondary voltage control

[B21ai-VORE-110144]

The Onshore Converter Station shall have the capability to contribute to Secondary Voltage Control (SVC or RST for its acronym in French Régulation Secondaire de Tension) thanks to the SVC controller of the HVDC system.

[B21ai-VORE-110146]

The SVC controller shall be able to:

- Receive a setpoint $U_{\text{refREMOTE}}$ every second, issued by RCC.
- Apply the $U_{\text{refREMOTE}}$ signal to U_{target} (primary controller's voltage) in less than 1s: maximum time between the instant when the information arrives at the interface with the link, and the instant when it is taken into account by the voltage controller;
- Simultaneously acquire the link's telemeasurements (voltage U_{PROD} , active and reactive power at PCC-AC Onshore P_{PROD} and Q_{PROD} , setpoint U_{CONS} applied to the primary voltage controller), obtained from measurements that are filtered in advance to reject frequencies above 0.05 Hz;
- Send this teledata to RCC.

[B21ai-VORE-110148]

For proper operation of Employer RST system, Employer use a remote command / remote signal (TC/TS) called “U.TCD”:

- The command (TC) U.TCD is used by Employer RST system to include the Converter Pole into the power infrastructures taking part into the SVC.
- The signal (TS) U.TCD is used by Employer RST system to determine the ability of the Converter Pole to execute voltage regulation setpoints.

When the command (TC) U.TCD = ON is received from the RCC, the Onshore Converter Station shall:

- Reject the command if the control of the Converter Pole is in LOCAL,
- Enable U-mode (U.MODE = ON) for the Converter Pole if U-mode is not already enabled,
- Answer with the signal U.TCD = ON if the Converter Pole is able to receive and execute voltage regulation setpoint from RCC. Such conditions are included as a minimum:
 - Reactive power is authorized (Q.AUTOR = ON),
- And** U-mode is enabled (U.MODE = ON),
- And** There are no contingencies identified by the C&P affecting the ability for the Converter Pole to perform voltage control.
- Answer with the signal U.TCD = OFF if the Converter Pole is not able to receive and execute voltage regulation setpoint from RCC, if any of the following conditions are met:
 - Reactive power is not authorized (Q.AUTOR = OFF),
- Or** U-mode is disabled (U.MODE = OFF),
- Or** There is one contingency identified by the C&P affecting the ability for the Converter Pole to perform voltage control.

The signal (TS) U.TCD shall be sent to RCC from ON to OFF in the following cases:

- Contingency identified by the C&P affecting the voltage controller (e.g.: faulty equipment, etc.; Further situations to be highlighted by the Contractor and discussed with Employer during design stage);
- The command to disable reactive power is received from RCC (Q.AUTOR = OFF);
- The command to disable U-mode is received from RCC (U.MODE = OFF);
- The control of the Converter Pole is switched from REMOTE to LOCAL.

Further details on precise implementation will be given by the Employer to the Contractor during design stage.

It has to be noted that even if the command U.TCD = ON has not been received from RCC, the station C&P shall execute the setpoint $U_{refREMOTE}$ received from RCC if the following conditions are met:

- Control of the Converter Pole is in REMOTE,
- Reactive power is authorized (Q.AUTOR = ON),
- U-mode is enabled (U.MODE = ON),
- There are no contingencies identified by the C&P affecting the ability to perform voltage control.

[\[B21ai-VORE-110149\]](#)

The signal U.TCD shall remain ON even if:

- $U_{refREMOTE}$ is out of the limits of the primary voltage control. (However in this case, the primary voltage control setpoint shall be clipped to the limit)
- The Converter Pole limits are reached.

[\[B21ai-VORE-110152\]](#)

Should the limits of the Converter Pole be reached (e.g.: maximum or minimum voltage, reactive power, etc.), the following signals shall be sent to the RCC independently for each Pole:

- BUT.UQ+: "LIMITED" if the maximum limit of the U/Q diagram of the Converter Pole is reached.

- BUT.UQ-: "LIMITED" if the minimum limit of the U/Q diagram of the Converter Pole is reached.

[\[B21ai-VORE-110153\]](#)

As soon as the limits are relieved, the setpoint voltage value $U_{refREMOTE}$ shall be applied to the primary voltage controller again. The corresponding telesignal ("BUT.UQ+" or "BUT.UQ-") shall then be positioned in the "NOT LIMITED" state.

A hysteresis and time delay mechanism shall prevent the telesignals BUT.UQ+/- from being affected by beating phenomena.

8.6.3.2.3 Operational Voltage Range (OVR)

[\[B21ai-VORE-110155\]](#)

This range limits the voltage setpoint both in the primary and secondary control.

[\[B21ai-VORE-110156\]](#)

OVR requirements:

- The OVR shall be settable in [kV] within a range of 300 and 500 kV and in steps of at least 0.1 kV.
- In SPM mode it shall be possible to set one independent OVR for each Converter Pole while in BPM OVR shall be common for both Converter Pole.
- U_{cons} value shall be always within the OVR, any change of settings that implies U_{cons} out of the OVR shall be blocked by the control.

8.6.3.3 Reactive power control (Q-mode)

[\[B21ai-VORE-110158\]](#)

In Q-Mode, a reactive power value ($Q_{setpoint}$) shall be entered by the Operator from the HMI. It applies at PCC-AC Onshore. It can be set at different values independently for each Onshore Converter Pole.

Once Q-Mode is activated, $Q_{setpoint}$ shall be initialized to the current measurement of Q_{PROD} so that there is no sudden change of reactive power delivered or absorbed by the Converter Pole at the activation of Q-mode.

When Q-mode is enabled, when transiting the Onshore Converter Pole control mode from LOCAL to REMOTE, $Q_{setpoint}$ shall be maintained to the last active setpoint value entered from the HMI.

[\[B21ai-VORE-110159\]](#)

$Q_{setpoint}$ requirements:

- $Q_{setpoint}$ shall be settable in Mvar in steps of 1 Mvar within the capacity Q_{min} and Q_{max} defined in 8.3.1. In BPM, these values correspond to the values of Onshore Converter Station and in SPM, these values correspond to the values of Onshore Converter Pole.
- A new $Q_{setpoint}$ could be sent and accepted every 5 second.
- Activation time of $Q_{setpoint}$ shall be less than 200 ms (including the latency time).
- The tracking error time shall be less than 666 ms.
- Rise time to 90% for a change of $Q_{setpoint}$ shall be more than 0.5 second and less than 1 second.
- Steady-state static error shall be less than 0.2% of Q_{max} .

[\[B21ai-VORE-110160\]](#)

Q_{ramp_rate} requirements:

- Any $Q_{setpoint}$ modification shall be done according to Q_{ramp_rate} .
- Q_{ramp_rate} shall be settable within a defined range established between 1 Mvar/min and maximal reactive power in one second with a resolution of 1 Mvar/min.
- The Operator shall be able to stop the ramping process so the control can freeze the change at any time. The delay between the signal reception and the freezing process should be no longer than 1s.

- The Operator shall be able to restart the ramping process with the same or different settings than the original one.
- The ramp is started from zero when the Onshore Converter Station goes into operation and is ramped down to this power level before the Onshore Converter Station is blocked.
- The reactive power measured and the margin to the capacity limits shall be available to the Operator.

8.6.3.3.1 Operational reactive power range (OQR)

[\[B21ai-VORE-110161\]](#)

This range limits the reactive power setpoint at PCC-AC Onshore.

[\[B21ai-VORE-110162\]](#)

OQR requirements:

- OQR values shall be set in [Mvar] within the capacity of the Onshore Converter Station defined in 8.3.1 and in steps of 1 Mvar independently for each Converter Pole.
- The OQR shall be set through a maximum and minimum reactive power value not necessarily symmetric, for instance [-100, +200] Mvar.
- In BPM, the control shall aim to balance the reactive power between the two poles and the OQR is common. In SPM, it shall be possible to set on independent OQR for each Converter Pole.
- Q_setpoint value shall be always within the OQR, any change of settings that implies Q_setpoint out of the OQR shall be blocked by the control.

8.6.3.3.2 Automatic reactive power adaptation after trip of one HVDC Pole Link

[\[B21ai-VORE-110165\]](#)

In case of trip of a HVDC Pole Link, the total reactive power of both Converter Poles shall be maintained and the resultant Q shall be the sum of Q_setpoint of both Converter Poles before the trip (as long as the maximum reactive capacity or the remaining Converter Pole or its OQR is respected).

The automatic adaptation can be enabled/disabled.

The adaption shall be by default disabled in case of SPM with 2 electrical nodes on Onshore Converter Station.

In U-mode, reactive power injection in case of the trip of a Converter Pole shall be adapted by voltage control.

8.6.3.4 Power oscillation damping through reactive power modulation (POD_Q)

[\[B21ai-VORE-110166\]](#)

The control system shall improve the damping of power oscillations in a frequency range from 0.1 to 5 Hz by means of reactive power modulation.

[\[B21ai-VORE-110167\]](#)

POD_Q requirements:

- It shall be possible to activate POD_Q both in Q_mode and U_mode.
- POD_Q should not lose efficiency when activated in U_mode.
- POD-Q shall be available in STATCOM mode.
- The Contractor shall propose a control structure and the possible range of control parameters to be validated by the Employer after performing dynamic stability assessment studies. The Employer shall be able to activate any of them if needed during the lifetime of the HVDC (after setting the parameters by its own). These parameters shall include a settable limit in the contribution of POD-Q in reactive power.

8.6.4 Offshore converter station control

8.6.4.1 General

[\[B21ai-EMCO-110595\]-N3](#)

The Offshore Converter Station control shall have the following capabilities:

- Collecting the power of OWF toward the PCC-AC Onshore.
- Controlling the AC network offshore voltage and frequency within normal operating range at PCC-AC Offshore.
- Coordinate the breaker operations in the Offshore Converter Station.

[\[B21ai-EMCO-110596\]-N2](#)

The Offshore Converter Pole shall withstand a voltage difference among the electrical nodes (i.e. among MV windings of the same or different converter transformers) of 0.05pu and in any case, the difference needed in compliance with the energisation strategy of the OWF.

[\[B21ai-EMCO-110597\]-N3](#)

The Offshore Converter Station shall provide stable voltage waveform at the PCC-AC Offshore.

[\[B21ai-EMCO-110616\]](#)

The offshore AC switchyard breaker operation shall be possible by an Operator of the HVDC Link.

8.6.4.2 Offshore converter station start-up and offshore grid energization

[\[B21ai-EMCO-110598\]-N3](#)

The Offshore Converter Station shall be able to energize the wind farms including the Inter Array Cable. A connection of an Inter Array Cable/string shall not lead to voltage deviation larger than 0.05 pu with no wind turbines connected or during the operation of a part of the wind farm.

[\[CBOD-LAYO-100272\]](#)

The circuit breaker for each IAC string shall be able to perform cable charging-breaking current of 250A.

8.6.4.3 Offshore frequency control

[\[B21ai-EMCO-110600\]-N1](#)

Contractor shall ensure the frequency at PCC-AC offshore stays within the limits stated in section 8.2.1. In any case, frequency at PCC-AC offshore shall be stable and at 50Hz in steady state.

[\[B21ai-EMCO-110602\]-N1](#)

Dynamic response times shall be the following in case of a frequency measure f_{ac} variation or active power measure P_{ac} variation:

- The activation time shall be as fast as possible and less than or equal to 0.5 second.
- Rise time to 90% shall be less than 1s
- Settling time to 95% shall be less than 2s
- Steady-state static error shall be less than 20mHz

f_{ac} : actual frequency offshore (Hz).

Should the dynamic response times be different from the values specified above to grant network stability, the Contractor shall justify it to the Employer for approval.

[\[B21ai-EMCO-110604\]-N2](#)

Frequency regulations shall be linear (no steps permitted).

8.6.4.4 Offshore voltage control

[\[B21ai-EMCO-110605\]-N3](#)

Contractor shall ensure the voltage at PCC-AC offshore stays within the limits stated in section 8.2.1. In any case, voltage at PCC-AC offshore shall be stable.

[\[B21ai-EMCO-110608\]-N2](#)

Dynamic response times shall be the following in case of a reactive power variation or voltage U_{ac} variation:

- The activation time shall be as fast as possible and less than or equal to 0.5 second.
- Rise time to 90% shall be less than 200 ms
- Settling time to 95% shall be less than 1 s
- Steady-state static error shall be less than 0.2 %

U_{ac} : actual voltage at PCC-AC Offshore (kV).

Should the dynamic response times be different from the values specified above to grant network stability, the Contractor shall justify it to the Employer for approval.

[\[B21ai-EMCO-110609\]-N2](#)

Voltage regulations shall be linear (no steps permitted).

8.6.5 Frequency control

[\[B21ai-EMCO-110539\]](#)

The OWF will be compliant with (Art_8_3_1). As such, all frequency services (e.g.: FSM (Frequency Sensitive Mode), LFSM (Limited Frequency Sensitive Mode), etc.) will be implemented by the WFO.

The HVDC Link shall stay connected during any frequency related service provided by OWF in stable conditions. As such, the HVDC Link shall manage the power variations due to the before mentioned services.

[\[B21ai-EMCO-110540\]](#)

Contractor shall study the dynamics linked to the frequency services and the impacts on the HVDC Link and consider them in order to adjust the design of the Converter Stations.

In case of interactions, Contractor shall cooperate with WFO to solve the issue. The Contractor shall take into account any communication time delays (such as onshore frequency measurements transmitted to the OWF) during the design and factory acceptance tests as part of the cooperation with the WFO.

8.6.6 Stability under dynamic conditions

8.6.6.1 Fault ride through capability (FRT)

[\[B21ai-STAB-110207\]](#)

For overvoltage and undervoltage ride through profiles, the term block/blocking used here below refers to the condition of a valve when all IGBT¹s are turned off accordingly with (IEC62747) section §8.19 "valve blocking state".

[\[B21ai-STAB-110208\]](#)

The Converter Station shall have the ride through capabilities described below, even during control transitions.

Moreover, the Converter Station shall be able to withstand reclosing on permanent faults, symmetrical or asymmetrical faults close to the Converter Station.

8.6.6.1.1 Undervoltage Fault Ride Through (UVRT)

[\[B21ai-STAB-110209\]](#)

The Converter Stations shall have a fault ride through capability at PCC-AC as reported in Figure 13 for positive sequence voltage profile. It shall operate stably without blocking, tripping or triggering abnormal situations or alarms during both symmetrical and asymmetrical faults.

[\[B21ai-STAB-110210\]](#)

The Offshore Converter Station shall withstand at least 3 voltage drops with 5 seconds between each. This requirement holds for any initial condition within the AC voltage ranges defined in 8.1.1.

¹ IGBT or other similar device (such as BIGT) could be used

[B21ai-STAB-110591]

The Onshore Converter Station shall withstand 3 voltage drops with 5 seconds between each as long as the thermal limits of the DBS is not exceeded. Only a protective action with regard to thermal overload of the DBS shall lead to a blocking or trip situation of the HVDC Link. This requirement holds for any initial condition within the AC voltage ranges defined in 8.1.1.

[B21ai-STAB-110211]

In addition to the non-blocking and non-tripping capability of the Converter Station, the Converter Station protection system shall be coordinated for the same capability and shall be set to the widest possible technical capability of the Converter Stations.

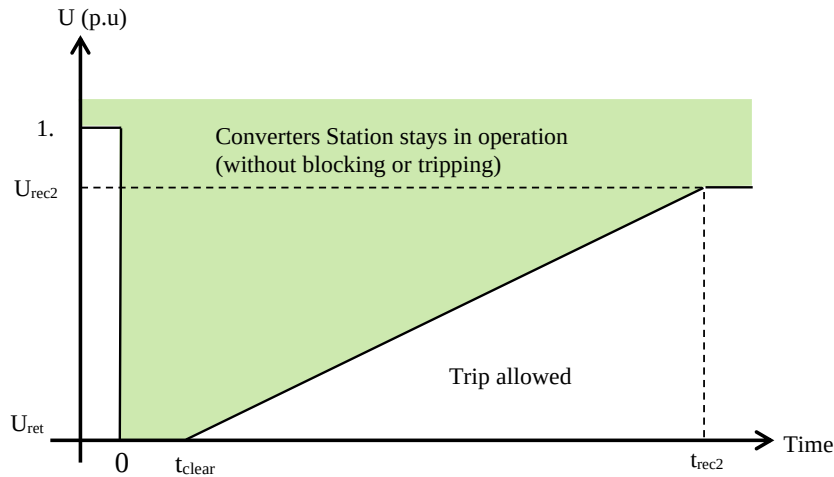


Figure 13 – Required UVRT capability under an AC undervoltage drop (positive sequence)

Voltage parameters [pu]		Time parameters [seconds]	
Symmetrical faults			
U _{ret}	0	t _{clear}	0.25
U _{rec2}	0.85	t _{rec2}	3
Asymmetrical faults			
U _{ret}	0	t _{clear}	0.25
U _{rec2}	0.85	t _{rec2}	3

Table 20 – Voltage and time parameters

Parameters of Table 20 are defined in Figure 13 and in (NC_HVDC).

Please note that UVRT capability includes all the green area of the Figure 13.

[B21ai-STAB-110212]

For any fault outside the UVRT profile, the Contractor shall inform the Employer about:

- The blocking conditions (profile): with voltage outside of UVRT profile, it is preferred that the Converter Station blocks instead of tripping; the Contractor shall inform about the possible blocking capability. In case of blocking of the Converter Station due to a voltage outside of UVRT profile, it shall deblock in less than 20 ms once the network voltage is re-established to a post-fault value within UVRT profile.
- The deblocking conditions (voltage, duration).
- The tripping conditions (profile).
- The reclosing conditions (minimum elapsed time after tripping occurrence).

8.6.6.1.2 Overvoltage fault ride through (OVRT)

8.6.6.1.2.1 Onshore converter station Over Voltage Ride Through (OVRT)

[B21ai-STAB-110213]

The Onshore Converter Station shall have an OVRT capability at PCC-AC. This capability shall enable the station to withstand the voltage profile depicted in Figure 14 (positive sequence).

The Onshore Converter Station shall continue stable operation without tripping, blocking or causing abnormal situations or alarms during both symmetrical and asymmetrical overvoltages under the following conditions:

- Starting from any steady state voltage level within the range of 0.95 pu to 1.05 pu
- Voltage follows the profile depicted in Figure 14

For voltages are in between 1.35 and 1.45 pu, blocking is allowed but not tripping as long as deblocking conditions described in this section are respected.

[B21ai-STAB-110214]

Onshore Converter Station shall withstand 3 consecutive overvoltage profiles with 5 seconds between each. This requirement holds for any initial condition within the AC voltage ranges defined in section 8.2.1.

[B21ai-STAB-110215]

In addition to the non-blocking and non-tripping capability of the Converter Station, the Converter Station protection system shall be coordinated for the same capability and shall be set to the widest possible technical capability of the Converter Station.

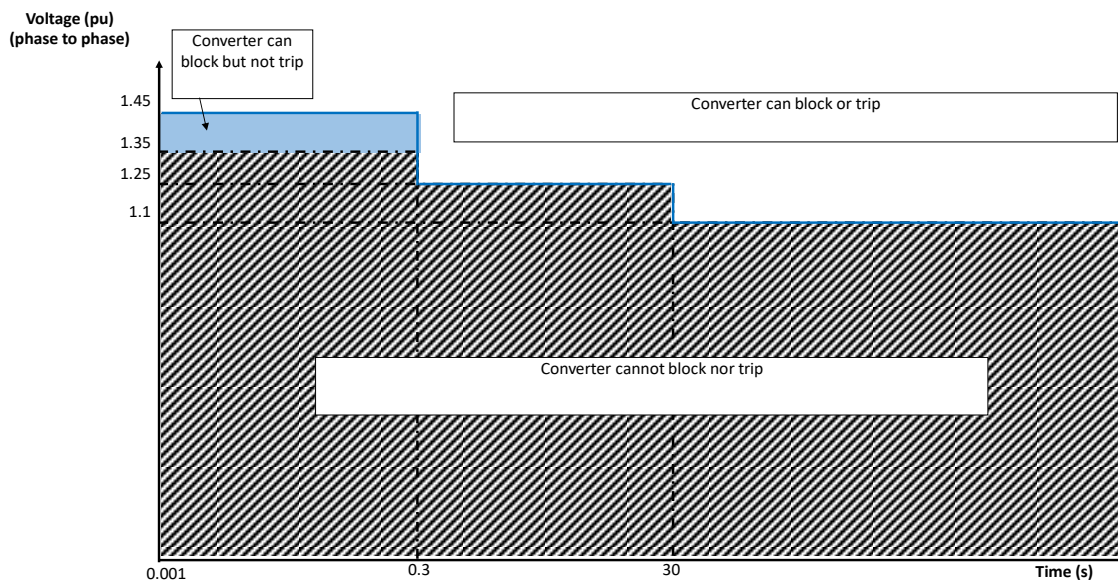


Figure 14 – Required FRT capability under an AC overvoltage for Onshore Converter Station.

[B21ai-STAB-110216]

In case of blocking of the Converter Station due to a voltage between 1.35 pu and 1.45pu, it shall deblock in less than 20 ms once the network voltage is re-established to a value lower than 1.35 pu.

[B21ai-STAB-110217]

In addition, after completing this recovery sequence (deblocking):

- Injection of reactive current shall be started in less than 50ms after deblocking and fulfil the requirements of section 8.6.6.1.3
- The Converter Station shall recover to the pre-fault active power transfer level with the same time requirement as in section 8.6.6.2.

8.6.6.1.2.2 Offshore converter station Over Voltage Fault Ride Through (OVRT)

[B21ai-STAB-110625]

Each Offshore Converter Pole shall have an OVRT capability at PCC-AC Offshore to withstand the voltage profile (positive sequence) as depicted in Figure 15.

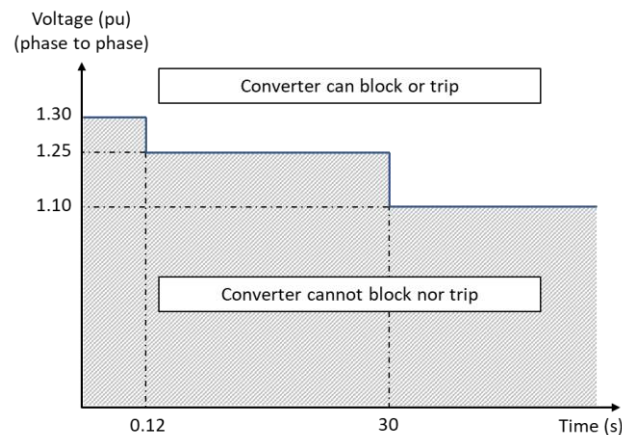


Figure 15 – Required FRT capability under an AC overvoltage for Offshore Converter Pole

[B21ai-STAB-110626]

Moreover, Each Offshore Converter Pole shall have the capability to withstand to all fault clearing situations. Contractor to define the worst-case scenarios. Following faults shall be considered at least:

- Solid 1 phase, 2 phases, 3 phases, 3 phases-ground faults
- Locations: at WTG, in IAC cable, at IAC feeder, on busbar of GIS
- For any nominal operating condition (including with only one power transformer)

[B21ai-STAB-110627]

In addition to the non-blocking and non-tripping capability of each Offshore Converter Pole, the Converter Station protection system shall be configured to fully align with the widest possible technical capability of the Converter Station.

Requirements related to current injection of OWF are stated in [C11-ACGR-010152].

8.6.6.1.3 Onshore converter station voltage support

[B21ai-STAB-110218]

Either the entire Onshore Converter Station and each Onshore Converter Pole shall be capable of providing an additional current contribution at the PCC-AC Onshore (i.e. fast current injection) during the period of fault, voltage drop or overvoltage (section 8.6.6.1.1 and 8.6.6.1.2). The current contribution could include positive and/or negative sequence components depending to the type of voltage drop or overvoltage.

[B21ai-STAB-110219]

The magnitude of the additional positive (negative) sequence current shall be proportional to the network positive (negative) sequence voltage drop (increase) at the PCC-AC Onshore.

[B21ai-STAB-110220]

For reactive current contributions, the reactive current is additional to the reactive current set point of the Q mode / U-mode control (see section 8.6.2). In no case the total reactive current shall be negative (absorbing) during voltage drops below the threshold setting (ΔU^-) or positive (generating) in overvoltage above the threshold settings (ΔU^+).

[B21ai-STAB-110221]

The following values shall be settable for both active and reactive current: maximum contribution ($\Delta I^+/\Delta I^-/I_2^-$) of this additional current, voltage range in which the additional current is varying ($\Delta U^+/\Delta U^-/U_2^+$), as well as voltage threshold ($DB^+/DB^-/DB_2^+$) below which the additional current injection starts.

[B21ai-STAB-110222]

The fast current injection function to be implemented in **Figure 16**:

- ΔI (pu) is the additional current to be provided by the Onshore Converter Station in case of fault, voltage drop or overvoltage.

- U (pu) is the PCC-AC Onshore positive sequence voltage.
- The values $\Delta U+$ shall be settable by the Employer within the range [0.05; 0.5] and $\Delta U-$ within the range [-0.05;-0.4] by steps of 0.01pu.
- The value of $DB-$ and $DB+$ shall also be settable by the Employer within the range [-0.2; 0]/[0, 0.2] respectively by steps of 0.01pu.
- The value of $\Delta I+$ shall be settable by the Employer within the range [0; 2] and $\Delta I-$ within the range [-2; 0] by steps of 0.01pu.
- For ΔU and DB pu values, base is 400kV; for ΔI pu values, base is the Onshore Converter Station maximum current rating. The origin point of axis is $\Delta I = 0$ pu and $U = 1$ pu.

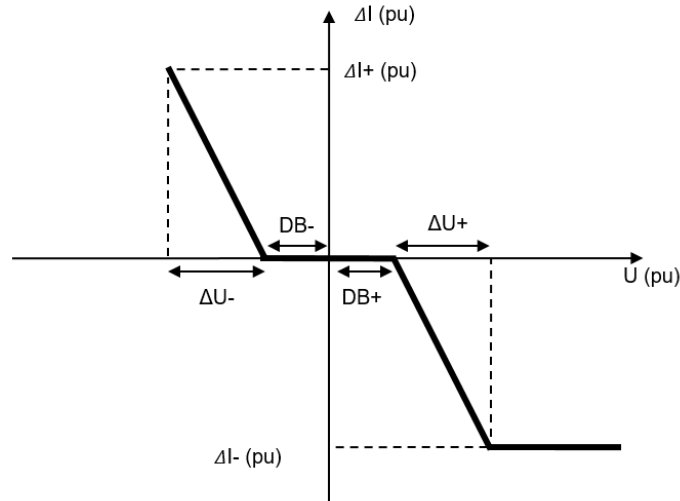


Figure 16 – Characteristic of the fast current injection (positive sequence component)

[\[B21ai-STAB-110223\]](#)

In asymmetric voltage drops or asymmetric overvoltages, the Onshore Converter Station must provide negative sequence component in quadrature to the negative sequence voltage in order to reduce the asymmetry. The characteristics of the negative current control are provided in Figure 17:

- I_2 (pu) is the additional current to be provided by the Onshore Converter Station in case of voltage asymmetry.
- U_2 (pu) is the AC negative sequence voltage at the PCC-AC Onshore.
- The values ΔU_2+ shall be settable by the Employer within the range [0.05; 0.5] by steps of 0.01pu.
- The value of DB_2+ shall also be settable by the Employer within the range [0; 0.1] by steps of 0.01pu.
- I_2- shall be settable by the Employer within the range [-2, 0] in steps of 0.01pu.
- For ΔU_2 and DB_2 pu values, base is 400kV; for I_2 pu values, base is the Onshore Converter Station maximum current rating. The origin point of axis is $I_2 = 0$ pu and $U_2 = 0$ pu.

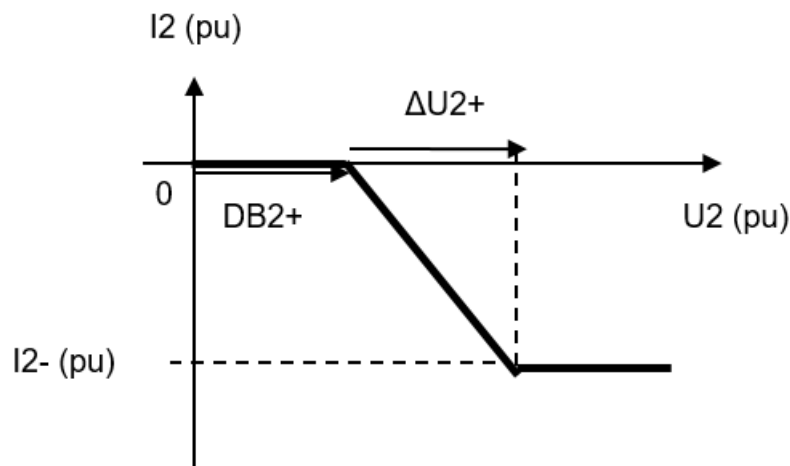


Figure 17 – Characteristic of the fast current injection (negative sequence component)

The initial negative sequence voltage is theoretically null. The control input shall be the value of the negative sequence voltage. Contrary to positive current sequence, the negative current sequence shall not be additive to any other negative sequence injection as there shall be no other control injecting a negative sequence.

During asymmetrical faults, negative sequence current injection shall not generate over-voltages in the healthy phases leading to a voltage profile outside the OVRT capability defined in §8.6.6.1.2).

[\[B21ai-STAB-110224\]](#)

The total current limit shall not be less than the Onshore Converter Station maximum current rating at the rated voltage level.

[\[B21ai-STAB-110225\]](#)

When the Onshore Converter Station provides a fast additional current:

- Priority is given to reactive current.
- The active power of the HVDC Link may be reduced if relevant.
- The reactive power operational limits (OQR in section 8.6.3.3.1) are not applicable.

The current limitations as well as the priority rules must be clearly described by the Contractor and approved by Employer.

[\[B21ai-STAB-110226\]](#)

The dynamic of fast current injection shall be the following (refer to section 8.6.1):

- Activation time: less than 10 ms.
- Rise time to 90%: less than 30 ms.
- Settling time to 95%: less than 60 ms.

[\[B21ai-STAB-110227\]](#)

After fault clearing, the voltage support function shall be stopped quick enough to avoid over-voltages exceeding the OVRT profile as defined in §8.6.6.1.2. Moreover, the fast voltage support function shall not lead to any temporary overvoltage over 1.35pu at PCC-AC Onshore for any network condition defined in §8.1.2.

[\[B21ai-STAB-110228\]](#)

It shall be possible to activate or disable independently both positive and negative sequence voltage support function manually or automatically (external signal).

[\[B21ai-STAB-110545\]](#)

When Grid Forming mode is activated (see section 8.6.7.1):

- during an AC fault, the supplied short-circuit current at fault initiation should be instantaneous (less than 5 ms – measurement shall be done in compliance with (IEC61400_21_1)) and shall follow the $U=K \cdot I$ rules after 30-60ms.
- After voltage recovery, the active power should reach the initial value in less than 150 ms (if possible) maintaining the regular operating mode performance. The Converter Station should contribute to restore the voltage within normal range after the fault clearance.

When the Grid Forming mode is deactivated, requirement [\[B21ai-STAB-110226\]](#) shall apply and the voltage shall return to the normal operating range within 150 ms.

8.6.6.1.4 Offshore converter station voltage support

[\[B21ai-STAB-110546\]](#)

Either the entire Offshore Converter Station and each Offshore Converter Pole shall be capable of providing AC voltage support at the PCC-AC Offshore during the period of fault, to limit the voltage drop or overvoltage as described in sections 8.6.6.1.1 and 8.6.6.1.2). The voltage support contribution shall include positive and/or negative sequence components depending to the type of voltage drop or overvoltage.

[\[B21ai-STAB-110547\]](#)

Offshore Converter Station shall support the AC grid voltage by supporting positive and as far as possible suppressing negative sequence voltage deviation until maximum output currents have been reached.

[\[B21ai-STAB-110548\]](#)

The AC voltage support during abnormal voltage range shall act instantaneously after the event (less than 5 ms – measurement shall be done in compliance with (IEC61400_21_1)).
The AC voltage support control must not lead to a switch of control mode.

[\[B21ai-STAB-110549\]-N1](#)

The supplied short-circuit current during the fault shall follow the $K \cdot I$ rules, 30-60ms after the fault ignition. K is a scalar parameter settable by the Employer within a range specified by the Contractor.

[\[B21ai-STAB-110550\]](#)

After fault clearing, the voltage shall return to the normal operating range within 150ms.

[\[B21ai-STAB-110551\]](#)

After fault clearing, no over-voltages shall be caused by the Offshore Converter Station. In addition, Offshore Converter Station shall do its best effort, within its maximum current limits, to limit the overvoltage at the PCC-AC Offshore caused by external systems below 1.15pu.

[\[B21ai-STAB-110552\]](#)

During voltage support, the total current limit shall not be less than the Offshore Converter Station maximum current rating at the rated voltage level.

[\[B21ai-STAB-110553\]](#)

AC voltage control philosophy during AC voltage support implemented by the Contractor shall be documented, including, but not limited to, positive and negative sequence prioritisation, current limit strategy, active and reactive power limitations.

8.6.6.2 Active power during the recovery after a fault clearance

[\[B21ai-STAB-110229\]](#)

For any fault which does not lead to a trip of the HVDC Link according to the UVRT or OVRT profiles of sections 8.6.6.1.1 and 8.6.6.1.2, the HVDC Link shall recover 90% (rise time to 90%, see section 8.6.1) of the pre-fault active power transfer level at the PCC-AC Onshore in less than 150 ms and have a settling time to 95% lower than 200 ms as soon as the network voltage is re-established to a value higher than 0.85pu, in compliance with Figure 18, under the conditions that the active power is injected by OWF.

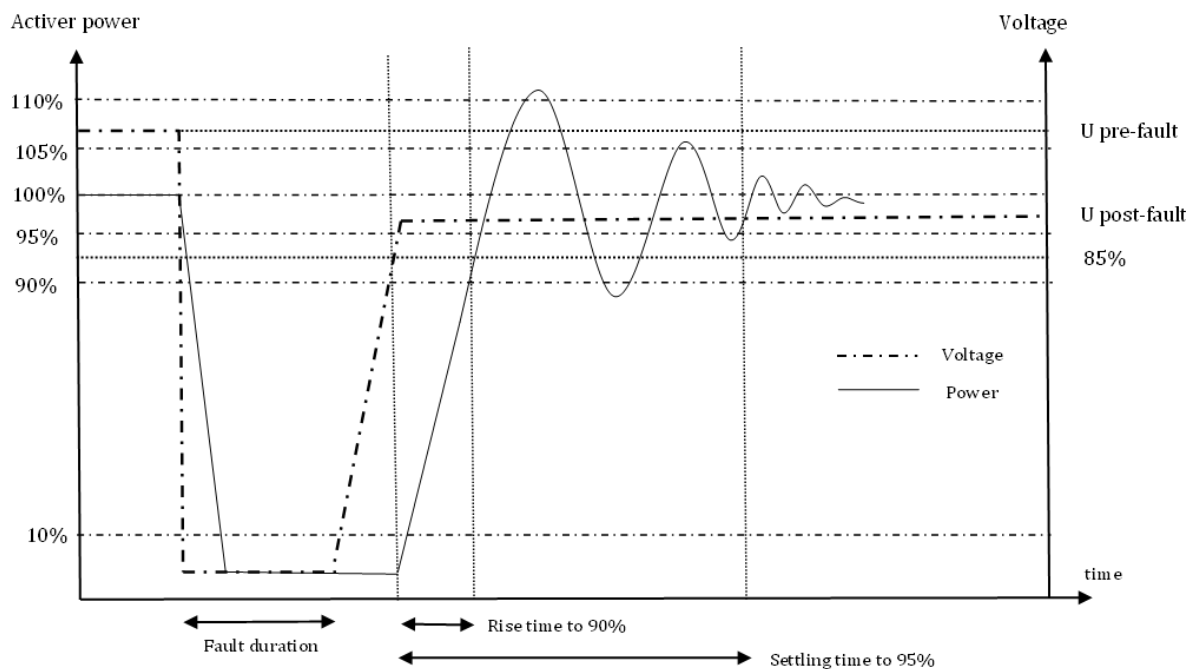


Figure 18 – Voltage and active power during fault and after fault clearance

8.6.6.3 Energisation of converter station

Requirements in section 8.4.6 related to the Start-up, Shutdown and Restart of the HVDC Link shall be applicable.

[B21ai-STAB-110232]

During the energisation or synchronisation of a Converter Station to the AC Network (onshore or offshore), the Converter Station shall have the capability in steady state to limit any voltage changes to 3% of the pre-synchronisation voltage at PCC-AC.

[B21ai-STAB-110557]

Synchronising device can be installed by Contractor on the Converter Stations' AC breaker. However, the Converter Station shall be designed in such a way that it can be started/operated without the use of any synchronising device on AC circuit breaker in the Offshore or Onshore Converter Stations, neither in the Employer's AC substation connected to the Onshore Converter Station

[B21ai-EMCO-110185]

The Offshore Converter Station shall possess the ability to initiate start-up autonomously, without reliance on external systems. This capability includes, but is not limited to, self-initialization of power-up sequences, system checks, and configuration processes. The energization of converter transformers and the capacitors charging in Converter are excluded.

[B21ai-COBA-110536]

During nominal operation conditions, all converter transformers of the Offshore Converter Station operate in parallel.

The offshore converter transformers shall be capable of being disconnected while the HVDC system remains in operation without negatively impacting the operation of the HVDC system and wind farms not directly connected to the converter transformer being disconnected.

8.6.6.4 Other network disturbance onshore

[B21ai-STAB-110234]

The Contractor shall ensure that whatever the Onshore Converter Station configuration and for any operating condition, the HVDC Link shall be able to operate during a nearby HV equipment energisation. For the design purpose, following scenarios shall be considered:

- power transformer energisation (600 MVA),
- capacitor bank switching (150 Mvar) and shunt reactors switching (200 Mvar).
- AC line energisation data will be provided at (Call_off).

Therefore, the active and reactive power variations shall be smooth until steady state and shall not amplify the disturbance.

8.6.7 Advanced functionalities

8.6.7.1 Grid Forming Mode

[\[CBOD-GFM-100273\]](#)

Grid Forming Mode (GFM) for the Onshore Converter Station is required.

For avoidance of doubt, the Onshore Converter Station shall be designed in both Grid Following Mode (GFL) and Grid Forming Mode (GFM). The GFM shall not lead to the overrating of the HVDC Link compared to the GFL.

FGM is the control mode by default.

[\[CBOD-GFM-100274\]](#)

Within the HVDC Link voltage, current and available energy limits, the Onshore Converter Station shall be capable of behaving as a controllable voltage source behind an internal impedance.

- If the HVDC Link capability, voltage, current and energy limits are exceeded, the current, voltage and energy shall be kept within their admissible limit values.
- The HVDC Link shall be capable of stable and bumpless transition when reaching the limits, without interruption, in a continuous manner and returning to the normal behavior as soon as the limitations are no more necessary.

[\[CBOD-GFM-100275\]](#)

GFM shall contain at least the following mandatory functionalities:

- Self-synchronization,
- Phase jump active power,
- Inertial active power,
- Positive damping power,
- Inherent reactive power.

[\[CBOD-GFM-100276\]](#)

Self-synchronization is capability to operate after the disconnection of the last synchronous connection with the grid. The GFM will provide sufficient voltage and frequency stability to allow for other generating units to operate.

The Onshore Converter Station shall be able to operate in islanded mode where the electrical island can either be a small area of power system containing only a few buses and generating units, or a larger portion of a synchronous area. With a sufficient DC source available, the Onshore Converter Station shall be able to supply its own ancillary load and other connected loads.

[\[B21ai-EMCO-110505\]](#)

Phase jump active power:

- Performance: The Onshore Converter Station shall behave as a Thevenin source (but with current limitation based on its nominal rated current) meaning that it shall inject/absorb active power instantaneously (in less than 5 ms as per the measurement) after the voltage phase angle changes. The converter will react as a voltage source, meaning that the current changes instantaneously.
- Withstand capability: The Onshore Converter Station shall stay connected to the AC grid and maintain the synchronism with AC grid when subjects to a phase jump of 60° à PCC_AC at rated power (assuming that the initial active power before fault is the rated power).

[\[B21ai-EMCO-110507\]](#)

Inertial active power:

The Converter Station shall inject/absorb active power instantaneously (in less than 5 milliseconds), within the HVDC Link voltage, current and available energy limits.

- Performance: the amount of the inertial power shall be proportional to the RoCoF and within the limits of the system rating.
- Withstand capability: The Onshore Converter Station shall stay connected to the AC grid and operate stably subject to the linear frequency change of ± 2 Hz/s RoCoF during 0.5 seconds.

The method used to assess the inertial support provided by HVDC system at its PCC_AC will be defined by Employer in (B21a_ii) (will be provided at (Call_off)) and in agreement with the Contractor.

[\[CBOD-GFM-100277\]](#)

Positive damping power is defined as the inherent capability of the GFM to respond to system sub-synchronous oscillations (resulting from inter-area mode, torsional interactions or control interactions).

The GFM converter shall be able to provide positive damping of sub-synchronous frequency oscillations at the point the point of connection or HVDC interface point. The start of provision of the positive damping power shall be immediately following the appearance of the sub-synchronous oscillations.

The method used to assess the damping capability will be defined by Employer in (B21a_ii) (will be provided at (Call_off)) and in agreement with the Contractor.

[\[B21ai-SGFM-110543\]](#)

Inherent reactive power / Voltage jump reactive power:

The Onshore Converter Station shall behave as a Thevenin source (but with current limitation based on its nominal rated current) meaning that it shall inject/absorb reactive power instantaneously in less than 5 ms (as per the measurement) after voltage amplitude jump/change.

[\[B21ai-SGFM-110542\]](#)

All functions and performances not mentioned in this section shall be fulfilled as per regular operating mode.

[\[B21ai-SGFM-110541\]](#)

It shall be possible to activate/disable the Grid Forming Mode.

The Grid Forming operation shall be possible from maximum SCL (see section 8.1.2) to very low SCL (less than 0.5)). This mode can be activated/disabled by manual switch initiated by the Operator (either through HMI or RCC) or by external Employer signal during the normal operation of the HVDC Link with bump less transition.

[\[B21ai-EMCO-110526\]](#)

Contractor shall be requested to perform offline and real time EMT simulations during DPS, FAT and commissioning (refer to PCS (B21a_ii) which will be provided at (Call_off)).

For the DPS, Contractor shall propose at least 40 simulations that covers all dynamic performances listed here below.

For FAT, Contractor shall propose a test matrix that validate DPS results and in addition test of change of parameters in the HMI.

For commissioning, Contractor shall propose a set of tests to be conducted. The Employer has the right to add up to 10 simulation test cases as additional cases to be conducted for each of DPS and FAT.

[\[B21ai-EMCO-110509\]](#)

In case of a sudden switch to a weak ($SCR \leq 1.5$) or passive network, the Onshore Converter Station shall adapt automatically and instantaneously the active and reactive power based on the remaining network (taking into account the limitation based on the Converter Station ratings and available active power).

[\[B21ai-EMCO-110516\]](#)

Manual switch initiated by the operator (either through HMI or RCC) from GFL to GFM and vice-versa should be possible during the normal operation of the link with bump less transition.

[\[B21ai-EMCO-110519\]](#)

It shall be possible for the operator (either in HMI or RCC) to manually activate/deactivate the POD function when the grid forming control is in operation.

[\[CBOD-GFM-100278\]](#)

Sink for voltage unbalances is the capability of the GFM converter to provide a negative sequence current (within its current limits) allowing to reduce the voltage unbalance factor at the PCC. The negative sequence current shall be sufficient to meet the load characteristics at the PCC.

The method used to assess the sink for voltage unbalances will be defined by Employer in (B21a_ii) (will be provided at (Call_off)) and in agreement with the Contractor.

[\[CBOD-GFM-100279\]](#)

Sink for harmonics is the capability of the GFM converter to provide passive, damping response in the harmonic frequency range (up to rank 13), allowing to reduce the harmonic voltage distortion (improve the voltage quality) at the PCC. Note that it would lead to harmonic current through the converter in the presence of network voltage background harmonic.

The method used to assess the sink for harmonics will be defined by Employer in (B21a_ii) (will be provided at (Call_off)) and in agreement with the Contractor.

[\[B21ai-EMCO-110523\]](#)

It shall be possible for the operator (either in HMI or RCC) to modify the main parameters of the grid forming control. An illustrative example of parameters is shown her below:

- P/F droop value "Kp",
- Q/V droop value "Dq",
- ramp-rates and limitations.

Changing the LP filters (or equivalent as synthetic inertia) shall be possible within a range, to be provided by Contractor. Obviously, this modification might impact the performances and the impacts shall be communicated by Contractor.

8.6.7.2 Blackstart

[\[B21ai-COBA-110568\]](#)

Black-start capability is not required.

8.6.7.3 Other considerations

[\[B21ai-COBA-110594\]](#)

Default parameters of control functions shall be determined by Contractor together with Employer during design stage.

8.6.8 System statuses and sequences

[\[B21ai-COBA-110089\]](#)

Each Converter Pole shall be able to reach independently the following system statuses and the status of each Converter Pole shall be determined independently (i.e., without having to refer to the status of other Converter Pole of same HVDC Pole Link).

The Acronyms used for system status and definitions in the table below are:

- DIS – Disconnected
- RFE – Ready for Energisation
- RFO – Ready for Operation
- POW - Power
- For switchgear positions
 - i. O – Open
 - ii. C - Closed

	AC Side			DC side			Converter	Reactive power control
System status	DS	ES	CB	DS	ES pole	ES cable		
EARTHED	O	C	O	O	C	O/C ¹	De-energised / Blocked	Not available
DIS	O	O	O	O	O	O/C ¹	De-energised / Blocked	Not available
RFE POW	C	O	O	C	O	O	De-energised / Blocked	Not available
RFE STATCOM*	C	O	O	O/C ²	O	O/C ¹	De-energised / Blocked	Not available
RFO POW	C	O	O/C ³	C	O	O	Energised / Blocked	Not available
RFO STATCOM*	C	O	C	O/C ²	O	O/C ¹	Energised / Blocked	Not available
POW	C	O	C	C	O	O	Energised / de-blocked	Available ⁴
STATCOM*	C	O	C	O/C ²	O	O/C ¹	Energised / de-blocked	Available ⁴

DS: Disconnecter switches; ES: earthing switches; CB: circuit breakers

* Only applicable for Onshore Converter Station

¹: both switching states are possible (with relevant interlocks implemented to comply with [\[B21ai-COBA-110466\]](#))

²: at least one of the DC disconnectors on each side of the HVDC Link shall be open

³: Open for the Offshore Converter Station and closed for the Onshore Converter Station

⁴: It shall be possible to enable/disable the reactive power control once the relevant Converter Pole is in POW/STATCOM status.

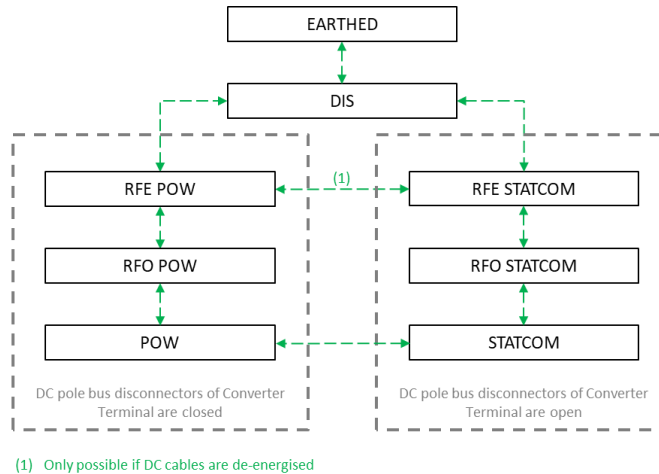
Each Converter Pole of Converter Station shall only be in one of the following states:

- in a status
- in a sequence
- in an unknown status in case the sequence failed without reaching the desired status.

In other words, the status shall be lost as soon as the command to execute a sequence is received, and the final status shall be reached only once the sequence is finished.

[\[B21ai-COBA-110091\]](#)

It shall be possible to execute automatically (either for each Converter Pole individually and for both Converter Poles in the same time) following sequences in each Converter Station independently. RFE STATCOM, RFO STATCOM and STATCOM are applicable only for Onshore Converter Station.



Precisions shall be added for:

- RFE POW to RFO POW. The Contractor shall precise the maximum time delay from the time the Onshore Converter Station has reached RFO to the time of start of the execution of the sequence at the Offshore Converter Station.
- RFO POW to POW. The Contractor shall precise the maximum time delay from the time the Onshore Converter Station has reached POW to the time of start of the execution of the sequence at the Offshore Converter Station.
- POW to RFO POW
 - In the case where the reactive power is not at 0 MVAR prior to start of execution of the sequence, ramping to 0 MVAR shall be included as part of the sequence with a ramp settable by the Operator.
 - In case where the active power is not at 0 MW prior to start of execution of the sequence, a warning shall appear and the sequence shall not be executed.

[B21ai-COBA-110092]

When inter-station communication is available, it shall be possible to execute additional automatic sequences. It shall be possible to request the execution of these sequences only from the Onshore Converter Station and it shall be possible to start this execution only once the Operator of the Offshore Converter Station has given its agreement to it:

- DIS to POW: this sequence shall bring both Converter Stations to the system status POW.
- POW to DIS:
 - This automatic sequence shall bring both Converter Stations to system statuses DIS.
 - It shall be possible to execute this sequence only once active power is at 0MW. In the case where the active power is not at a 0MW prior to start of execution of the sequence, a warning shall be raised and the sequence aborted.
 - In case reactive power is not at 0Mvar prior to start of execution of the sequence, ramping to 0 Mvar shall be included as part of the sequence with a ramp settable by the Operator.
- POW to STATCOM onshore: this automatic sequence shall bring the Onshore Converter Station to STATCOM and the Offshore Converter Station to DIS.
- STATCOM onshore to POW: this automatic sequence shall bring the Onshore Converter Station from STATCOM to POW and the Offshore Converter Station to POW.

For POW to STATCOM onshore and STATCOM onshore to POW, voltage controls or reactive power control shall remain active during the transition for Onshore Converter Station.

In case of trip occurring in the Offshore Converter Station due to offshore AC grid fault (or Offshore Converter Station fault) while in POW, the Onshore Converter Station shall automatically transition to STATCOM.

[B21ai-COBA-110531]

When selecting the RFE POW/RFO POW/POW status for the Offshore Converter Station, the offshore converter transformers to be energized shall be selected by the Operator.

[\[B21ai-COBA-110534\]](#)

The sequence switching shall include automatic start-up and switch to the relevant auxiliary power sources. As such, when switching the Offshore Converter Station to RFE, diesel generators shall be automatically started offshore to ensure auxiliary power supply as soon as the supply from the offshore grid is stopped.

[\[B21ai-MSGE-110272\]](#)

The Contractor shall provide a detailed description of all sequences and statuses. The description shall include:

- The conditions that must be fulfilled to start a sequence
- Timing of the different unitary steps during the sequence
- Logical flow diagram and illustrations with schematics
- Status of HV equipment, cable and converter station (e.g.: energised, connected, on/off, etc.)

The sequences shall be described under the presence and absence of telecommunication.

[\[B21ai-COBA-110094\]](#)

If an interlock prevents the execution of a sequence, the failed condition shall be indicated to the Operator as required in [\[B22av-HMIO-670330\]](#) of (B22a_v).

In addition, in case of an error or abortion during a sequence, the failed step with relevant information shall be indicated in an appropriate window of the OWS. The Operator shall be able to restart from the stage at which the problem occurred or go back to the previous operating state by using the sequences. It shall also be possible to switch to manual mode.

[\[B21ai-COBA-110095\]](#)

Closing the AC bus bar breaker in the 400kV Employer AC substation will require the authorisation of the corresponding operator.

During the design stage, the Employer will confirm if the breaker of the Employer AC substation can be manoeuvred directly from the Contractor's control system.

[\[B21ai-COBA-110474\]](#)

In case of abnormal situation on the Contractor main AC Circuit Breaker (e.g.: low gas pressure) in the Onshore Converter Station, it shall be possible to disconnect the Onshore Converter Station from the AC network without damaging the equipment through a given procedure to be validated by Employer.

[\[B21ai-COBA-110560\]](#)

In case of abnormal situation on the Contractor AC Circuit Breaker (e.g.: low gas pressure) in the Offshore Converter Station, it shall be possible to isolate the relevant AC circuit breaker (e.g.: by opening the disconnectors around it) minimizing the energy unavailability and without damaging the equipment through a given procedure to be validated by Employer.

[\[B21ai-COBA-110561\]](#)

It shall be possible to isolate the AC Circuit Breaker (meaning both sides switched off) in less than 20 minutes if open, 2 hours if closed. In both situations, the breaker shall not be operated as long as it is not isolated.

[\[B21ai-COBA-110618\]](#)

Contractor shall detail the operating procedure to be implemented to avoid capacitor overvoltage in case the Converter Station is in blocked status for an extended duration, to be defined and provided by the Contractor.

8.6.9 Loss of inter-station telecommunication

[B21ai-COBA-110108]

The Converter Stations protection and performances (fault ride through capability, etc.) shall not be impaired in case of loss of interstation telecommunication. The Contractor shall describe the consequences of loss of inter-station telecommunication of control functions which are entirely dependent on the availability of communication i.e. rely on measurements available on both Onshore and Offshore Converter Stations.

[B21ai- COBA-110469]

Step by step procedures shall be described by Contractor to reach the different system statuses from an initial status (e.g.: POWER from DIS, DIS from POWER, etc.) in case of loss of inter-station communication.

[B21ai-COBA-110086]

In case of loss of interstation communication, a logic shall be established in all controls functions in order to change/adapt/disable, etc. depending on previous state and possible recovery of communications keeping security and stable conditions of operation.

8.7 Electromagnetic Compatibility (EMC)

[CBOD-EMC-100236]

The Employer's requirements relating to the electromagnetic compatibility detailed in (B21a_iv) shall be applied. Studies on disturbances (section 3.4 of (B21a_iii)) shall be carried out by the Contractor.

8.7.1 Radio Frequency Interference (RFI)

[CBOD-RFI-100237]

The radio interference due to the presence of HVDC Link shall be limited. The following limits shall be applied for the Radio Frequency Interference (RFI) for onshore and offshore converter stations.

	Around the line	Onshore converter station	Offshore converter station
Frequencies between 9 kHz and 150 kHz	- RFI field at 200 m from the closest active part of the substation and at one third of 200m from the closest conductor of the line: Values reported in table F.1 or F.3 limit 4 of (CIGRE_TB391) - RFI field at 4 km from the Onshore converter station and at 30 m from the closest conductor of the line: values reported in table F.1 or F.3 limit 4 of (CIGRE_TB391)	Values reported in table F.1 or F.3 limit 2 of (CIGRE_TB391) at 200 m	(NF_EN_50121_2) at 10 m
Frequencies between 150 kHz and 1 GHz			Table 16 of (CISPR_11) at 30 m

Table 21 – Limit of RFI. The measuring points are referred to Figure 12.1 of (CIGRE_TB391) for onshore converter station and line.

8.7.2 Electromagnetic emissions and electrostatic fields limits (EMF)

[CBOD-EMF-100239]

Electromagnetic emissions and electrostatic fields caused by the HVDC Link shall be limited, so personnel may not be harmed. The limits shall be applied for electric and magnetic fields at onshore and offshore converter stations in compliance with table 1 of the article Art. R.4453-4 of (Decret2016_1074), article 12bis (Arrete_17_2001) and (EN50647).

	Inside converter station	At surrounding the converter station fence	At the reactor fence
Electric fields	35 kV/m at 50 Hz Workers shall be informed if the electric field is above 20 kV/m Public limitation: 5 kV/m	5 kV/m at 50 Hz.	
Magnetic field	The levels of magnetic field shall comply with occupational low action values (in particular with table 1 of the article Art. R.4453-4 of (Decret2016_1074)) and (EN50647). For 50Hz, this value is 1000 μ T. The magnetic field function of frequency should satisfy the equation of (IEC_61786_2), section sum of weighted magnitudes.	100 μ T at 50 Hz	500 μ T at 50 Hz for onshore 1000 μ T at 50 Hz for offshore

Table 22: Some examples of limit of electric and magnetic fields

8.7.3 Emission and immunity inside and near the converter station

[\[B21ai-SSDO-110308\]](#)

The Contractor has the full responsibility for the desired and error-free operation of all equipment within the Converter Stations under all normal and abnormal conditions in the Converter Stations thus also during electromagnetic disturbances.

Reference shall be made to (CIGRE_TB535). Other standards shall be considered, in particular (IEC_61000_6_1), (IEC_61000_6_2), (IEC_61000_6_4), (IEC_61000_6_5), (NF_EN_61000_1_2) and (NF_EN_62271_1).

8.8 Audible noise level

[\[B21ai-NOIS-110041\]](#)

The noise emission to the environment cannot exceed the specified requirements at any time during construction and operation. Hence, the most critical operation modes throughout the complete operating range must be considered.

8.8.1 During construction

8.8.1.1 Onshore

[\[B21ai-NOIS-110042\]](#)

Noise compliance requirements during construction are specified in (HSE). The Contractor is responsible for respecting possible restrictions from local or State authorities regarding noise emission during construction which may apply to the administrative area in which the Onshore Converter Station is built.

8.8.1.2 Offshore

[\[B21ai-NOIS-110578\]](#)

Contractor shall minimize as much as possible the generated noise during installation, in compliance with section "noise mitigation" of (HSE). In addition, (ERS_TI) shall be applicable. The Contractor is responsible for respecting possible restrictions from local or State authorities regarding noise emission during construction which may apply to the administrative area in which the Offshore Converter Station is built.

8.8.2 During operation

[\[B21ai-NOIS-110043\]](#)

During operation the noise requirements shall be based on the assumption that the noise has a single-tone character.

8.8.2.1 Indoor noise level

[\[B21ai-NOIS-110574\]](#)

Labour Code shall be respected.

8.8.2.1.1 Onshore

[\[B21ai-NOIS-110044\]](#)

The maximum spatially averaged sound pressure level everywhere within buildings may not exceed the following requirements with all supplies in operation:

In resting and office room	50 dB (A)
In control and pause room	50 dB (A)
In protection, equipment and telecommunication room	70 dB (A)
In locker room, cleaning store, showers and toilets	50 dB (A)
In battery and pressure supply room, workshop and store	70 dB (A)
In rooms for cooling equipment and fans, etc. (unless where hearing protection is to be required)	80 dB (A)

Table 23 – Sound Pressure

Peak values of the C-weighted sound pressure level of more than 130 dB (A) re 20 µPa shall not occur in the working area.

The noise requirements shall be met under all normal operating conditions, i.e. at full load as well as at part load.

In areas where enduring work is to be performed, the noise should be limited to 80 dB (A). In areas with noise level above 80 dB (A), signs shall be provided and hearing protection shall be easily accessible.

8.8.2.1.2 Offshore

[\[B21ai-NOIS-110575\]](#)

Requirements of (ERS_2_3) section “Noise, vibration and acoustic requirements” state the general principles of compliance for the rooms and shall be applicable. Requirements (ERS_6_3) section “Application” state limits for specific rooms or outside and shall be applicable.

8.8.2.2 Outdoor noise level

8.8.2.2.1 Onshore

[\[B21ai-NOIS-110045\]](#)

The Order of 26 January 2007, modifying the Order of 17 May 2001, shall be applicable: For all the operating conditions of the HVDC Link, the acoustic pressure levels produced by the global site made of the Onshore Converter Station and the Employer Existing AC substation, measured inside the nearest dwelling houses to the Onshore Converter Station shall meet at least one of the two criteria:

- General level: the measured ambient noise, including the noise from the electrical installations, must be less than 30 dB (A) or;
- General sound differential: the maximum sound differential values are 5 dB (A) during the day (from 7am to 10pm) and 3 dB (A) during the night (from 10pm to 7am), with respect to the existing noise level without [the Onshore Converter Station and Employer Existing AC Substation].

If acoustic pressure levels produced by the Employer Existing AC substation exceeds the noise limits in some measurement points, the Onshore Converter station should not produce any sound differential on these measurement points.

[\[B21ai-NOIS-110380\]](#)

In the case of noise of single tonal character, the noise level has to be 5 dB (A) lower than the required limits. Background noise must also be taken into account.

[\[B21ai-NOIS-110046\]](#)

During execution of the Contract, the Contractor shall collaborate with Employer to respect the global noise emission set as per [\[B21ai-NOIS-110045\]](#). The Contractor shall take into account all possible noise mitigation measures in order to reduce the noise generated by the Onshore Converter Station if limits are exceeded.

If noise reduction measures if any like noise screens are foreseen, then they shall be designed for easy dismantling and re-mounting in order to facilitate maintenance and to reduce the risk of damage. Description of such dismantling/remounting procedures shall be included in the maintenance instructions.

8.8.2.2.2 Offshore

[\[B21ai-NOIS-110047\]](#)

Contractor shall minimize as much as possible the generated noise during operation and prove with a study that the noise emitted is negligible in the offshore environment. If noise emissions are not negligible and if required by impact studies or authorizations, additional mitigation measures shall be studied and implemented by Contractor.

8.9 Station layout and HV switchgear

[\[B21ai-HSDB-110014\]-N3](#)

The Contractor is responsible for the design of the power circuit arrangement and layout for the Converter Stations considering all component required for the function of HVDC Link.

[\[B21ai-HSDB-110524\]-N2](#)

If an AC harmonic filter is necessary, a dedicated circuit breaker shall be integrated in the design so that the Converter Station can be operated without the filter (in particular in case of failure of that filter).

[\[B21ai-HSDB-110569\]-N3](#)

The required performances of the HVDC Link shall be met without the use of redundant units (e.g.: valve submodules, cooling fans, etc.) and for all operating and network conditions, except if stated otherwise.

[\[B21ai-HSDB-110580\]](#)

There shall be a common spare converter transformer for Onshore Converter Station for HA1 and HA2.

There shall be a hot-standby spare converter transformer for Onshore Converter station for GLC.

There shall be one common spare converter transformer to both Offshore Converter Stations for le HA1 and HA2.

There shall be one spare converter transformer for Offshore Converter Station for GLC.

The spare transformer (onshore and offshore) for BNC will be confirmed at *(Call_off)*.

The offshore spare converter transformers shall have capability to substitute for any of the offshore converter transformer of HA1, HA2, GLC and BNO projects.

[\[CBOD-HDSB-100017\]-N3](#)

Direct pole to pole faults during the operation of the HVDC Link shall be prevented through appropriate layout design.

[\[CBOD-HSDB-100018\]](#)

During a converter transformer or equipment replacement affecting only one Converter Pole, the unaffected Converter Pole shall stay in operation.

[\[CBOD-HSDB-100235\]](#)

Electrical switchgears shall be compliant with enforced F-gas regulation 2024/573 at the Signing Date. As such, their functioning shall not use or rely upon fluorinated greenhouse gases in insulating or breaking medium. Contractor may use equipment using or relying upon fluorinated greenhouse gases in insulating or breaking medium with a GWP higher than 1 if and only if he duly justifies that no other

equipment met the relevant technical requirements after conducting a tender procedure for such equipment, as per article 13, sections 11-12-13 of the F-gas regulation.

[\[B21ai-GIS-110069\]](#)

All equipment shall be compliant with the PFAS related regulation at Taking Over.

[\[C11-DCHA-010147\]](#)

It shall be possible for personnel to access the HVDC Cable Link termination DC halls including HVDC cable DC halls and DMR cable hall and perform inspection and maintenance while the HVDC Link is in operation.

If a specific arrangement for the converter room is offered (for ex: combination of DC hall & DC termination with the valves) which does not allow access online, then a system shall be proposed:

- to allow adequate visual inspection, covering more than 90% of the equipment in the non-accessible room, such as: DC termination (including gas monitoring, cleats, etc.), reactors, etc. See (B22a_iv) for valve hall visual inspection.
- to allow adequate maintainability both for preventive and curative. This should be duly justified with case studies, e.g. to verify handling gas refilling especially for DC cable termination.

Cable Contractor will be responsible for the provision of remote refilling system. Converter Contractor shall consider the interface in execution stage, in order to integrate the remote refilling system of Cable Contractor from an accessible area during operation until the cable termination (e.g.: pipes).

Same remote refilling with pipes shall be possible for other non-accessible equipment during operation.

- to allow adequate temperature, humidity and dust control (should be fully justified with CFD type analysis for temperature control especially).

8.9.1 Onshore station

8.9.1.1 General requirements

[\[B21ai-HSDB-110525\]-N3](#)

The Contractor shall design, procure and install on the grid side of the converter transformer the 400kV circuit breakers and associated disconnectors and earthing switches if needed.

[\[B21ai-HSDB-110017\]-I](#)

The connection means (OHL or HVAC cable) from the Onshore Converter Stations to the Employer' transmission grid and associated data are provided by Employer in (Call_off).

[\[C11-ACGR-010079\]](#)

The equipment rated short circuit current for the 400 kV Onshore Converter Station HV equipment including busbars and switching devices (1 second) shall be 63 kA.

Short-circuit current of 63 kA at PCC-AC Onshore shall be considered for the rated short circuit current of the Onshore Converter Station transformer.

[\[B21ai-HSDB-110484\]-N2-M](#)

Grounding points shall be located so that maintenance can be carried out in accordance with (NFC18_510_1) and the Employer's implementation of the before mentioned standard. Grounding shall be possible by means of a grounding switch or by means of a temporary grounding point.

Grounding switch shall be used at least in the following cases:

- (i) Interlock for the purpose of providing access to enclosed spaces: the interlock shall prevent entry to a space if the equipment within it is not electrically grounded. In addition, it shall be possible to visually check the grounding of the zone in question (either from the outside or by means of cameras).
- (ii) Grounding action cannot be performed from the floor without lifting devices with a temporary grounding point.

Grounding switches shall be installed directly at the primary and secondary sides of the Onshore Converter Transformer in any case.

[\[B21ai-HSDB-110485\]-N2](#)

In case of presence of “hot standby” converter transformer, the Contractor shall use a bus bar structure for the delta (or wye) coupling of the tertiary windings of the converter transformers. This is to allow easier and faster connection/disconnection of the terminals in case of a transformer swap.

[\[B22ai-FUNC-420013\]](#)

In addition, in case of the spare transformer is a hot spare unit “ready to use”, considering that all prerequisites in order to start the intervention on site are met (necessary tools, etc.), the time necessary for such intervention shall be limited for 30 hours of work considering a team of two technicians and one supervisor. In addition, after replacement of the faulty unit with the spare unit it shall be possible to remove from the converter station the faulty unit with the Converter Station in operation. This means that sufficient clearances shall be granted between the faulty transformer and live parts even when sliding beams are fitted below the transformer.

[\[B22ai-FUNC-420375\]](#)

To facilitate transformer swap, no low voltage wiring tasks (for CTs, C&P IO, monitoring, alarms, message ...) shall be necessary during the transformer swap procedure. The swap shall be performed via switch systems with an appropriate interlock (for e.g. only accessible when converter station is Earthed and by using a latchable key coming from the faulty transformer).

[\[B21ai-HSDB-110498\]-N2](#)

The converter side bushings of the converter transformers shall not penetrate the valve hall or any other building.

[\[CBOD-LAYO-100271\]](#)

The height of the converter transformer fire and blast barriers shall at least 40 cm above the level of transformer conservator.

[\[C11-DCHA-010148\]](#)

Regarding DC capacitors (if any)

- To ensure personnel's safety, no access to the DC capacitor bank is permitted during operation.
- If a protection relay is insufficient or not feasible to safeguard individuals and nearby equipment, a physical protection measure must be implemented to restrict access to any area that projectiles may reach.

8.9.1.2 Cable termination hall onshore

8.9.1.2.1 Design

[\[C11-DCHA-010110\]](#)

HVDC Link Cable Terminations including the HVDC cable and DMR terminations shall be installed indoor within cable termination DC Halls.

Oil insulation is prohibited for AIS cable terminations.

Requirements and interfaces are described in (SOW03) and (BOD_02_Interface_Basis_of_design).

[\[C11-SYST-010106\]](#)

As part of the commissioning tests, the Cable Contractor will perform a voltage test on the HVDC cable link. This test will be performed on each pole of the complete link separately, at $-U_{TP1}$ ($-1.45 \cdot U_0 = -761$ kV) for 1 h with a dedicated generator.

[\[CBOD-SYST-100019\]](#)

The installed DMR cable system will be subjected to a negative DC voltage test that will be determined through agreement between Cable Contractor and Employer. The tested voltage value will not be lower

than the highest of either $2.5 U_{RC,DC}$ or -60 kV. This voltage will be applied between conductor and metallic sheath of DMR for 1 h.

[\[C11-DCHA-010138\]](#)

Transition across the building wall, from the trench ducts outside building to the termination pit inside building, shall ensure waterproofing of building whilst complying with power cable maximal admitted compression.

[\[C11-DCHA-010139\]](#)

Top of the termination pit shall be covered in order to recreate flatness with the building zero level. This cover shall withstand a minimal load of 400 kg/m².

[\[C11-DCHA-010141\]](#)

Frame structures (cable termination frame structure or dedicated one) and cable layout shall enable the installation of current transformers (closed toroid) around power cable, especially by:

- freeing space around cable for current transformers installation – depending of toroid dimensions,
- fixing of the current transformers – depending of toroid fixing arrangements,
- bearing of the current transformers – depending of toroid weight,

8.9.1.2.2 Installation requirements

[\[C11-DCHA-010145\]](#)

HVDC Cable Link termination DC Halls roof structure shall be equipped for installation and maintenance purpose of hooks for lifting, one directly above each HVDC Cable Terminations including DMR termination. These hooks shall withstand a minimum load of 2 T and shall enable the hanging of a pulley block – depending of pulley block characteristics.

[\[C11-DCHA-010146\]](#)

Access door shall be planned and shall be adequately sized by Converter System Contractor to allow equipment (such as crane) necessary to the replacement of any indoor cable termination DC hall apparatus (cable termination, or else) to enter/exit the hall and make such replacement possible.

8.9.2 Offshore station

8.9.2.1 General requirements

[\[B21ai-HSDB-110527\]-N2](#)

Contractor shall follow AC GIS offshore Single Line Diagram as illustrated in Figure 19, with notably the following characteristics:

- Double busbar type (as illustrated in zone 1 in Figure 19)
- AC CB to couple the busbar sections to allow connection/disconnection of the converter transformer while the HVDC Link is in operation. Contractor may propose other configurations (e.g.: disconnectors) to fulfil this requirement (as illustrated in zone 2 in Figure 19)
- In total 14 connections for IAC strings (as illustrated in zone 3 in Figure 19). The number of connections shall be confirmed in (Call_off). The power difference across AC offshore electrical node will be less than 30 MW.
- One line disconnector for each IAC between the AC CB and the IAC.
- Disconnector to connect the busbar sections to optimize power flows coming from the available IAC to one HVDC Converter Pole during commissioning (as illustrated in zone 4 in Figure 19). The Contractor shall propose the suitable interlocking system to prevent the inadvertent connection of all four busbar sections during any situation other than commissioning.

In addition, Contractor shall design the AC GIS fulfilling the following requirements:

- Ensure availability/reliability during the lifetime of the Converter Station as required in section 7.2 and minimize the energy unavailability.
- Ensure that the Guaranteed Active Power Link Capacity can be exported from PCC-AC offshore to PCC-AC onshore.

Contractor shall prove the above requirements are fulfilled in a study provided to Employer.

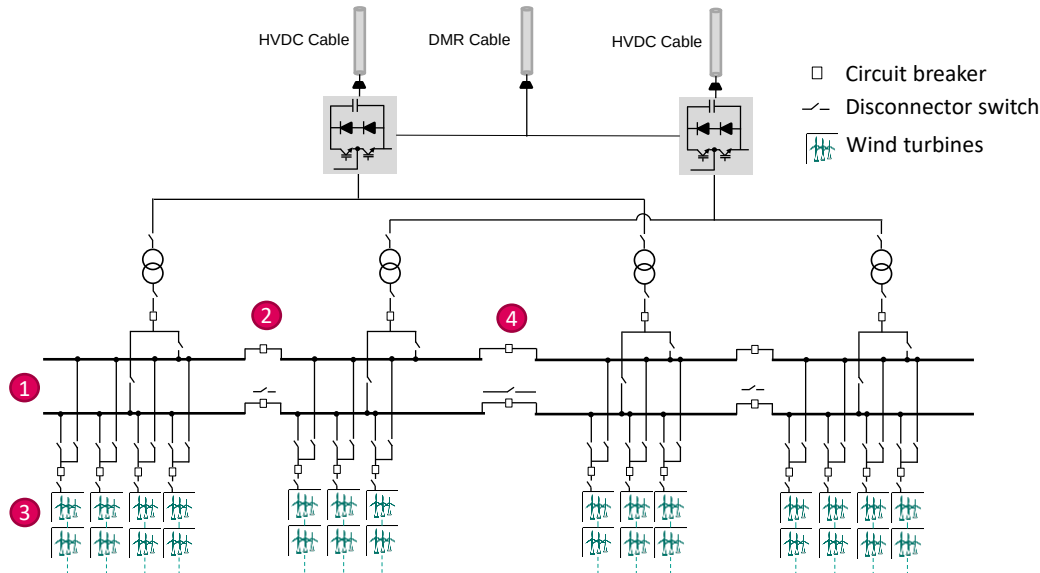


Figure 19: Overview of offshore Single Line Diagram. Zone 1: busbar structure, zone 2: coupling between different electrical nodes, zone 3: IAC strings, zone 4: coupling between nodes for commissioning purpose

[\[B21ai-HSDB-110579\]-N2](#)

Each of the four converter transformers of the Offshore Converter Station shall be rated for a minimum of 25 % of Guaranteed Active Power Link Capacity continuously. This shall be for the active parts. The non-active parts (e.g. bushings) shall be designed for 30% of Guaranteed Active Power Link Capacity continuously.

In addition, each converter transformer of the Offshore Converter Station shall be designed with an overload capability according to (IEC60076_7).

Besides, all other equipment where the 25% of power flows through in normal operation shall be rated to withstand the 30% of Guaranteed Active Power Link Capacity continuously.

[\[B21ai-GIS-110623\]](#)

If cables are used between the 132 kV GIS and the converter transformers, oil cable boxes shall be used.

[\[CBOD-HSDB-100020\]](#)

The converter transformers shall be connected to the converters either via AIS or GIS (HV cables are not permitted). If GIS is used, it shall be a DC GIS.

The DC switchgear for the Offshore Station shall be designed in air-insulated or in GIS.

8.9.2.2 DC cable termination hall offshore

[\[C11-DCHA-010160\]](#)

Oil insulation is prohibited for AIS cable terminations (including DMR cable termination).

[\[C11-DCHA-010161\]](#)

Requirements and interfaces are described in (SOW03) and (BOD_02_Interface_Basis_of_design).

8.10 High voltage equipment

8.10.1 General requirements

[CBOD-HVEQ-100057]

For every component, type tests shall be performed on a complete unit corresponding exactly to the type used in the Onshore and Offshore Converter Stations and under the conditions required for operation. Type and routine tests shall be carried out in accordance with relevant standards (e.g. IEC). Additional tests (e.g. based on Employer standards – see dedicated equipment specifications) shall also be performed.

- For reactor and DC capacitor, type test shall be repeated if the provided type test report is not referring to the exact same design. No comparison reports are accepted. For short-circuit test, a comparison report may be provided if the tested reactor has the exact same design and withstands the same or a higher stress.
- For Employer's qualified equipment (see (B22a0)), type test is not requested as they are deemed already passed.
- For all other equipment: Evidence of type tests that have already been performed according to the relevant standard can be accepted in lieu of new type tests. However, this acceptance is contingent upon the tested unit precisely corresponding to the equipment deployed in the projects. Any exceptions shall be reviewed with the Employer in the context of the type of equipment and its location. In addition, a Type Test verification report is required for the Employer' evaluation for each case.
- The list of qualified/preferred equipment/system is provided in (B22a0)

[B22a-CHANG-410023]

Test reports relevant to Type/special Tests carried out by third parties can be submitted to the Employer only if the tests have been certified by a qualified independent laboratory complying with Standard (ISO_IEC_17025).

[B22a-CHANG-410024]

Test reports relevant to Type/special Tests carried in the Contractor's premises can be submitted to Employer only if a Qualified Inspector from an independent body certified according to (ISO_IEC_17065) has witnessed the tests.

[CBOD-HVEQ-100021]

Particular protection shall be provided for high voltage equipment against dirt and the effects of weather (e.g. salt spray) considering the respective site locations of the Converter Stations and the operating environment where the equipment is installed. Hermetic housing of all high voltage equipment shall be ensured in the Offshore Station; any exceptions shall be reviewed with the Employer in the context of the type of equipment and its location.

The minimum equipment corrosion protection shall be:

Onshore			Offshore		
Outdoor / Uncontrolled indoor	Indoor Semi controlled*	Indoor Full controlled**	Outdoor / Uncontrolled indoor	Indoor Semi controlled***	Indoor Full controlled***
C5-H	C4-H	C4-H	CX	C4-H	C4-H

* Semi-controlled: controlled temperature (min, max), dust and salinity (1S stage air filtration with at least ePM10(60%) as per ISO16890). Environment shall be controlled in a way to avoid condensation on the surface of equipment.

** Full controlled: controlled temperature (min, max), dust and salinity (2S stage air filtration with at least ePM10(60%) and at least ePM1(80%) as per ISO16890), maximum relative humidity.

*** Corrosion protection of offshore equipment shall meet at least the requirements described in (ERS_Corr).

[CBOD-HVEQ-100058]

Employer general requirements related to high voltage equipment stated in (B22a) shall be applied.

8.10.2 Insulation coordination

8.10.2.1 General requirements

[\[C11-SYST-010055\]](#)

The insulation coordination of the HVDC Link shall be carried out by Converter System Contractor for the respective AC and DC side of the stations in accordance with (IEC60071_1) and (IEC60071_11). The protection levels shall be obtained from the insulation coordination study. Surge arresters in adjacent onshore substations shall be considered in the design.

The Converter System Contractor shall supply all AC and DC surge arresters to protect the equipment of the Converter Stations, the HVDC Cable including DMR cable, and the Inter Array Cables against overvoltages in the AC systems, in the Converter Stations themselves or in the DC sections.

[\[C11-SYST-010158\]](#)

Insulation failure shall not lead to an additional failure of any kind including additional insulation failures, failure from excessive heat caused by current stress or a combination of current and voltage stress on the HVDC equipment.

The converter station shall withstand the inadvertent opening of a breaker in the last line connecting the HVDC station. However, if it is identified only during design phase or later that the Converter Station may not meet this requirement, it is only tolerated if damage is limited to surge arresters. The rest of the station shall withstand the stress. Note that for breaker opening in the substation where the HVDC is connected, a signal will be sent.

Proof that sufficient margin is considered shall be given by Contractor to Employer on critical cases (e.g.: valve short circuit).

[\[CBOD-100022\]](#)

To avoid doubt, the definitions of AC and DC sides of both Onshore and Offshore Converter Stations follow the description in Figure 8 of (IEC60071_12).

8.10.2.2 Minimum insulation levels

8.10.2.2.1 AC withstand level (AC grid side)

[\[C11-SYST-010059\]](#)

The AC withstand levels shall be compliant with (IEC60071_1) with the minimum values as stated in Table 24.

Description		Value	
		Onshore	Offshore
Phase-Earth	Switching Impulse Withstand Level (SIWL)	1050 kV pk	/
	Lightning Impulse Withstand Level (LIWL)	1425 kV pk	650 kV pk
	Short-term AC voltage	520 kV rms	275 kV rms
Phase-Phase	Switching Impulse Withstand Level (SIWL)	1575 kV pk	/
AC disconnector	Across open contact SIWL	900 + 435 kV pk	/
	Across open contact LIWL	1425 + 240 kV pk	750 kV pk
	Short-term AC voltage across open contact	610 kV rms	315 kV

Table 24: Minimum insulation levels for HV equipment connected to the onshore and offshore AC grid

8.10.2.2.2 Converter station withstand levels (DC side)

[\[C11-SYST-010060\]](#)

Insulation coordination shall be made in accordance with (IEC60071_11). Insulations margins between withstand voltage and protective level indicated in table 3 of (IEC60071_11) shall be used.

8.10.2.3 Creepage distance

[\[C11-SYST-010065\]](#)

Minimum creepage distance shall be determined according to the respective base voltage for creepage distance multiplied by the unified specific creepage distance (USCD) in mm/kV. The required creepage distance shall be adjusted for the insulator diameter as described in (IEC60815_4). The base voltages

are specified in (IEC60071_11) section 8.2, except for RLC, the base voltages are specified in [\[B22aiii-GENE-440003-N3\]](#) of (B22a_iii).

The pollution classes characterizing Site Pollution Severity (SPS) are referred to (IEC60815_1).

The minimum USCD specified in Table 25 shall be applied.

	AC side			DC side				
	Outdoor / Uncontrolled indoor	Indoor Semi controlled*	Indoor Full controlled**	Outdoor / Uncontrolled indoor	Indoor Semi controlled*	Indoor Full controlled**	Valve hall	Converter tower
USCD (mm/kV) Composite	53.7 for HA1, HA2 For BNO and GLC: call-off	34.7	27.8	45 53.7 for converter transformer bushings	30	20	20	14
USCD (mm/kV) Porcelain	53.7 for HA1, HA2 For BNO and GLC: call-off	34.7	27.8	60	30	20	20	14

* Semi-controlled: controlled temperature (min, max), dust and salt (1S stage air filtration with at least ePM10(60%) as per ISO16890). Environment shall be controlled in a way to avoid condensation on the surface of equipment.

** Full controlled: controlled temperature (min, max), dust and salt (2S stage air filtration with at least ePM10(60%) and at least ePM1(80%) as per ISO16890), maximum relative humidity.

Table 25: Minimum USCD. Final values shall be confirmed at (Call_off). The correlation between USCD and SPS level is following (IEC60815_2) and (IEC60815_3).

8.10.2.4 Clearance distance

8.10.2.4.1 AC air clearance distance

[\[B21ai-SITE-110019\]](#)

For the AC side of onshore and offshore converter stations, national regulations and (IEC60071_1) and (IEC60071_2) shall be respected.

In addition, the following distance shall be respected. These distances are related to Employer security rules (CCG_P).

	Nominal voltage (kV)	63	90	132	225	400
	Rated voltage (kV)	72.5	100	*	245	420
DM	Phase to earth clearance distance (m) (a) Conductor / Structure (b) Point / Plane or point / structure	0.66	0.92	*	2.14	2.90 (a) 3.50 (b)
DP	Phase to phase clearance distance (m) (c) Conductor / conductor (d) Point / conductor	0.76	1.06	*	2.47	4.00 (c) 4.20 (d)
	Height of firt insulating section (m) Vertical operator clearance	2.25	2.25	2.25	2.25	2.25
D1 DLV	Vicinity limit distance D1 (m)	3.00	3.00	3.00	3.00	4.00
DMA	Minimum Approach Distance DMA (m)	0.80	1.00	1.20	1.60	2.50
DMAC	Corrected minimum approach distance between phase and earth DMAC (m)	0.95	1.10	*	1.85	3.10
DMS	Minimal distance from ground DMS (m)	3.20	3.35	*	4.39	5.75
D3	Comfort distance (m)	1.60	1.80	*	2.40	3.50
Z1	Operator development zone (m)	2.00	2.00	2.00	2.00	2.00
Z2	Operator development zone (m)	1.20	1.20	1.20	1.20	1.20
DT1	Working distance in one direction (m)	5.00	5.00	5.00	5.00	6.00
DT4	Working distance in 3 other directions (m)	2.80	3.00	*	3.60	4.70

* These values will be provided at (Call_off).

8.10.2.4.2 DC air clearance distance and/or mixed AC/DC clearance distance

[\[B21ai-SITE-110020\]](#)

For the DC side of Onshore and Offshore Converter Stations, the Contractor shall design the minimum air clearances depending on its experience and design practice considering the environmental

conditions stated in these specifications, (IEC60071_11) shall be used as a reference. The onsite test of HVDC cable shall also be considered in the determination of air clearance distance.

The Contractor shall submit technical documentation highlighting the adopted calculation criteria and methodology.

8.10.3 Maximum allowable overvoltages and currents on HVDC cable terminations

8.10.3.1 DC temporary overvoltage (DC TOV) for HVDC cables

[C11-SYST-010061]

The DC temporary over voltage (DC TOV) on the HVDC cable termination shall never exceed the voltage profile illustrated in Figure 20 for all fault scenarios (type and location). The parameter values are given in the table integrated in Figure 20.

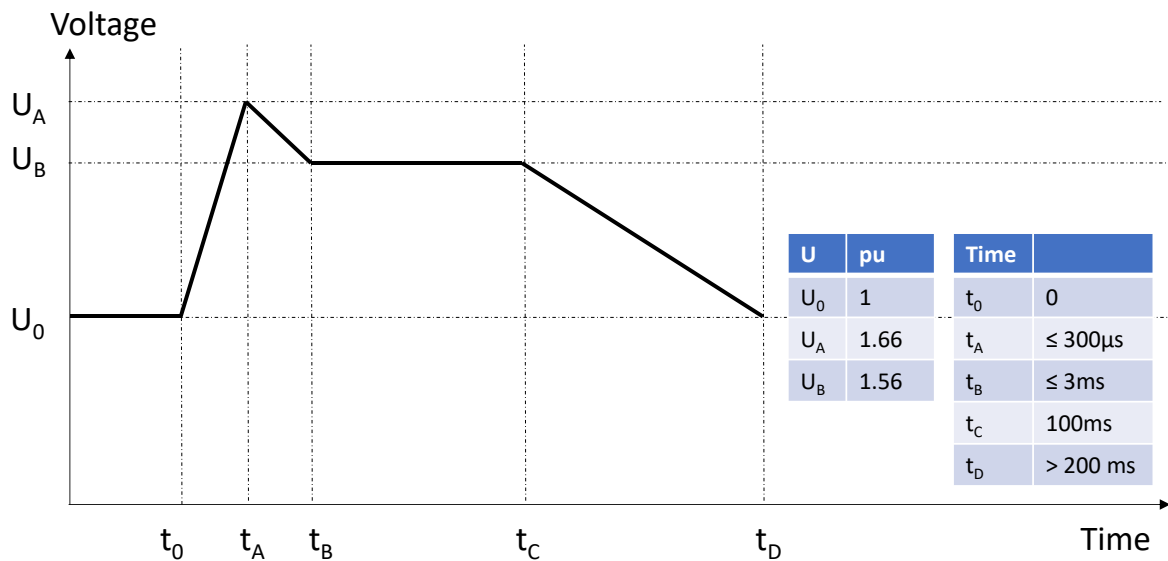


Figure 20: DC TOV profile. U_0 equals to 525 kV.

[CBOD-SYST-100023]

The Converter System Contractor shall design the system to ensure that, in any case of fault, on the HVDC cable termination, opposite voltage oscillation U_c , if any, shall never exceed $0.85 U_0$.

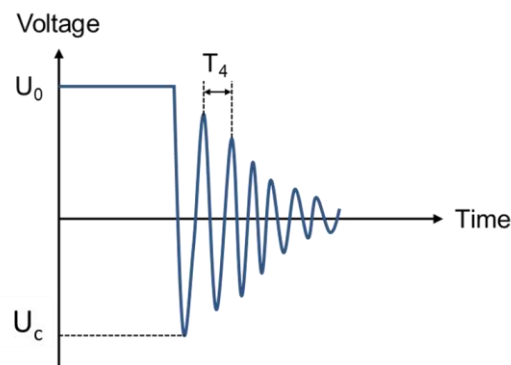


Figure 21: Zero-crossing damped temporary overvoltage

8.10.3.2 Withstand voltage level for DMR cable

[CBOD-SYST-100024]

The insulation coordination of DMR cable shall be studied by the Contractor and submitted to the Employer for validation.

In any case of fault, the over voltage on the DMR cable termination shall never exceed

- Equivalent to switching impulse withstand level (SIWL) 200 kV
- Equivalent to DC TOV (profile defined in (CIGRE_TB852)): $U_1 = 150$ kV

[\[C11-SYST-010064\]](#)

All HVDC Cable Link including DMR cable must be protected from lightning impulse voltage.

8.10.3.3 DC short circuit current

[\[C11-SYST-010045\]](#)

Converter Station shall be designed such that the fault currents are less than the values stated below.

- Maximum fault current: 45kA during 100ms.
- Maximum fault current in screen: 25kA during 100ms.
- Maximum fault current in core: 25kA during 100ms.

[\[C11-SYST-010046\]](#)

During project execution, the Converter System Contractor shall confirm the DC short-circuit current profile (values versus duration) on the basis of its own converter design, the AC short circuit levels given in this document and the provisional and final HVDC Cable Link design data provided in (BOD_02_Interface_Basis_of_design).

8.10.4 Converter transformers

[\[CBOD-TRANS-100060\]](#)

The Employer's requirements for converter transformers detailed in (B22a_i) shall be applied.

As part of Employer's requirements, the following requirements shall be applied as minimum.

- **Design**
 - The converter transformers shall be compliant with IEC standard, (IEC_60076_series)
 - [\[B22ai-FUNC-420023\]](#): No windings with a rated voltage below 10 kV are allowed in the converter transformer.
 - [\[B22ai-FUNC-420025\]](#): The use of internal varistors such as ZnO is forbidden.
 - [\[B22ai-FUNC-420026\]](#): No additional active parts such as auxiliary transformers or series reactors (even in tertiary) are allowed in the converter transformer tank. Nevertheless, if a series reactor was deemed absolutely necessary it could be placed in a separate tank attached to the converter transformer tank or in any other suitable location.
 - [\[B22ai-FUNC-420044\]](#): Neutral end of windings either HV or MV with non-uniform insulation shall have a minimum U_m of 100 kV and be tested with the relevant insulation levels. The minimum insulation levels and test voltages of converter transformer shall be according to (IEC60076_3) and (IEC60076_22) standard. If Contractor duly justifies it with an exhaustive study (considering in particular unbalanced faults), depending on the neutral impedance, neutral insulation level U_m of 72.5kV may be considered.
 - [\[B22ai-FUNC-420046\]](#): The Contractor shall determine and guarantee the noise level of converter transformer in case of no load and in case of short-circuit. The noise levels to be guaranteed shall be in sound power ($L_w(A)$) and the following values shall be guaranteed.
 - No load sound level with the converter transformer energized at rated voltage and rated frequency with and without coolers in operation
 - Short-circuit sound level with the converter transformer supplied with rated current, for this it will be energized at short-circuit voltage and rated frequency,
 - [\[B22ai-CONS-420394\]](#): Relevant design shall be made and justified for offshore converter transformer erection. Vibration shall not generate malfunctions in the long term of adjacent equipment. Vibration pads shall be installed in compliance with section "Vibrations from transformers" of (CIGRE_483).
 - [\[B22ai-TEST-420337\]](#): Dynamic short-circuit withstand tests shall be performed by Contractor, in accordance with category III of the standard (IEC60076_5). If dynamic

short-circuit withstand tests shall not be performed, all detail of the converter transformer shall be supplied to Employer to check the calculation of proof of withstand. All the data requested by Employer are mentioned in section 7.2.3.9.2 of (B22a_i).

- [\[B22ai-CONS-420056\]](#): The Contractor shall submit a study of the transport from the factory to the installation sites, highlighting the critical issues (maximum dimensions and weights) and the modalities detected to solve them. If required by Employer, Contractor shall give all the relevant documents and accesses to its Marine Warranty Surveyor
- [\[B22ai-CONS-420132\]](#), [\[B22ai-CONS-420135\]](#), [\[B22ai-EQPT-420272\]](#): The tap changer shall be installed in a dedicated chamber and be vacuum type for low maintenance. The use of fuses is forbidden for motor drive of tap change, auxiliary and control cabinet.
- [\[B22ai-CONS-420066\]](#): The converter transformers shall be “core-type” with concentric winding.
- [\[B22ai-CONS-420098\]](#): The tank and the turrets shall be designed to withstand internal arc without blowing away or leaking in the valve hall. The Contractor shall demonstrate this ability taking into account the tank flexibility (m^3/kPa), an arc energy of 10 000 kJ and a dynamic amplification factor. The tank can be subject to plastic (permanent) deformation in this calculation.
- [\[B22ai-CONS-420082\]](#): The insulating oil shall be unused (type TVAI – 30°C) and in accordance with IEC 60296 standard for the class “transformer oil”. The oil shall offer enhanced biodegradability properties i.e. more than 40 % of biodegradability according to OECD 301.
- [\[B22ai-CONS-420103\]](#): Bolts and nuts shall be stainless steel A4-80 or higher. Where bimetallic gudgeons are used, the non stainless steel (lower) end shall be protected against corrosion as the main tank. The use of A4-70 may be allowed at Employer's discretion and upon written deviation request for bolts not subject to force. All heavy duty bolting must be A4-80.
- [\[B22ai-CONS-420084\]](#): The tank for the converter transformers shall be bell type with welded joint and flange designed to allow at least four de-welding and re-welding operations without modifying the flange.
- **Testing**
 - [\[CBOD-TRANS-100061\]](#) For Factory Acceptance Tests, Employer's additional recording and measuring requirements as mentioned in section 7.2 of (B22a_i) shall be applied
- **O&M**
 - [\[B22ai-CONS-420170\]](#): The converter transformer must be equipped with a fixed safety ladder compliant with standard NFE 85 016 placed on one of the sides of the converter transformer. The base of the ladder must be fitted with a closing door that can be maintained in the closed position by means of a padlock. The device must lock the first rungs of the ladder up to a minimum height of 1.90 m.
 - [\[B22ai-FUNC-420371\]](#): To limit the noise, noise enclosure shall be provided by Contractor if needed. The noise enclosure can be supported by the tank. In that case, it shall allow access to the equipment necessary for operation and for maintenance by a simple door mounted on hinges and opened by horizontal rotation. These doors inside the enclosure shall be identified and guarantee total opening to allow access to the equipment it protects. In all cases, the sound enclosure shall allow easy access to the necessary equipment for operation and maintenance.

8.10.4.1 Onshore converter transformers

[\[B22ai-FUNC-420021\]](#)

The onshore converter transformers are single phase units intended to be connected in three-phase bank. Auto-connected windings are not allowed. The tertiary connection shall be capable of supplying the converter station's auxiliary supply. Auxiliaries shall be supplied through a dedicated auxiliary transformer on the MV side.

All neutral points shall be brought out.

[\[B22ai-FUNC-420022\]](#)

The three phase bank connections are:

- HV YN
- MV To be defined by the Contractor
- LV d11

8.10.4.2 Offshore converter transformers

[\[B22ai-FUNC-420440\]](#)

The offshore converter transformers shall be three phase units. Auto-connected windings are not allowed. Yoke is not allowed.

All neutral points shall be brought out.

8.10.5 Auxiliary supply transformer (AST)

8.10.5.1 Onshore Auxiliary supply transformer

[\[CBOD-TRANS-100062\]](#)

The Employer's requirements for onshore auxiliary supply transformers (AST) detailed in (B22a_viii) shall be applied.

As part of Employer's requirements, the following requirements shall be applied as minimum.

- [\[CBOD-TRANS-100063\]](#) The auxiliary transformers shall be compliant with IEC standard, (IEC_60076_series). For routine and type tests, Employer's additional recording and measuring requirements as mentioned in section 8.3 and 8.4 of (B22a_viii) shall be applied.
- [\[B22avi1-ASTS-680042\]](#): The value of the rated voltage (Ur) onshore of each AST or Neutral Earthing Auxiliary Transformer winding is indicated below:
 - HV: 20 kV / LV: 400 V
 - The rated voltage of the winding that may be necessary to obtain the zero-sequence impedance of the Neutral Earthing Auxiliary Transformer shall be determined by the Contractor. This winding shall be brought out and shall be able to be used for testing purposes.
- [\[B22avi1-ASTS-680043\]](#): The cooling shall be of ONAN type.
- [\[B22avi1-ASTS-680044\]](#): The phase displacement indice shall be 11.
- [\[B22avi1-ASTS-6800258\]](#): The minimum insulation levels and test voltages of the AST shall be in accordance with (IEC60076_3). For 145 kV level, the requested full wave lightning impulse, chopped wave lightning impulse, switching impulse and are withstand are 650 kV, 715 kV, 540 kV and 275 kV respectively.
- [\[B22avi1-ASTS-680060\]](#): AST shall be hermetically sealed and shall have expandable tank or radiator.
- [\[B22avi1-ASTS-680130\]](#): The on-load tap changers shall be vacuum type.
- [\[B22avi1-ASTS-680149\]](#), [\[B22avi1-ASTS-680157\]](#): The use of fuses is prohibited for low voltage circuit of tap changer and for control cabinet.
- [\[B22avi1-ASTS-680077\]](#): All of the fasteners shall be made from stainless steel of quality A4-70 or higher. If bimetallic welded studs are used, the bottom part of the stud shall be protected against corrosion like the tank. For accessories consisting of threaded parts (some boxes for example), the tapping may be unprotected if it is equipped with a screw during transport, storage and on-site operation.
- [\[B22avi1-ASTS-680177\]](#): The bushing installed on the high voltage side of the AST shall be plug-in, have an outer cone and comply with standard (NF_EN_50180).
- [\[B22avi1-ASTS-680093\]](#): If cable boxes are installed, they
 - Shall have at least IP 54 and may have vertical walls for separation of each phase,
 - Shall be coated on the inside with anti-condensation paint,

- Shall be fully equipped,
- Shall have a removable wall to be able to perform a cable connection,
The LV cables shall enter the cable box through cables glands fitted on a wall tilted downwards.
- [\[B22avi1-ASTS-680053\]](#): The proof of short-circuit withstand shall be provide to Employer. If short-circuit withstand tests shall not be performed, the Contractor shall submit a theoretical evaluation of the ability of the Auxiliary Transformer to withstand the dynamic effects of short circuit for acceptance from the Employer. This calculation shall be made in accordance with (*IEC60076_5*) and comparison made to similar units.

8.10.5.2 Offshore auxiliary supply transformer

[\[CBOD-TRANS-100064\]](#)

The Employer's requirements for offshore auxiliary transformers detailed in (ERS_5_6) shall be applied.

8.10.6 Gas Insulated Switchgear

[\[CBOD-GIS-100065\]](#)

The employer requirements for AC/DC Gas Insulated Switchgear detailed in (B22a_vii) shall be applied. DC Circuit breaker is excluded by Employer.

As part of Employer's requirements, the following requirements shall be applied as minimum.

- **Design**
 - [\[CBOD-GIS-100066\]](#) The Contractor shall propose GIS in compliance with (*IEC62271_203*). For AC GIS circuit breaker, the electrical, capacitive and mechanical classes shall be E2, C2 and M2 respectively to ensure the sustainability of the product.
 - [\[CBOD-GIS-100067\]](#) To ensure the reliability of DC GIS if any, tests according to (*IEC62271_318*). Additionally, to guarantee long-term performance, long-term energized test compliant with (*IEC62271_318*) is mandatory.
- **O&M**
 - [\[CBOD-GIS-100068\]](#) The GIS bays shall be designed in such a way to be compliant with Employer's reliability and availability (section 7.2.2.1). The SLD with gas compartments shall be presented to Employer for validation.
 - [\[B.2.2.a.vii-GIS-460088\]](#): The GIS shall be partitioned to avoid the risk of discontinuity of service. Working on pressurized insulators is not permitted, even if the conductor is mechanically fixed. In case of fault in a compartment, for safety reasons, gas pressure in compartments adjacent to the fault compartment shall be reduced to 0.3 bar relative before any operation in fault compartment.
 - [\[B.2.2.a.vii-GIS-460072\]](#): During operation of GIS, it shall be possible to perform gas filling operations. The refilling system for GIS shall be stored in Offshore Station.
 - [\[B.2.2.a.vii-GIS-460090\]](#): Each IAC cable/GIS termination shall have a removable link for onsite dielectric tests and cable fault search.
- **Monitoring**
 - [\[CBOD-GIS-100069\]](#) The GIS monitoring system shall be installed by Contractor shall follow Employer's requirements provided during execution. For information, they are reminded in Appendix 26 of (B22a_v).
 - [\[B.2.2.a.vii-GIS-460058\]](#): For gas pressure monitoring system, density monitor shall be equipped with an electronic interface, which enables continuous monitoring of gas density, gas pressure and gas temperature (additional gas humidity monitoring is preferred by Employer). It shall be possible to check the signal output of a density monitor while the GIS is in operation without changing the gas pressure in the monitored gas compartment and without removing the density monitor.
It shall be possible to locate the internal arc fault but, in any case, arc sensors with optical detection methods are forbidden.
 - [\[CBOD-GIS-100070\]](#): For online partial discharge monitoring, Contractor shall be responsible for optimizing the locations and the number of PD sensors. However, a PD sensor shall be installed as close as possible to each interface cable/GIS termination.

8.10.7 AC/DC equipment

[\[CBOD-ACDC-100025\]](#)

The Employer's requirements for AC/DC equipment including surge arrester, post insulator, bushing, AC circuit breaker, disconnect switch, voltage and current instrument transformers detailed in (B22aii) shall be applied.

[\[B22aii-GENE-430284\]](#)

For outdoor composite housing equipment, weather ageing test equipment needs to be performed in accordance with the 5000 hours multiple stress test protocol mentioned in (IEC/TR 62730). The test on similar equipment or similar design can also be submitted to Employer for validation.

[B22aii-GENE-430287\]](#)

The composite housing shall be polymeric hydrophobic (HTM) uniformly, in compliance with (IEC_TS_60815_3) clause 5.

8.10.7.1 Surge arrester

[\[CBOD-SURA-100071\]](#)

The AC and DC surge arresters shall be designed and tested in compliance with the (IEC60099_4) and (IEC60099_9) respectively.

8.10.7.2 Post insulator

[\[CBOD-INSU-100072\]](#)

The post insulators shall be designed and tested in compliant with relevant IEC standards.

[\[B22aii-INSU-430035\]](#)

Ceramic insulators are preferred. In case of composite, HTV (High Temperature Vulcanisation) material shall be used.

8.10.7.3 Wall bushing

As part of Employer's requirements, the following requirements shall be applied as minimum.

- [\[CBOD-WALB-100073\]](#) AC and DC wall bushing shall be designed and tested in compliance with (IEC60137) and (IEC_IEEE_65700_19_03) respectively
- [\[B22aii-WALB-430062\]](#): All bushing shall be of dry type (RIP or RIS) and without any liquid to reduce the fire risk. Gas filled bushings are not allowed.
- [\[B22aii-WALB-430074\]](#): The DC bushing shall have a test tap according to (IEC60137) clause 3.38.
- [\[B22aii-WALB-430070\]](#): Insulation test level for BIL, SIL and power frequency, whatever the bushing type is, shall be chosen depending on U_m in the table 3 of (IEC60137). The DC test voltages are defined according to (IEC_IEEE_65700_19_03).
- [\[B22aii-WALB-430090\]](#): As part of type test, even and/or uneven wetting DC voltage tests are required in case the bushing is partially sheltered. Contractor can justify the omission of this test by the design and mounting arrangement of the bushing.
- [\[B22aii-WALB-430093\]](#): Partial discharge measurement in DC voltage during routine tests shall follow the Employer's requirement.

8.10.7.4 AC circuit breaker

[\[CBOD-ACCB-100074\]](#)

The AC AIS circuit breakers shall be compliant with (IEC62271_100) and shall be mechanical class M2 and capacitive class C2.

[\[B22aii-ACCB-430113\]](#)

To ensure the reliability of circuit breaker, each operating mechanism shall have 2 independent opening coils on different magnetic systems and at least 1 closing coil. The opening circuits shall be redundant with separate auxiliary equipment and wiring. Additionally, circuit breaker status shall be readable by means of mechanical indication on the breaker. The indication shall be visible during operation.

8.10.7.5 AC/DC disconnect switch

[\[CBOD-DISC-100075\]](#)

The AC/DC disconnect switches shall be compliant with (IEC62271_102) and shall be at least mechanical class M1.

[\[B22aii-DISC-430263\]](#)

In case of outdoor installation, ceramic insulators shall be provided for AC/DC disconnect switches.

[\[CBOD-DISC-100026\]](#)

For DC disconnect, the contractor shall propose and realise a test where 5 successive closing of the DC disconnect, under the highest possible voltages which can be reached under operating conditions plus 5%. The disconnect cannot be replaced and shall remain the same for the 5 switching, however it is accepted that a change of setting is applied to the disconnect during the test. These settings shall be described in the test report. The test will be considered as successful and validate under the basis of a visual control of the active parts of the equipment.

The Contractor shall describe and propose a test procedure, for review and approval of the Employer.

8.10.7.6 Instrument transformers

As part of Employer's requirements, the following requirements shall be applied as minimum.

- [\[CBOD-DISC-100076\]](#): The AC/DC instrument transformers (conventional and LPIT) shall be compliant with the relevant IEC standard of (IEC_61869_series).
- [\[B22aii-CTVT-430289\]](#): The AC/DC instrument transformers shall be compliant with multiple (600) impulses tests as mentioned in (IEC_61869_1) and the Employer's mechanical tests.
- [\[B22aii-CTVT-430168\]](#): For conventional current transformer, Contractor shall propose top core design.
- [\[B22aii-CTVT-430196\]](#): For conventional current transformer, Contractor shall limit the voltage value when secondary winding is open. A voltage limit device on the output shall be provided.
- [\[B22aii-CTVT-430169\]](#): For conventional transformer (CT and VT), Contractor shall propose equipment in compliance with Employer internal arc test requirement. For 420 kV voltage level, fault duration of 525 ms for internal arc test is requested. Fault duration may be reduced if duly justified by Contractor.

8.10.8 Resistor, Reactor and Capacitor (RLC)

[\[CBOD-RLC-100077\]](#)

The Employer's requirements for resistor, reactor and capacitor (RLC) detailed in (B22a_iii) shall be applied.

As part of Employer's requirements, the following requirements shall be applied as minimum.

- [\[B22aiii-GENE-440201-N3\]](#): For Onshore station, fiber-reinforced concrete as supporting structure shall be used to avoid vegetation to grow and to avoid metallic parts forming closed loops inside reactor enclosures. Contractor may propose alternative solution which fulfils the same purpose.
- [\[B22aiii-GENE-440003-N3\]](#):

Reactors, Capacitors and Resistors shall be designed to withstand the most onerous current and voltage transients that could appear, determined in the system studies.

In the presence of harmonics, in the presence of harmonics, both background harmonic levels and future background harmonics as described in section "Background harmonics" of (CBOD) shall be considered.

- The rated current shall be calculated as the square root of the sum of squares of fundamental and harmonic currents.
- The rated voltage for different equipment shall be calculated as follow:

	Rated voltage across equipment	Creepage across equipment*	Insulator inside equipment	Insulator to earth	Other inter- frame insulators	Bushing
AC filter capacitor (HF filters included)	a)	a)	-	a)	a)	a)
AC filter reactor	b)	c)	-	RSS	-	-
Resistor (including AC filter, PIR, ...)	d)	d)	d)	RSS	RSS	d)
Smoothing reactor	b)	b)	-	RSS	-	-
Converter reactor	b)	b)	-	RSS	-	-
DC capacitor	d)	d)	d)	d)	d)	d)
Other reactor	b)	c)	-	RSS	-	-

* Turn to turn / between R or C elements

- a) Linear sum the fundamental and harmonic RMS voltages up to rank 50 considering the values provided in (61000_3_6) table 2.
- b) Linear sum of fundamental and the 5 harmonics with the highest magnitudes + RSS sum of the other ranks up to rank 50 considering the values provided in (61000_3_6) table 2 + 25 % margin as required in the specification.
- c) Linear sum of fundamental and the 5 harmonics with the highest magnitudes + RSS sum of the other ranks up to rank 50 considering the values provided in (61000_3_6) table 2.
- d) Linear sum of 25 first harmonics + RSS of the others ranks up to rank 50 considering the values provided in (61000_3_6) table 2.

- **Reactor**

- [\[CBOD-REAC-100078\]](#): Contractor shall provide list of type and routine tests to be performed according to relevant IEC standards, with detail of standard applied for each use case. It should be noted that the new standard to be issued will simplify the use case requirement as the use cases will be clarified.
- [\[B22aiii-REAC-440018\]](#): Short-circuit test shall be performed on each series reactor type.
- [\[B22aiii-REAC-440032\]](#): Contractor shall provide study showing the iso-flux lines. For each reactor, a magnetic field plot showing at least the iso-flux lines 126 μ T, and for public and worker exposure for all the normal operating conditions (including all the current harmonics); the iso flux lines 1260 μ T shall be provided for the rated short-circuit current.
- [\[B22aiii-REAC-440017\]](#): The use of Mylar (or Polyethylene terephthalate) as an insulation material for reactors is not permitted.

- **Capacitor**

- [\[CBOD-RLC-100259\]](#) Contractor shall provide list of type and routine tests to be performed according to Employer's requirements (section 4.1.4 and 4.1.5 of (B22a_iii)). For endurance test, overvoltage and ageing (IEC60871_2_1999), test detail to be agreed with The Employer.
- [\[B22aiii-RLC-440180\]](#): The protection of AC and DC capacitors against internal and external faults including protection for people and goods shall be provided by Contractor. A protection system shall be supplied to protect against internal and external fault. For DC Capacitors, if a protection relay is insufficient or not feasible to safeguard individuals and nearby equipment, additional requirements shall be considered.

- [\[B22aiv-FUNC-450037\]](#): VSM capacitors shall be designed and tested in line with (IEC_61071). The capacitor units shall be of internal fuse or self-healing design. Dry type capacitors shall be used and no gas filled capacitors shall be used. A capacitor failure detection system shall be provided

8.10.9 HV and MV cable inside converter station

[\[CBOD-CABL-100079\]](#)

The Employer's requirements for HV or MV cables inside Converter Stations (if any) detailed in (B22e) shall be applied.

As part of Employer's requirements, the following requirements shall be applied as minimum.

- [\[CBOD-CABL-100080\]](#): The short-circuit constraints shall be computed by the Contractor based on its own studies. The cables shall withstand such short circuit constraint.
 - Contractor is responsible to provide report for validation.
- [\[CBOD-CABL-100081\]](#): For medium voltage cables $U_m \leq 132$ kV
 - The electrical part of the extension prequalification tests as per (IEC60840) section 13.3.2.3 shall be carried out on a cable system which validates all ranges of approval stated in section 3 of (IEC60840).
 - Contractor is responsible to prove certificate for validation.
- [\[CBOD-CABL-100082\]](#): For high voltage cable $U_m > 225$ kV, if any
 - The prequalification test as per (IEC62067) section 12 shall be carried out on a cable system which validates all range of approval stated in section 3 of (IEC62067).
 - Contractor is responsible to prove certificate for validation.
- [\[CBOD-CABL-100083\]](#): For cable terminations
 - Contractor is responsible to provide type test certificate on termination to be provided.
 - Otherwise, type test shall be performed during the project.
- [\[CBOD-CABL-100084\]](#): For complete cable system
 - Type test sequence on complete cable system loop including accessories shall be provided and demonstrated by Contractor.
 - Type test sequence on selected cable sample shall also be demonstrated by Contractor.

8.10.10 Converters

[\[CBOD-CONV-100260\]](#)

The Employer's requirements related to Converters detailed in (B22a_iv) shall be applied.

As part of Employer's requirements, the following requirements shall be applied as minimum.

- **Design**
 - [\[B22aiv-TEST-450246\]-N3](#): The converter shall be designed using Modular Multilevel Converter (MMC) technology in compliance with the requirements described in (IEC62501).
 - [\[B22aiv-FUNC-450061\]-N3](#): The Contractor shall state during design phase the power transfer capability reduction (run-back) due to a decrease of cooling capability or due to an ambient outside air temperature exceeding the maximum specified temperature.
- **O&M**
 - [\[B22aiv-FUNC-450138\]-N3-M](#): The method to identify and procedures to replace the faulty components shall be described in dedicated maintenance documents.

- [\[B22aiv-FUNC-450139\]-N3-M](#): All maintenance activity shall be possible without needing to climb inside the Valve Structure. In case a ladder is required inside the Valve Structure, it must be included in the valve design.
- [\[B22aiv-FUNC-450141\]-N3-M](#): Both access and handling equipment (special tools) for each constitutive part of the VSM (capacitor, auxiliary circuit, power electronics or VSC Valve Level) replacement during maintenance shall be as such that maintenance staff never has to lift or carry heavy weight at arm's length. The type of access equipment shall be rated to lift to people and all tools and test equipment necessary for the most onerous maintenance task plus an additional weight of 250 kg.
- [B22aiv-FUNC-450145\]-N3-M](#) to [\[B22aiv-FUNC-450150\]](#): During maintenance or outage, there shall be an external dedicated mechanism to earth the valves and this mechanism shall be compliant with Employer's requirements. The grounding mechanism shall:
 - be designed in accordance with (IEC62271_102)
 - guarantee the continuity to the ground on all connection points. At least one terminal of each single capacitor of the VSC valves and the entire Valve Structure shall be earthed by the grounding mechanism.
 - be activated remotely (outside of the valve hall) after a discharge time ensuring the complete discharge of the VSM d.c. capacitors. A manual activation shall be possible as a backup.
 - be lockable by means of a manual lock. The lock key will be part of the station interlock system.
 - Be designed such that the continuity to earth can always be observed visually by maintenance staff.

• Monitoring

- [\[B22aiv-FUNC-450241\]-N3](#): The Contractor shall propose an innovative video system (or mobile robotic) solution allowing to optimize O&M inspections and interventions for both the offshore and onshore converter station. These solutions shall take the form of either fixed cameras (CCTV + IR + sound) and/or a mobile robotic system equipped with cameras (CCTV + IR + sound). In particular, any room which is not accessible while the substation is in operation (e.g. HVDC converter hall ...) shall be equipped with this video system solution. The video system solution shall cover at least 90% of the converter hall equipment.
- [\[B22aiv-FUNC-450242\]-N3](#): The cameras / mobile robotic system shall also allow for visual inspections after a fault has occurred (corrective maintenance).
- [\[B22aiv-FUNC-450243\]-N3](#): The video feed (CCTV + IR + sound) shall be accessible live by the remote operators which will be located onshore. The video feed (CCTV + IR + sound) of each inspection (may they be automated or not) shall be recorded and stored on a dedicated Network Attached Storage.

• Testing

- [\[B22aiv-FUNC-450246\]](#): The design of the converter valve shall be subjected to all the type tests specified in (IEC62501).
- [\[B22aiv-FUNC-450249\]](#): The Contractor shall perform Production tests as described in Clause 14 of (IEC62501). the Contractor shall submit, for approval by the Employer, a detailed description of the test procedures proposed to meet the production test objectives
- [\[B22aiv-FUNC-450166\]](#): Test certificates and technical reports with a very detailed description of the test object shall be provided to the Employer for acceptance.
- [B22aiv-TEST-450204\]-N3](#): The Contractor shall propose a clear and well-defined method for assessing complete valve losses based on measured losses on a test object such as defined in (IEC62501). The Contractor shall perform such measurement as part of special test and applying to the fullest extend the (CIGRE_TB844). The objective is to validate the Contractor's losses assessment methodology as per (IEC62751).

- [\[B22aiv-TEST-450250\]-N2](#): The Contractor shall perform a Routine maximum continuous operating duty test on all VSMS, similar to the type test of subclause 6.4 in Standard (IEC62501), but for a reduced time period of 5 minutes ("burn in").

8.10.11 Replica

[\[CBOD-REPL-100085\]](#)

The Employer's requirements related to Replica detailed in (B22d) shall be applied.

[\[INTR-1\]](#)

With the objective of performing dynamic studies and testing the Control and Protection (C&P) system of the HVDC link, the Contractor shall supply the Employer with a replica of the C&P system, which is a copy of the one to be installed on site (including redundancy).

[\[B21ai-MISC-110588\]](#)

As the Converter Stations shall be technically identical, the Contractor shall provide only one replica (Onshore Converter Station and Offshore Substation) common for all HVDC Links. It shall be possible to change the configuration to represent the right HVDC Link in less than 2 hours.

As part of Employer's requirements, the following requirements shall be applied as minimum.

[\[DESC-3\]](#) The provided replica shall incorporate the same components integrated in the on-site C&P system, including but not limited to:

- Control and protection cubicles that are identical to the original cubicles at the on-site converter stations with the same interfaces, software and configurations
- Redundant equipment implemented in the on-site converter cubicles.
- Equipment related to telecommunication as well as relevant interfaces and configurations

Certain simplifications can be proposed by the Contractor, which shall be subject to the Employer's approval prior to implementation. For example, the valve control cubicles might be simplified if and only if complete and accurate representation of the functional and dynamic behavior of the system is not compromised. The method used must be agreed with the Employer.

- [\[DESC-6\]](#) The supplied replica shall ensure connectability to C&P replicas from wind farm systems. This indicates that the replica shall include all communication interfaces with the offshore system (e.g., circuit breaker status, trip signals, interface with the power park controllers, etc.) that are relevant for the operation of the Control and Protection system.
- [\[TEST-1\]](#) Replica shall be commissioned in Employer's lab at least 4 months before first energisation on site.

8.10.12 Dynamic Breaking System (DBS)

[\[CBOD-DBS-100027\]](#)

In case of onshore load rejection, the DC side of Onshore Converter Station shall be equipped with two Dynamic Braking Systems (DBS) – one for each Converter Pole.

[\[B21ai-STAB-110592\]](#)

Each Dynamic Braking System (DBS) of each pole shall be rated to 1pu at Guaranteed Active Power Pole Capacity during 2.4 seconds.

Further requirements on the DBS detailed in (B22a_iv) shall be applied.

8.11 General Control & Protection System

[\[CBOD-CPSY-100154\]](#)

A central control and protection system (C&P) shall be designed to perform equipment sequencing, monitoring, control, protection, status display, sequence of event recording and data reporting. The C&P system should be as far as practicable standardized and aligned with the Employer's existing control and protection systems on previous HVDC projects. The control system shall be designed for unmanned operation, i.e. the Onshore and Offshore Stations are usually not manned and are only controlled, monitored and trouble-shot via the remote control and remote access.

[\[CBOD-CPSY-100155\]](#)

At the end of the Defect Notification Period, no equipment, software or operating system shall have become obsolete. All hardware components shall be approved by the Employer.

[\[B21ai-COBA-110075\]](#)

Employer requirements related to C&P detailed in following documents: (B22a_v), (B22av_2), (B22av_3), (B22av_4), (B22av6) and (ERS_5_4).shall apply.

8.11.1 General Requirements

8.11.1.1 Redundancy

[\[CBOD-CPSY-100156\]](#)

The single-fault principle (n-1 principle) shall apply for all control and protection systems. A fault in one system shall not lead to disturbance or failure of another system. Errors and a loss of redundancy shall be automatically detected and shall lead to an uninterruptible redundancy switchover without interruption or reduction of the power transmission. Redundancy shall be provided for all components necessary.

[\[CBOD-CPSY-100185\]](#)

It shall be possible to test and maintain the redundant control system in stand-by mode without disturbing normal operation. After installation of a component, the replaced overall system shall automatically return to the set operation point.

8.11.1.2 Control cabinet requirements

[\[CBOD-CPSY-100157\]](#)

In the Onshore Converter Station, all control, protection and auxiliary power cabinets shall be placed on a raised floor. Control cabinets located in the Onshore Converter Station and Offshore Substation shall be equipped with rodent protection. Control and protection cabinets shall be earthed according to section 2.4.2 of (B22a_v). The cabinet shall be coordinated with the Employer and approved by the Employer. Cables entering the building shall be installed with airtight penetration systems (Multi-Cable Transit system), which shall be EMC compliant.

[\[CBOD-CPSY-100158\]](#)

Wiring of control cabinets shall be done according to Employer requirements expressed in section 2.4 of (B22a_v), in particular dimensions of cubicles, spare place, IP index, wiring of VT and CT secondary circuits (equipped with specific models for test plugs, short-circuit device and fuse holder).

If possible, maintenance-free hardware components shall be used. The usage of moving and rotating parts (e.g. fans, etc.) shall be avoided.

[\[CBOD-CPSY-100159\]](#)

The requirements of electromagnetic compatibility and mechanical durability shall be implemented in the design, and the respective product and technical standards shall be applied.

8.11.2 Control Requirements

[\[CBOD-CPSY-100160\]](#)

A control system shall be developed for the HVDC Link, which is centralized to the highest degree in terms of operation and monitoring. The number of subsystems with their own control system shall be minimized for auxiliary systems.

[\[CBOD-CPSY-100240\]](#)

The control architecture of the HVDC Link is shown in Figure 22 and shall be used as a design basis. The CPU load and the memory requirement for applications shall not exceed 70 %. All control and protection equipment shall be self-monitoring ("device watchdog") and alarmed.

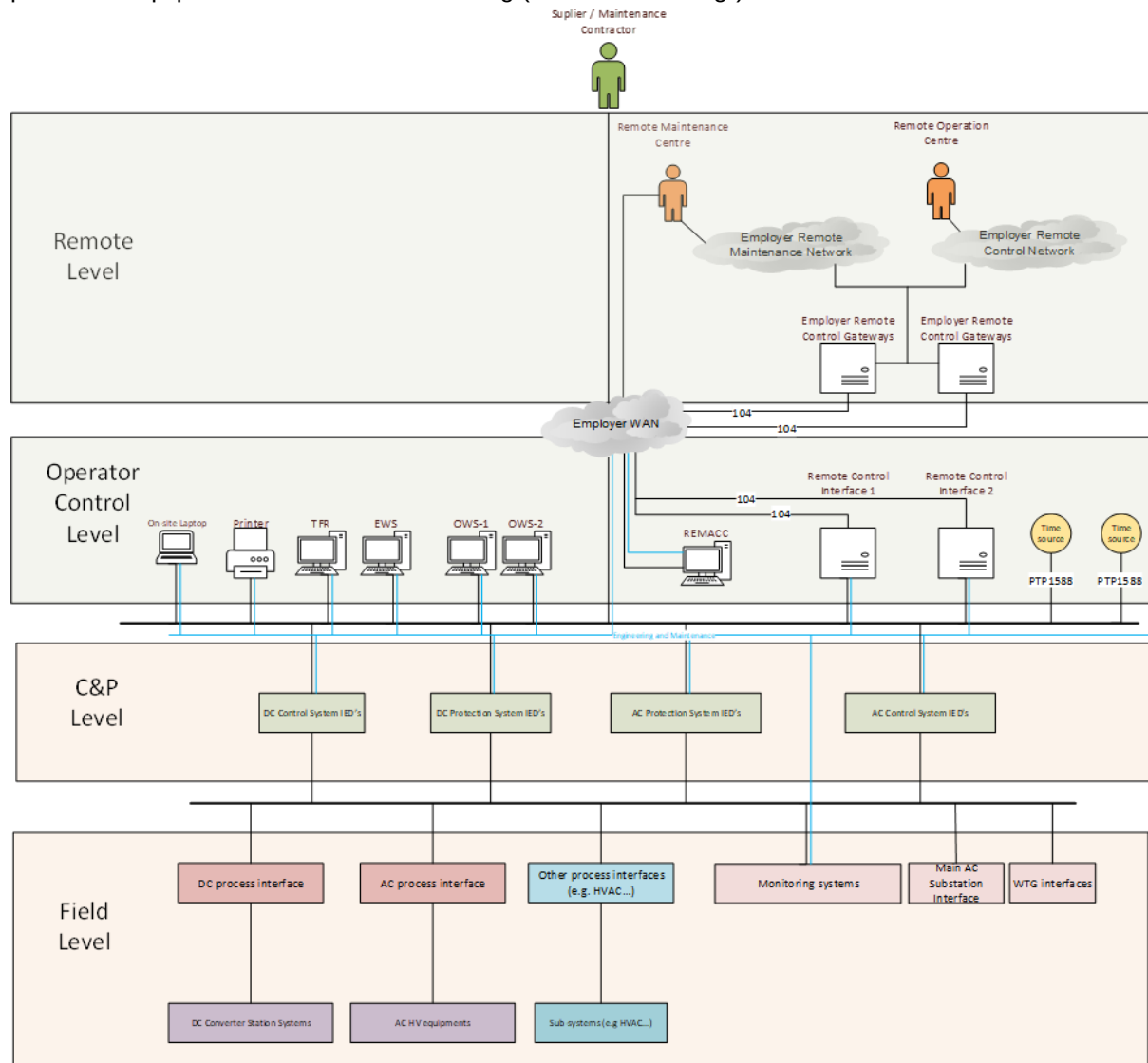


Figure 22 – Control architecture of the HVDC Link

The following requirements apply for the stated control level and corresponding control locations.

8.11.2.1 Bay Control Level (C&P Level)

[\[CBOD-CPSY-100241\]](#)

The bay controllers shall be designed with a sufficient reserve of at least 20 % with respect to the I/O level for each card module.

[\[CBOD-CPSY-100242\]](#)

The control and signal voltages connected to the bay controllers shall be monitored by the bay controller. If a control voltage is interrupted, a corresponding alarm shall be generated and any commands shall be negatively acknowledged.

8.11.2.2 Control Concept

Control Locations

[\[CBOD-CPSY-100243\]](#)

The following control locations shall be implemented:

- Local control of the bay (C&P Level) or directly at the switchgear or at the control units of auxiliary systems (Field Level),
- Station HMI (Operator Control Level),
- Remote Control Centers (RCC) (Remote Level)

[\[CBOD-CPSY-100244\]](#)

All control locations shall be subordinated to a control location concept with control authority with requests and releases.

Local Control (C&P level)

[\[CBOD-CPSY-100245\]](#)

Components and devices shall be able to be manually controlled directly at the boundary to the bay level. Local control shall not be integrated into the concept of control authorities. The following options shall be provided for the local control of primary switchgear:

- Local controls, un-interlocked, directly at the switchgear (emergency operation level of the local control),
- Local control via bay controller, operating via the bay controller's display or via a switch.

[\[CBOD-CPSY-100246\]](#)

The activation of the local control shall be reported to the Station HMI and to RCC.

[\[CBOD-CPSY-100247\]](#)

The local control of individual components shall limit the scope of operation in the associated control authority, but still allow the release and transfer of control authorities. The activation of the local control shall block the control of superordinate control locations and shall not be able to be revoked by them.

Station HMI (Operator control level)

[\[CBOD-CPSY-100248\]](#)

The Station HMI shall be the overall control location within the control rooms and shall include the complete range of functions for operating and monitoring the HVDC Link. The Station HMI shall be in French language and shall be able to be switched to English language.

[\[CBOD-CPSY-100249\]](#)

Operational functions are performed from OWS (Operator Work Station) and maintenance, parameters and configuration functions are performed from EWS (Engineering Work Station).

[\[CBOD-CPSY-100250\]](#)

The operation and visualization of the Station HMI shall be based as far as practicable on the systems already in operation at the Employer and be approved by the Employer. The Contractor shall comply with HMI interface requirements described in section 4 of (B22a_v).

[\[CBOD-CPSY-100251\]](#)

Two separate client systems shall be provided for each control room in the Onshore Converter Station and Offshore Substation and shall be hierarchically on the same level of control authorities. "Select-before-operate with enhanced security" as described (IEC61850) shall be implemented. The redundant Station HMI servers shall be virtualized. The local control system shall be designed with high performance for delay-free operability. Different user groups with different rights shall be implemented.

[\[CBOD-CPSY-100252\]](#)

As a minimum, the Onshore Converter Station control room shall be equipped with a dedicated OWS providing the same operational functions as the OWS on the Offshore Substation (including the possibility to place commands).

Remote Control Centers

[\[CBOD-CPSY-100253\]](#)

The remote control shall allow the control of all systems from Remote Control Centers (RCC). The RCC shall be able to operate all measures and activities, which take place within the HVDC Link, and shall receive the relevant information and control options. The system operation shall comply with the basic principles of electrical grid management in the transmission grid of the Employer. Further requirements are defined in section 3.3.1.

[\[CBOD-CPSY-100254\]](#)

Communication with RCC shall be performed through RCI (Remote Control Interface).

[\[CBOD-CPSY-100255\]](#)

All events/alarms, measurements, commands and setpoints available within the C&P system shall be able to be transmitted to / received from the RCC. List of signals will be defined during the design phase.

Control Authority

[\[B22av-CSDP-670137\]](#) N3

An adequate hierarchy level-based system shall be provided to avoid simultaneous commands coming from different operation locations or workstations. At any time, only one operation location shall be able to send commands to the substation elements.

In order to achieve such control authority, Employer requires that the C&P system of each site include two modes:

- Site-operating mode: either set in local or remote
- Functional bay mode: either set in activated or deactivated

[\[CBOD-CPSY-100256\]](#)

Functional bays will be defined during the design stage based on Contractor's proposed functional and hardware architectures. It could consist, for example, of the following: "HVDC terminal bay", "GIS monitoring bays", "IAC feeder bays", "bus bar coupler bay"...

Depending on their status these two modes have effects on:

- Whether or not individual events or alarms are transmitted to RCC,
- Whether or not individual measurements are transmitted to RCC,
- Whether or not individual commands can be executed from local control (OWS),
- Whether or not individual commands can be executed from remote control (RCC),
- Whether or not alarms acknowledgment is required from OWS.

[\[CBOD-CPSY-100257\]](#)

Requirements detailed in section 3.1 of (B22a_v) shall apply.

8.11.2.3 Requirements for the Control System

Time Synchronization

[\[CBOD-CPSY-100161\]](#)

All components of the entire HVDC Link shall be redundantly supplied with the same, exact and synchronous time and date information at all times. The time stamping of the events, alarms, measurements and commands/set-points shall be synchronized at the source with the central time

servers. Redundant clock source based on GNSS shall be provided, according to Employer's requirements expressed in section 2.5 of (B22a_v).

[\[CBOD-CPSY-100162\]](#)

Furthermore, all control and protection equipment, which stores fault records and protocols in the internal memory, shall be equipped with a real-time clock that continues to operate even during failures of the external time synchronization. The operability of the system shall not be restricted by a failure of the external time synchronization. The internal time deviation over 24 h shall not exceed 1 ms for all units. In the event that the absolute time signal is available again, a transition from the internal time synchronization to the absolute time signal shall take place. Requirements regarding time synchronization are further detailed in section 2.5 of (B22a_v).

Mirror Port

[\[CBOD-CPSY-100163\]](#)

For possible testing or monitoring purposes, every equipment of network (switches, routers ...) shall support the port mirroring feature.

Signals

[\[CBOD-CPSY-100164\]](#)

The Contractor shall develop a suitable alarm system concept based on the Employer requirements expressed in section 4.4 of (B22a_v) that enables the respective operating personnel and the RCC to assess the respective situation easily and quickly. All events of the HVDC Link shall be logged in one central event signal list for the Onshore Converter Station and Offshore Substation. The event signal list and the signals of the HVDC Link should be similar to the systems currently in use by the Employer, both in terms of presentation and informational content. Signals shall be categorized as events, alarms, measurements, commands and set-points. Alarms shall be categorized into three distinct severity levels according to Employer's requirement [\[B22c-MAIN-910005\]](#) of (B22c). Additionally, an analyze function for the HVDC system shall be implemented to indicate which cause initially lead to the first trip of the system.

[\[CBOD-CPSY-100165\]](#)

The Employer specifies a standard template for the creation of the signal exchange list, in which the individual signals of the event signal list shall be integrated. Signals to the RCC shall be condensed via group messages. It shall be possible to have several group messages per system / zones. The group messages shall be sent again in case further single messages of the group are triggered.

[\[CBOD-CPSY-100166\]](#)

Alarm signals shall be classified as signals requiring acknowledgement and events shall not require acknowledgement. All signals requiring acknowledgement shall be acknowledgeable from the Station HMI only, in accordance with the control authority. When a site-operating mode is set to remote control, alarm signals shall automatically acknowledge on Station HMI (except for alarm signals associated with a functional bay set in intervention mode).

[\[CBOD-CPSY-100167\]](#)

All signals shall be time-stamped at their source with an update time of 1 ms and then transmitted to the control system. Signals shall be displayed at least 2 s after generation in the signal list. Invalid measurements and signals shall not be processed and shall be reported. The quality attribute "Invalid" (IV) shall be set when a flutter message is detected.

Interlocks and Other Protective Mechanisms

[\[CBOD-CPSY-100168\]](#)

To avoid switching operations and system controls and functions that can cause damage to the environment, equipment or danger to life and health, an interlocking system shall be set up for the entire system. Interlocks shall be set up in real time. Hardware interlocks shall be replicated in the control/automation system and displayed in the Station HMI.

[\[CBOD-CPSY-100169\]](#)

It shall only be possible to select functions and commands if the interlocking conditions are fulfilled. Interlocking conditions shall be displayed on the Station HMI. More generally, any pre-conditions preventing the execution of a command shall be displayed on the Station HMI.

[\[CBOD-CPSY-100170\]](#)

Mutual interlocking between the AC substation of the Employer and the HVDC system shall be taken into account, implemented and agreed with the Employer.

[\[CBOD-CPSY-100171\]](#)

The HVDC system shall be equipped with a door-interlocking system based on keys for preventing that anybody comes into the valve hall (and any other "not walk-through" areas) during energized conditions.

[\[CBOD-CPSY-100172\]](#)

Requirements regarding interlocking further detailed in sections 3.5 and 3.6 of (B22a_v) shall apply.

[\[B21ai-COBA-110533\]](#)

In addition, to avoid overloading equipment, an interlock shall prevent and warn the Operator from the following actions:

- Disconnect a converter transformer if too many OWF feeders are connected,
- Connect too many OWF feeders in a given time to be defined by Contractor that would create overvoltages,
- Any other action that would hinder the correct operation of the HVDC Link.

Transient Fault Recorder (TFR) and Fault Locator (FL)[\[CBOD-CPSY-100173\]](#)

Requirements regarding TFR detailed in section 7.10 of (B22a_v) shall apply.

[\[CBOD-CPSY-100174\]](#)

A transient fault and event recorder shall be provided in the Onshore Converter Station and Offshore Substations for all necessary systems and it shall record across all stations. It shall also monitor the power quality at the respective PCC. Faults and events are understood to be onshore and offshore grids and converters disturbances, disturbances of the auxiliary systems. The event recording shall be carried out as a combined solution of fault recorder and power quality monitoring. The automatic remote transmission of records to the Employer shall be coordinated during the engineering phase. TFR data from one Converter Station shall be available to the other converter station of the same HVDC Link).

[\[B22av-PROT-670660a\]](#) N3

The TFR record format shall be COMTRADE and shall follow the (IEC60255_24).

[\[CBOD-CPSY-100175\]](#)

The fault recorders shall function as multi-station fault recorders and exchange trigger signals. The parameterization of the trigger groups and the signals shall be coordinated with the Employer in the engineering phase. The fault recorder shall be equipped with a redundant power supply.

[\[CBOD-CPSY-100176\]](#)

As a minimum, all signals of the measuring system and all tripping signals of the protection functions of each protection zone shall be recorded. Wherever possible, digital interfaces between the individual systems and the fault recorder shall be utilized. A reserve of 20 % shall be planned for the input channels and digital interfaces.

[\[CBOD-CPSY-100177\]](#)

In addition to the TFR, a Fault Locator (FL) system shall be provided in order to estimate the distance of a defect affecting the HVDC cables. Requirements regarding FL are detailed in section 7.11 of (B22a_v).

General Site Function (TG – “Tranche Générale”) – Onshore Converter Station only

[\[CBOD-CPSY-100178\]](#)

Onshore site functions necessary for Employer personnel when working on site (“Tranche Générale”) shall be implemented and are described in Employer requirements expressed in section 8.2 of (B22a_v). These functions include for example: worker presence indication on site, danger alarm, technical alarms, management of phone ringing, doorbell, horns, outdoor lighting, doors positions, site mode, interface with Employer access control, video surveillance and perimeter protection systems.

Low Voltage and Telecommunication cables

[\[CBOD-CPSY-100179\]](#)

Requirements regarding Onshore Low Voltage and Telecommunication cables are detailed in section 2.4.12 of (B22a_v).

[\[CBOD-CPSY-100180\]](#)

Requirements regarding Offshore Low Voltage and Telecommunication cables are detailed in (ERS_5_5_LV_cable).

Remote Access

[\[CBOD-CPSY-100182\]](#)

Requirements regarding remote access detailed in section 6 of (B22a_v) shall apply.

[\[CBOD-CPSY-100181\]](#)

Remote access is different from Remote Control (see section 8.11.2.2) and is used to provide different services to the Employer's or O&M Contractor maintenance centers (or remote locations) such as: file transfer, remote configuration or other maintenance actions, HMI viewer, DNS server and terminal server...

[\[CBOD-CPSY-100183\]](#)

It shall allow a remote use of the C&P system HMI. However, operating functions are prohibited from the remote access (operating functions are either performed locally on OWS or remotely from RCC), except for some specific commands to be discussed during engineering stage.

[\[CBOD-CPSY-100184\]](#)

Remote access functions shall not require installing on Employer or O&M Contractor workstations specific software. If a webserver is accessed, it shall be compatible with HTML5/Chromium based browsers.

SNMP

[\[B22av-MAIN-650232\]](#)

A SNMP Agent shall be implemented and shall enable the transmission of important information on the status and supervision of the different equipment of the C&P system to Employer remote maintenance centers. It shall not be restricted to IT and network equipment.

Export of Alarms and Events

[\[B22av-MAIN-650327\]](#)

The Contractor shall make available to the Employer remote maintenance centers (through remote access), automatic and manual exports of the C&P system alarms and events logs. Requirements (names, format, automatic export frequency...) associated with these export files will be detailed by Employer during engineering stage.

8.11.3 Data Network Infrastructure and Software

[\[CBOD-CPSY-100187\]](#)

For the Offshore Substation, control and protection equipment and system related to the operation and maintenance of the electrical process (converter, cooling system, converter transformers, GIS...) are part of "PCM SCADA". All other equipment and system (auxiliary power supply, fire protection system, HVAC...) are referred as "Utility SCADA" and specified in (ERS_5_2).

[\[CBOD-CPSY-100188\]](#)

The main language of all software and related documentation shall be French; if technically not possible, English shall be used. This applies to computer systems as well as to operating units of components, e.g. of bay control units and protection devices.

[\[CBOD-CPSY-100189\]](#)

If the Contractor is not the manufacturer of the software, this shall be indicated by the Contractor at an early stage. The Contractor shall be responsible for adapting the software to the latest status until commissioning at their own expense. If there is no alternative for a more up-to-date compatible software until the time of commissioning, this shall be indicated by the Contractor at an early stage.

8.11.4 Protection Requirements

[\[CBOD-CPSY-100190\]](#)

Requirements regarding protection systems are detailed in section 7 of (B22a_v).

[\[CBOD-CPSY-100191\]](#)

The Contractor shall provide all protection systems necessary to ensure complete protection of all equipment within the scope of the HVDC Link.

[\[CBOD-CPSY-100192\]](#)

The protection system shall have the following characteristics: redundancy, diversity, speed, selectivity, no under- or over-functioning.

[\[CBOD-CPSY-100193\]](#)

Faults and faulty components within the HVDC Link shall be recognized and cleared so that all components are protected, the stability of the Employer's transmission grid, the offshore AC systems and the HVDC Link is ensured at all times and for every grid status. All relevant faults for the protection design study shall be approved by the Employer.

[\[B22av-PROT-670608\] N3](#)

The design of the protection systems shall ensure that the whole HVDC Link operates safely even in case of telecommunication loss between converter stations (onshore/offshore).

[\[CBOD-CPSY-100194\]](#)

The protection concept and protection study shall be developed by the Contractor and shall be approved by the Employer. It shall be possible to test the protection system regularly. The Contractor shall develop a test concept for the protection system and hand it over to the Employer.

[\[CBOD-CPSY-100195\]](#)

It shall be possible to perform maintenance or change the parameters on one set of redundant protections while the other sets remain in service, and the transmission power shall not be disturbed. If filters are used, they shall be protected in a separate protection zone and shall be able to be switched off selectively.

[\[CBOD-CPSY-100196\]](#)

Furthermore, the following requirements shall be implemented:

- Separate DC supply: All protection devices shall have two power supplies,

- Separate cabinets, current transformer cores, voltage transformer circuits, cables and control circuits, trip coils and trip circuits: External tripping (e.g. from converter protection) always affects protection systems 1 and 2 and therefore both trip circuits,
- Trip circuit supervision,
- Self-monitoring system.

[\[B22av-PROT-670613\]](#)

There shall be strict separation of control & protection tasks, in particular the protection functions shall use separate processors, IEDs and bus connections. I/O boards can be shared between control and protection functions.

8.11.4.1 Fault clearing time request

[\[C11-ACGR-010083\]](#)

Any fault between PCC-AC Offshore and PCC-AC Onshore shall be eliminated in less than 75 ms including breaking time of the circuit breaker of 40 ms. The Breaker Failure Protection shall be included both in the Onshore and Offshore Converter Station C&P system. The total fault clearance with such protection shall be below 200 ms and this time shall be taken into account in the design. For the onshore breaker failure protection only one pole failure is considered, thus only one phase is temporized.

For the Onshore Converter Station, Converter System Contractor shall take into account that the breaker failure protection of the main AC circuit breaker in Onshore Converter Station if any shall be coordinated with Employer AC system such as Breaker Failure Protection and inter-trip tele-protection.

8.11.4.2 Protection of AC equipment

[\[C11-ACGR-010084\]](#)

The Converter Stations protection system shall be designed to cover all AC equipment including those between converter transformers and the PCC-AC for both converter stations.

8.11.4.3 Protection of Inter Array Cables

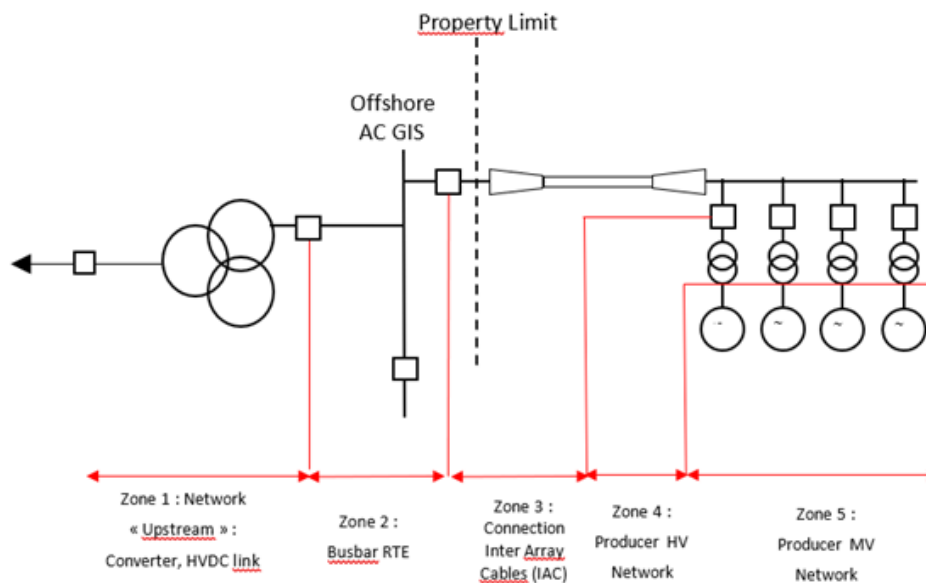
[\[C11-ACGR-010163\]](#)

Converter System Contractor shall supply all the necessary equipment to protect the Inter Area Cables of the Producer (Zone 3, below diagram), including instrument transformers and protection devices. This includes the protections but also the relevant surge arresters rating.

The protection system shall be capable of:

- Detecting any short-circuit on the Inter Array Cables, whether the OWF are injecting or consuming active power.
- Detecting faults in all configurations provided by Producer for the Inter Area Cables, including temporary configuration for commissioning or maintenance.

Protection system shall be operational even if the main AC network is unavailable. Protection plan shall be coordinated with Producer who should accept the protection system.
Automatic reclosing of a circuit breaker by a protection relay is not allowed.



8.11.4.4 Offshore GIS Protection Requirements

[CBOD-CPSY-100198]

Employer requires specific protection functions in section 7.7 of (B22a_v) regarding protection of some Offshore Substation zones (converter transformers, GIS, IAC feeders, auxiliary supply transformers...). In particular it has to be noted that for GIS, busbar differential protection shall be redundant.

8.11.4.5 Other Protection Requirements

[CBOD-CPSY-100199]

Employer requirements for specific protection functions (e.g. DC current transformer protection, DC overload cable protection, GIS monitoring, capacitive voltage transformers protection, sub synchronous oscillations protection...) are expressed in section 7.6, section 7.7 and section 7.8 of (B22a_v).

8.11.4.6 Protection setting and selectivity

[B21ai-STAB-110235]

The protection system under the Contractor's responsibility shall clear the faults:

- On the HVAC link, between the Employer Existing AC Substation's breaker and the Onshore Converter Station, in an appropriate time specified by the Employer after the fault apparition, including a breaker breaking time given by the Employer.
- From the converter transformer HV side to the PCC-DC Onshore and on the DC Cable, in an adequate time that shall be provided and explained by the Contractor to Employer.

[B21ai-STAB-110236]

In case of AC faults on any Employer's equipment situated outside the Onshore Converter Station zone (phase-to-earth, phase-to-phase, phase-to-phase-to-earth, three-phase, fault inception), the protection system of the Onshore Converter Station shall not send any trip order before Employer's protections (i.e. shall fulfil the UVRT and OVRT profiles as defined in sections 8.6.6.1.1 and 8.6.6.1.2).

Anyhow as backup solution (to ensure all AC faults are cleared), an undervoltage protection shall be implemented by the Contractor which shall trip the converter if the voltage is lower than a settable value longer than a settable time.

8.11.5 Testing Requirements (FST/FAT)

[CBOD-CPSY-100200]

Specification (B22av_2) provides Employer requirements for the Factory System Tests (FST) and the Factory Acceptance Tests (FAT) of the C&P system of the Offshore Station and Onshore Converter Station.

[B22av2-GENE-620004] N3

Objective of FST tests is the verification of the complete Control and Protection System (control, protection, measurements, monitoring and telecommunications) functionalities and performance (steady-state and dynamic) according to the specifications.

[CBOD-CPSY-100201]

They are performed by Contractor following completion of its internal developments and validations. FST tests shall be performed on final software version. Results shall demonstrate that the complete Control & Protection System has been successfully tested and comply with Employer specifications. Employer will attend some of the FST tests but will provide its comment mostly on FST test results which shall be provided periodically by the Contractor during FST testing.

Objective of FAT tests is to witness by the Employer re-run of selected parts of the FST tests.

[B22av2-ACHI-620041] N3

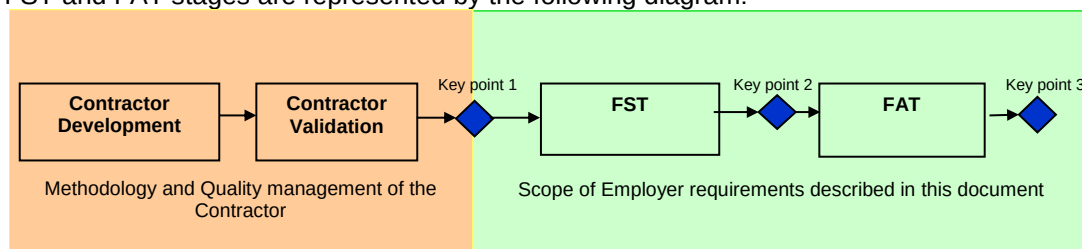
All FST / FAT tests shall normally be performed on the same software (including equipment firmware) and hardware versions.

[B22av2-STAG-620014] N3

During FST / FAT, if software or hardware modifications are deemed necessary, Contractor shall propose a list of non-regression tests (with associated procedures) to be performed in order to demonstrate that results of previous testing are still valid. Any modifications shall be properly documented as part of the change management process.

[B22av2-STAG-620017] N3

The FST and FAT stages are represented by the following diagram:



[CBOD-CPSY-100202]

All the different FST / FAT tests shall be listed in a Test Plan Outline proposed by the Contractor to be approved by the Employer during design stage/detailed engineering phase. Minimum tests are listed in requirement [B22av2-TEST-620012] of (B22av_2).

[CBOD-CPSY-100203]

Entry requirements of the FST stage (Key point 1) are listed in section 2.1 of (B22av_2).

Entry requirements of the FAT stage (Key point 2) are listed in section 2.3 of (B22av_2).

Conditions of acceptance of the FAT (Key point 3) are listed in section 2.6 of (B22av_2).

FAT acceptance from Employer is required before shipment of equipment to site (for Onshore Converter Station) and to yard (for Offshore Substation).

Test platform requirements are expressed in section 3 of (B22av_2).

[B22av2-STAG-620018/620027] N3

For the whole duration of the FST / FAT tests, Contractor shall provide means for Employer specialists to attend remotely to the tests.

8.12 Auxiliary systems

8.12.1 Auxiliary power systems

8.12.1.1 General requirements

[CBOD-AUXS-100086]

All redundant systems shall be designed with identical electrical and mechanical ratings, the amount of different components shall be minimized. Redundant systems and related cable routes shall be located in different spaces where technically feasible.

[CBOD-AUXS-100087]

The outage of a single Main Switch Board (MSB) or a single Auxiliary Supply Transformer (AST) or a single DC UPS or a single AC UPS or multiple outages of one of these systems shall not interrupt or reduce the transmission capacity of the HVDC Link and operation of auxiliary systems.

[CBOD-AUXS-100088]

The outage of all Auxiliary Supply Transformer (AST) and a single Diesel Generator (DG) (for the Offshore Converter Station only) shall not limit the operation of auxiliary systems, except for the cooling systems. The operation of the UPS systems and the DG (also including the DG CB) shall not depend on any other systems such as external cooling, external voltage supply, external controls.

[CBOD-AUXS-100089]

All auxiliary power systems and all connected consumers shall be able to operate without limitations within electrical tolerances depending on the primary systems (HVDC system and connected OWFs) and the DG. Major faults (e.g. short circuits) of any single system of DG, AST, UPS, MSB, sub-distribution (SD) boards, CB, energy supply points (ESP) or cabling between these systems shall not damage any further systems.

[CBOD-AUXS-100090]

All consumers with a system redundancy $> n-1$ shall be connected in a way, that during outage of a main switchboard a minimum of two redundant systems are available including all related auxiliaries (e.g. valves, flaps, controls).

[CBOD-AUXS-100091]

All auxiliary power systems (e.g. DG, UPS, MSB, SD) with internal control units and protection relays shall be time-synchronized from a central clock. Messages (alarms and events) generated by these units, shall be time-stamped accordingly. This time-stamp shall be indicated in the local operator panels (if available) and shall be transferred and indicated at the overall control system message system (Station HMI).

8.12.1.2 Onshore station auxiliary power system

[B21ai-AUXP-110064]

Contractor shall provide LV and MV switchgear, auxiliary transformers, diesel generators, cables and accessories, DC and UPS distributions (batteries, rectifiers, inverters, switchgears), control, protection, metering and all other equipment required for the AC auxiliary power system. All equipment shall be in accordance with applicable IEC and EN standards. Employer requirement in (B22a_vi) shall be applied.

The Contractor shall apply following standards: NFC 15-100 ($\approx$ equivalent to IEC 60364), NFC-13-100 and NFC 13-200.

Power supply changeover between sources philosophy and operation of auxiliaries is described in (B22a_vi).

[\[B21ai-AUXP-110065\]](#)

Preventive measures shall be taken to minimize induced and other disturbances in the Auxiliary Power System. All redundant equipment shall be located in rooms separated by fire barriers.

[\[B21ai-AUXP-110066\]](#)

The auxiliary power system shall be designed to accommodate:

- Variation in AC auxiliary system voltage within +10 % to -15 % of rated voltage
- Variation in AC grid voltage and frequency variation as specified in section 8.2.1
- Voltage imbalance as in (IEC61000_2_4)
- Harmonic distortion as in (IEC61000_2_4)
- Transition between auxiliary power sources
- Voltage Transients as specified in this document, section 8.6.6

The Onshore Converter Station shall have three independent auxiliary power supply sources in decreasing priority order:

- [\[B22avi1-DESC-680005\]](#) N3 **Source 1:** main auxiliary power supply source supplied from the Converter Station. The Contractor can propose its own preferred solution for Employer review and approval.

In case of a bipolar configuration, Source 1 shall consist in two independent sources for each Converter Pole (called Source 1a for Converter Pole #1 and Source 1b for Converter Pole #2). In this configuration, each Source 1a and Source 1b rated power shall be higher than the consumption of all the Converter Station's auxiliary services with a minimum 15% margin and shall also withstand low power factors.

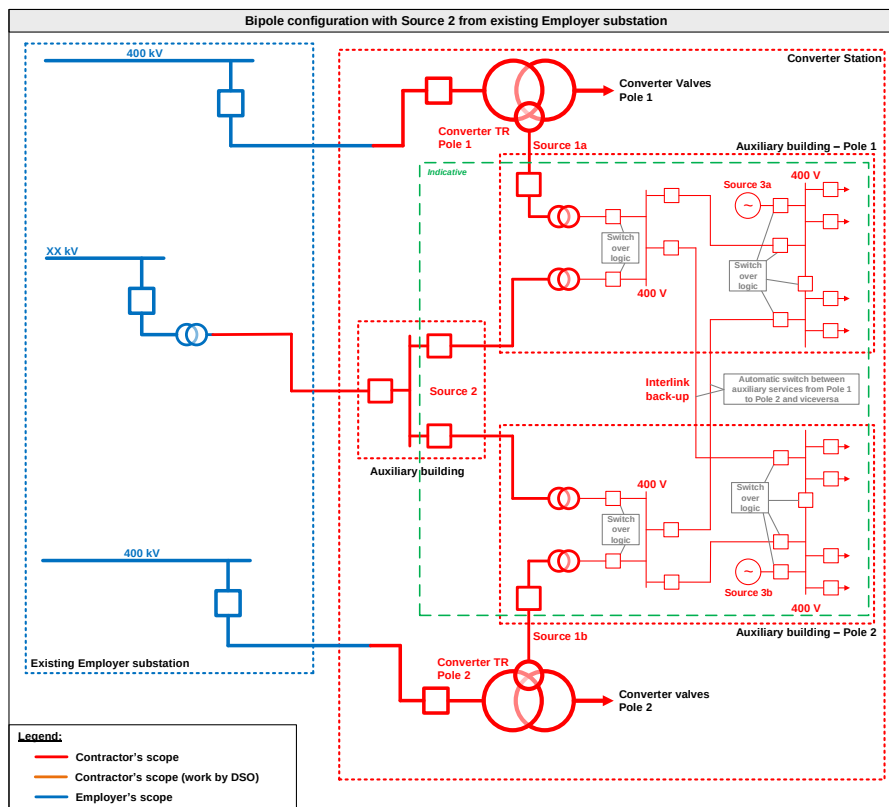
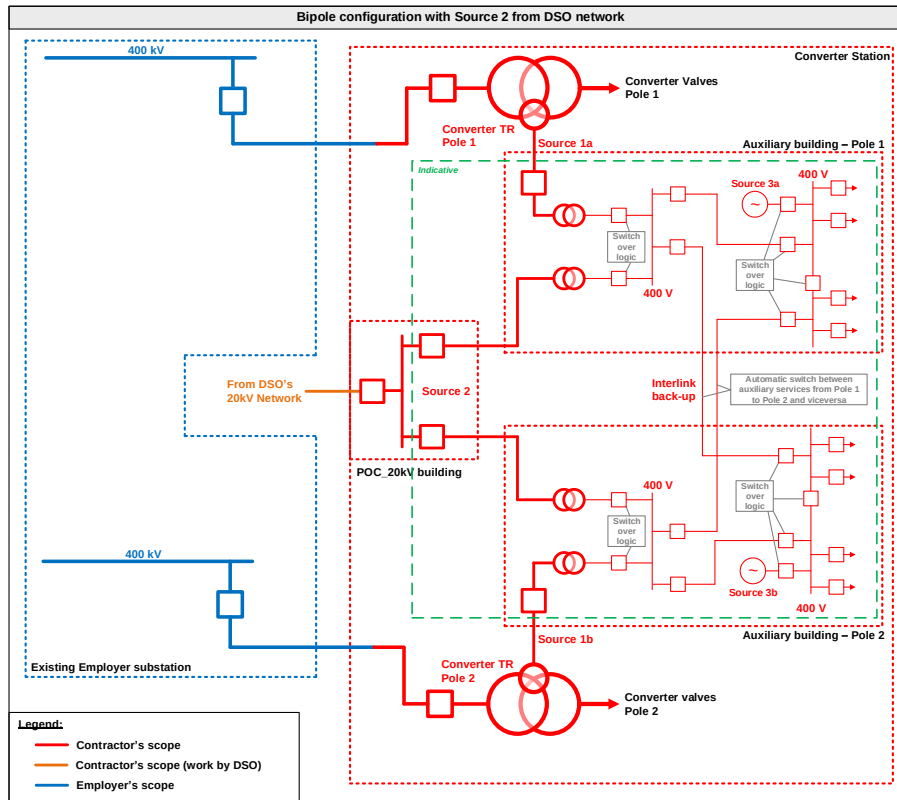
- [\[B22avi1-DESC-680007\]](#) N3: **Source 2:** back-up auxiliary power supply source supplied from an AC network different than Source 1. This can be a separate MV line running from the existing Employer AC Substation or a separate MV line running from the nearest DSO substation (choice will be made by the Employer in *(Call_off)* stage). The Source 2 power supply shall cover the consumption of the Converter Station's auxiliary services with a 15% margin.
- [\[B22avi1-DESC-680008\]](#) N3: **Source 3:** For emergency supply, diesel generator units shall be supplied by the Contractor. For a bipolar configuration, one diesel generator unit shall be supplied per Converter Pole. Its minimum required power shall cover all the essential consumption with a 15% margin (battery of the UPS system shall not be considered in the sizing), including black-start capabilities (if any).

[\[B22avi1-OPER-680023\]](#) N3

In any case, parallel coupling of power supplies Source 1, Source 2 and Source 3 is strictly prohibited.

[\[B22avi1-DIAG-680015\]](#) N3

The following figures are intended to illustrate the above principles regarding the different sources and the scope of work of the Contractor. Contractor can propose alternative architecture (see area shown as "indicative" on the figures below). For bipolar configuration, both Converter Poles shall have a separate low voltage distribution network.



[B22avi1-SPEC-680010] N3

Auxiliary supply shall be designed in a way that the Converter Station withstands a trip of the HVDC Link without interruption to the auxiliary supply. For a bipolar configuration, or when two symmetrical monopoles are present on the same site, auxiliary supply shall be designed in a way that the converter station withstands a simultaneous trip of both Converter Poles/links without interruption to the auxiliary power supply (this could occur in the AC grid, for example in case of busbar fault longer than "Fault Ride

Through" with both Converter Poles/links connected to the same busbar at the existing Employer substation).

[\[B22avi1-SPEC-680011\]](#) N3

In the event that diesel generators fail to start after trip of the HVDC Link, the Converter Station shall not be damaged.

Medium-Voltage Switchboards

[\[CBOD-AUXS-100092\]](#)

Requirements regarding Medium-Voltage auxiliary power components (including MV switchboards) are expressed in section 8 of (B22a_vi).

[\[B22avi1-MVAX-680251\]](#) N2

The MV switchboards shall be air insulated type (ambient air pressure). SF₆ or other gas-insulated switchgear is forbidden.

[\[CBOD-AUXS-100094\]](#)

The MSB busbar sections shall be designed spatially separated, the cable routes shall also be realized separately. Each busbar section shall be able to supply the entire Onshore Station at maximum power output. To ensure high availability, the essential loads (e.g. cooling system, technical building equipment) shall be connected to both busbar sections and, if necessary, provided with their own automatic changeover device.

Low-Voltage Switchboards

[\[CBOD-AUXS-100093\]](#)

Requirements regarding Low-Voltage auxiliary power components (including LV switchboards) are expressed in section 9 of (B22a_vi).

[\[B22avi1-OPER-680024\]](#) N3

Low voltage network shall use TN-S earthing system.

Auxiliary Supply Transformers (AST)

[\[CBOD-AUXS-100204\]](#)

The Employer's requirements related to onshore AST are mentioned in section 8.10.5.1. A motorized OLTC shall be installed on each AST. Each step shall be clearly identifiable. Remote, on-site and emergency hand crank control shall be possible.

[\[B22avi1-PHSH-680232\]](#) N3

When installing the Auxiliary Supply Transformers, the Contractor shall take into account the phase shifts of Employer's installed transformer or of DSO network if applicable.

Diesel Generators (DG)

[\[CBOD-AUXS-100095\]](#)

Requirements regarding diesel generators are expressed in section 9.4 of (B22a_vi).

For testing purposes, an additional load bank shall be provided, which shall be switched via a 400 V switchgear as part of the DG in the container.

The on-site breakdown time for the DG shall not exceed 4 h when replacing common wearing parts.

[\[B21ai-AUXP-110586\]](#)

The diesel generator shall be designed with sufficient fuel capacity to permit a minimum of 24 hours operation at the maximum essential supplies load (including cooling during de-energisation at maximum

temperature as defined in section 2.2, power for all control room's power supply, C&P system, etc of (B22a_vi).

LV cables

[\[CBOD-AUXS-100096\]](#)

Requirements regarding Onshore Low Voltage and Telecommunication cables are detailed in section 2.4.12 of (B22a_v).

Onshore Uninterruptable Power Supply

[\[CBOD-AUXS-100097\]](#)

For the Onshore Station, two identical UPS systems, each consisting of 125 V_{DC} UPS system with separate battery systems and 230 V, 50 Hz AC UPS system, shall be designed and installed.

Both UPS batteries' capacity shall, as a minimum, be designed for 4-hour operation for the Onshore Station, assuming that all connected systems (e.g. control units, communication systems, control & protection systems, alarm systems, emergency and escape lighting, plant surveillance systems etc.) are in operation. All CB (e.g. HV CB and MSB CB) shall be fully controllable during this time.

A minimum of two (redundant) 48 VDC converters shall be connected to the DC UPS or AC UPS in order to supply some of Employer equipment.

DC power supply (48V_{DC} and 125V_{DC}) shall use IT earthing system.

The AC and DC distribution shall be designed with 20 % power reserve. Low-maintenance closed lead-acid batteries of the cell shall be used.

8.12.1.3 Offshore station auxiliary power system

[\[B21ai-EMCO-110556\]](#)

The requirements for Auxiliary Supplies detailed in (ERS_5_6) shall be applied.

8.12.2 Auxiliary support systems (onshore & offshore)

8.12.2.1 HVA/C

[\[CBOD-HVAC-100098\]](#)

Employer's requirements stated in (ERS_6_3) and in section 3.3 of (B22a_iv) shall be applied.

8.12.2.2 Drain systems for Onshore Converter Station

[\[CBOD-DRAI-100099\]](#)

The aim of planning drain systems is to drain precipitation water from buildings and sealed surfaces and allow it to run to specific areas for percolation. Drainage shall therefore be planned and implemented by the Contractor as a separation system. Employer's requirements stated in section 5.1.1 of (B22b_i) shall be applied.

8.12.3 Fire protection (onshore & offshore)

[\[B22bi-UTIL-810036\]](#)

The Contractor shall design and install all the firefighting systems according to French regulations and applicable standards for Onshore Converter Station.

The Contractor shall design and install all the firefighting systems (fire detection system, fire protection and firefighting). Fire firefighting measures shall fulfil French regulations. Technical fire protection, which covers provision of fire water, hydrants, areas for firefighters, firefighting with water and inert gas, fire

detection, gas detection and fire containment shall be set up in accordance with applicable requested regulations and considering the state of the art.

[B22aiv-FUNC-450116]-N3

For valve hall, a fire detection system shall be designed and provided by the Contractor to alert of incipient risk of fire and trip the converter if required.

[B22aiv-FUNC-450115]-N3

For valve hall, full redundancy is required for the fire detection system, including the fire protection system PLC.

Fire protection panel with built-in redundancy (both power supply and the PCBs) is accepted.

[B22aiv-FUNC-450100]

For valve hall, the best materials against fire risk shall be selected for all the converter components (e.g. self-extinguishing materials, with a low emission of dark, dense and toxic smoke).

[CBOD-FIRE-100100]

Further Employer's requirements related to firefighting system stated in (B22b_i) for Onshore Converter Station, (ERS_6_3) for the Offshore Converter Platform and (B22a_iv) for the valve hall shall be applied.

8.13 IT and information security (Cyber security)

[CBOD-CYBE-100262]

The Security requirements specified in this document apply to all the components of the solution, including devices and software necessary to remote operation and maintenance. The Employer's requirement stated in (ERS_5_7) shall be applied.

[CBOD-CYBE-100263]

The Contractor's Information Security Standards shall be consistent with the (ISO_IEC_27001) standards and (IEC_62243) series of standards as appropriate. The solution proposed by the Contractor shall comply with common cybersecurity "good practices" during in the whole life cycle, including design, supply, installation, operation, maintenance, and decommissioning phases.

[CBOD-CYBE-100264]

The following requirements shall be applied as overall cybersecurity design principles for all systems and components in the Contractor's scope of delivery:

- Confidential information shall be encrypted for storage and transmission.
- Cloud storage allowed but, in a datacenter located within the European Union, with no cloud maintenance sub-contracted outside the EU.
- An identity and access management policy shall be implemented, in compliance with cyber standards (ISO 27001, ANSSI...)
- Cybersecurity's responsibility over networks that the Contractor considers to be necessary for the O&M Contractor's operation (ie: Internet access network on the OSS).

[CBOD-CYBE-100265]

As part of Employer's requirements stated in (ERS_5_7), the following requirements shall be applied as minimum.

- **ERS 5.7 – 03** The Contractor's Information Security Standards shall be consistent with the ISO 27001 standard and IEC 62443 series of standards as appropriate.
- **ERS 5.7 – 07** Any access to a system resource, either by a user or by an automatic process, shall be protected by an authentication process.
- **ERS 5.7 – 70** Systems with different sensitivities (PCM, onshore C&P, Utility, Internet), which may eventually be managed by different entities in the future, shall be on physically separate networks.

- **ERS 5.7 – 71** Each system shall be split into homogenous sub-systems having the same security requirements and functions. These sub-systems are separated at least logically from each other and from external systems.
- **ERS 5.7 – 68** All Control and Protection systems shall work properly without an Internet access.
- **ERS 5.7 – 25** The system shall ensure that all users of the solution have access only to systems and resources that are strictly necessary to carry out their tasks (principle of “least privileges”).
- **ERS 5.7 – 37** The system and the Contractor shall implement a security event log covering all hardware and software subsystems.
- **ERS 5.7 – 47** The system and the Contractor shall implement a virus and malware detection tools to prevent installation and propagation of malicious code on the solution.
- **ERS 5.7 – 53** The system and the O&M Contractor’s remote access to the system must pass through the Employer datacenter. The Contractor will install its access equipment in the Employer datacenter dedicated DMZ.
- **ERS 5.7 – 83** The Contractor shall maintain a continuous hardening process throughout the project phase and during applications entire life cycle.

8.14 Communication technology

Employer’s requirements stated in (ERS_5_4) shall apply.

8.14.1 Core concepts

The following are the customer’s core needs regarding the telecom networks.

Physical separation of networks

[ERS-5.4-960003]

The Employer requires the networks to enable his O&M strategy to be flexible. Therefore, to allow for the possibility that different parties may operate the Employer network, the onshore C&P network, the PCM network and the Utility network respectively, each network must be physically separate from the others. Therefore:

- The networks shall not have any physical element in common.
- The EC optical fibers connecting the onshore and offshore components of each network must be allocated for the sole use of that network.

Satellite backup for critical data services

[CBOD-COMM-100267]

The EPCI Contractor shall provide and install a satellite-based link to serve as backup for critical Utility data services.

The Employer will provide his own satellite link, separate from the one mentioned above, to serve as a backup for other critical data services; for this link, the EPCI Contractor shall only be in charge of its installation on the OSS.

Redundant design

[ERS-5.4-960004]

The networks shall be immune to the loss of any one single piece of equipment or any one single connection. Therefore, every network function must be implemented as a redundant pair of physical units. Each of the two physical units must be in a separate room.

8.14.2 Main telecom requirements

[ERS-5.4-960010]

The EPCI Contractor shall design, supply, install, configure, and test a backup satellite solution for transporting the following Utility services identified as critical for OSS operations in islanded condition:

- IEC104 RCI signals of Utility SCADA

- Helideck CCTV stream

[\[ERS-5.4-960011\]](#)

As soon as the OSS becomes islanded, the critical Utility services shall be automatically redirected to the backup satellite link.

[\[ERS-5.4-960034\]](#)

To enable different parties to operate the Employer network, the PCM network, the Utility network, and the onshore C&P network respectively, each of these networks shall be physically separate from the others (i.e., there shall be no physical network elements in common between any two of these networks). and shall use different fibres for connecting the onshore components of these networks to their OSS components. If it turns out in the design stage that hardwired exchanged signals are not sufficient to ensure communication between the Utility SCADA and the PCM SCADA, then an exception to this principle may be considered for approval by the Employer, using a gateway for data exchange and a firewall.

[\[ERS-5.4-960046\]](#)

The PCM network, Utility network and onshore C&P network shall be immune to the failure of any one piece of equipment or link. To achieve this, redundancy shall be factored into its design.

[\[ERS-5.4-960047\]](#)

The equipment of the PCM network, Utility network, onshore C&P network serving as redundancy backup shall be in a separate cabinet and in a separate technical room from the main equipment.

[\[ERS-5.4-960048\]](#)

The main and redundancy backup network equipment serving for the communication link between the OSS and the Onshore Converter Station shall connect to separate physical media having no common failure mode.

[\[ERS-5.4-960123\]](#)

The EPCI Contractor shall deliver a fully functional ready-to-use WLAN network no later than two weeks after the Employer has provided the EPCI Contractor with a fully working optical link to the onshore substation.

[\[ERS-5.4-960124\]](#)

The EPCI Contractor shall ensure (e.g., via a subscription to an Internet service provider) the WLAN provides users with access to the Internet from the moment the EPCI Contractor has delivered the ready-to-use WLAN until Taking Over Completion.

8.15 Electrical system studies and models

[\[CBOD-STUD-100101\]](#)

Employer's requirements stated in (B21a_iii) shall be applied.

8.15.1 Electrical system studies

8.15.1.1 Studies purposes

[\[CBOD-STUD-100102\]](#)

The contractor is required to conduct a comprehensive set of studies to ascertain:

- The equipment is designed to withstand all foreseeable stresses during its operational lifetime as mentioned in 7.2.
- The system is capable of functioning under all operational configurations specified within this document.
- The specified disturbance thresholds outlined in this document are respected.
- The system performances are aligned with the requirements specified in this document.

- In the situation of faults, studies must substantiate that all faults are identified and addressed. All conceivable faults shall be considered.

[CBOD-STUD-100103]

At least, the subsequent studies shall be undertaken. The detail of each study presented in (B21a_iii) shall be applied.

Design studies	Grid studies	Dynamic Performance Studies (DPS)	Perturbations studies	Overall System Studies
• Main circuit parameters • Power circuit arrangement • Reactive Power studies • Insulation coordination and overvoltages: • Insulation coordination design • AC temporary overvoltage • DC Temporary overvoltage • Short circuit studies : AC / DC • AC breaker study • Ferroresonance • Voltages and currents stresses for components rating, losses and audible noise • AC harmonic filter performance (if applicable) • Earthing system design onshore/offshore • Lightning and earthing protection concept	• Harmonic stability : onshore / offshore • Steady state AC harmonic onshore / offshore • Subsynchronous impedance (SSO mitigation performance if required)	• System studies • AC and DC faults - system protection • Management of HVDC system configuration • Connection of AC grid equipments • Step changes • Wind Farm frequency control • Grid code compliance studies • Active and reactive power capabilities • AC voltage and reactive power control - onshore • Behaviour during and after voltage disturbance - over and under voltage ride through • Frequency ranges • Grid forming • Power oscillation damping through reactive power modulation	• Electromagnetic compatibility • High frequency interference • Electromagnetic Field (EMF) • DC harmonic • Audible Noise	• Losses • RAMS • FMECA • HVDC System Life Expectancy analysis • Layout, building • Seismic, Transport, and Wind Loading Calculations • Other civil and structural requirements • Transportation, storage, construction and installation

Table 26: List of minimum studies

[CBOD-STUD-100104]

The Contractor is required to perform studies to identify and assess risks associated with all equipment and materials within their scope, in line with relevant standards. This includes analysing stresses due to overvoltages, overcurrents, and other transient phenomena (e.g., for circuit breakers: Transient Recovery Voltage (TRV), Rate of Rise of Recovery Voltage (RRV), and zero-miss effects).

[CBOD-STUD-100105]

The scope encompasses all DC and AC equipment of the Offshore Converter Station and Onshore Converter Station. For the DC side, even though requirements are specified at the DC terminal (interface) like DC TOV, DC Short Circuit Current, faults and measurements along the DC cables should be incorporated in the study parameters and results.

The studies shall be performed in all the AC grid connection configurations (including GIS offshore) system with all the operating modes mentioned in section 8.4.

8.15.1.2 Participation of Employer during the design phase

[CBOD-STUD-100106]

Prior to commencing the studies, a study outline (including the models and scenarios used) and the test matrix shall be shared with the Employer. The Employer reserves the right to advocate for the inclusion of additional scenarios when deemed necessary.

[CBOD-STUD-100107]

At the beginning of project, during design phase, the contractor is advised to utilize a OWF and OWF model based on their historical expertise. The main assumptions used for this model shall be discussed and approved by the Employer before any study commencement.

[\[CBOD-STUD-100108\]](#)

Upon completion of the studies: Every study must culminate in a comprehensive report elucidating the goals, methodology, and conclusions derived from the contractor's research. The employer shall have the right to review these studies. If deemed necessary, Employer shall be able to reproduce the study results with the information and models given by the Contractor. All results shall be provided in a digital format, compatible with mainstream commercial software.

8.15.1.3 Collaboration with Employer and third party

[\[CBOD-STUD-100109\]](#)

Due to the delayed availability of OWF models in the project timeline, studies with the most updated models shall be conducted late, and therefore, issues may be detected late in the execution phase.

Contractor shall repeat at least the studies defined below and the relevant cases if needed with the final OWF model once available. Contractor shall justify the studies that are relevant and those that are not repeated.

- offshore overvoltages,
- transient current stress,
- harmonic stability
- dynamic studies

Employer will perform network studies related to the integration of the HVDC Link into the onshore and offshore AC networks during all project phases using the model given by Contractor.

[\[CBOD-STUD-100119\]](#)

If Employer or Contractor identifies an issue during its tests or studies needing an adaptation of the control and design of the HVDC system, the Contractor shall collaborate to solve the issue especially if any updates of the C&P are needed. Depending on the circumstances, the Employer can undertake the responsibility for the new development as mentioned in [\[B21ai-MSGE-110249\]](#) of (B21a_iii).

The collaboration process shall be:

- First, all parties shall define the level of information to exchange.
- Second, the Contractor, shall suggest the most optimal mitigation solutions, including on the Converter Stations.
- Third, Employer with support of the Contractor and potentially the Producer, if relevant, shall perform additional network interaction studies to find the best suited fix.
- Fourth, all parties agree on the update
- Finally, if needed, the Contractor shall perform non-regressive assessment

8.15.2 Models

8.15.2.1 EMT models

[\[CBOD-STUD-100120\]](#)

The Contractor shall provide the models used for all studies. The Employer's requirements related to EMT models as outlined in (B21a_iii) – section 4 shall be applicable.

At a minimum, the EMT models shall be compliant with the following requirements.

- HV equipment model shall be open (no black box)
- [\[B21ai-MODE-110395c\]-C](#) All HVDC protections with duration lower or equal than 60 seconds shall be included in PSCAD and EMTP models delivered by Contractor.
- C&P models interfaces shall comply with the IEEE/CIGRE DLL interface
- EMT model provided by Contractor to be benchmarked with RTS and SAT. Models validation shall be confirmed through verification against the performance requirements mentioned in "Performance Checking Sheet" (comprising approximately 30 simulation cases). The "Performance Checking Sheet" will be provided by Employer in (Call_off).

- [B21ai-MODE-110393\]-C](#) The Contractor shall deliver the models based on the following time schedule:
 - Preliminary models 1: shall be provided at the start of the Dynamic Performance Study (DPS) as the latest.
 - Preliminary models 2: models shall be provided after DPS and 4 months before the start of FST.
 - Intermediate models shall be delivered after the FST and before the end of FAT of C&P
 - As-built models shall be delivered one month after beginning of trial operation.

The timeline for model update provision is detailed in [\[B21ai-MODE-110394\]](#) of (B21a_iii).

8.15.2.2 RMS model

[\[CBOD-STUD-100121\]](#)

The Contractor shall provide the models used for all studies. The Employer's requirements related to RMS models as outlined in (B21a_iii) – section 4 shall be applicable.

At a minimum, the RMS models shall be compliant with the following requirements.

- [\[B21ai-MODE-110415-416\]](#) Models shall be delivered either in open software or bloc diagram with detailed parameters allowing the Employer to model the HVDC Link in its preferred tool with accurate performances.
- [\[B21ai-MODE-110415-416\]](#) Models shall represent all control loops with a time constant above 1 ms (voltage control / POD ...)
- [\[B21ai-MODE-110415-415\]](#) Models shall properly represent fault behavior of HVDC Link.
- [\[B21ai-MODE-110415-417\]](#) Models shall fit to the performance requirements mentioned in "Performance Checking Sheet" (comprising approximately 30 simulation cases). The "Performance Checking Sheet" will be provided by Employer in (Call_off).

8.16 Onshore civil works

[\[CBOD-CIVW-100122\]](#)

Employer's requirement related to civil works detailed in (B22b_i) shall be applied.

8.17 Operation, preparation of operation and maintenance

[\[CBOD-MAIN-100123\]](#)

The Employer's requirements related to maintenance in design detailed in (B22c) shall apply.

[\[CBOD-MAIN-100124\]](#)

All special tools and access equipment required for operation, maintenance and repair shall be supplied along with sufficient information for their use and maintenance.

As part of Employer's requirements, the following requirements shall be applied as minimum.

8.17.1 Operation and maintenance documentation

A full set of O&M Manuals and associated technical documentation as specified in the following shall be provided.

8.17.1.1 Maintenance plan

[\[B22c-MAIN-910171\]-N3](#)

The Contractor shall provide a **Maintenance Activity List (MAL)** for the planned maintenance delivery. The MAL shall exhaustively list all maintenance activities to be performed during the whole life expectancy of the Converter Stations (40 years).

This document, in table format, shall be the basis on which the Contractor shall develop the tasks needed to properly maintain the Converter Stations for their whole life expectancy. The MAL shall be designed to ensure the continued operation and maintenance of the HVDC Link and HVAC system.

[\[B22c-MAIN-910187\]-N3](#)

Each of the tasks of the MAL shall have the characteristics requested by Employer's as mentioned in (B22c). If necessary, it shall be updated during the Defect Notification Period.

[\[B22c-MAIN-910172\]-N3](#)

Each Maintenance task of the MAL shall refer to a specific document (usually the Preventive Maintenance Handbook indicated in section 6.5 of (B22c)) that shall contain specific information regarding this maintenance activity as requested by Employer.

[\[B22c-MAIN-910021\]-N3](#)

The Contractor shall develop a detailed **Maintenance and Inspection Plan (MIP)**, based on the MAL, but focused on a rolling five-year period and updated annually during the Defect Notification Period, which shall be the basis for preventive maintenance planning for each equipment the Converter Stations (high and low voltage AC and DC equipment, building, auxiliaries, HVAC/Climate, cooling system, C&P system, Valves, etc...).

This MIP shall detail the preventive maintenance tasks to be performed during the rolling five-year period. It shall detail parallelization of maintenance tasks to fulfil the Scheduled Outage requirement described in section - . The MIP is to be reviewed and agreed by the Employer as mentioned in (B22c).

[\[B22c-DOCM-910140\]-N3](#)

The Contractor shall include in his MIP the maintenance actions that have to be carried out on the spare parts and maintenance kits.

8.17.1.2 Operational manuals

[\[B22c-DOCM-910141\]-N3](#)

The operational manual should be provided for **each system** (for example station HMI, station control, cooling, high ventilation air conditioning, remote control, auxiliaries) and should contain the system's description (with each equipment), relevant information on how each system works, has to be operated (start/stop sequence, parameter or status modification, ...) and interacts with other systems.

These manuals should also contain screenshots with step-by-step explanation on where to click and how to proceed for each sequence on the dedicated HMI.

8.17.1.3 Preventive maintenance handbook

[\[B22c-DOCM-910191\]-N3](#)

The Contractor shall provide an Earthing procedure that details how to perform the full earthing and lock-out procedure of the Converter Station (from de-energisation of the Converter Station to earthing the Converter Station), considering the interlocking system, and including earthing/lock-out of the necessary HV/MV/LV equipment (including the valves). Earthing procedures shall comply with (NFC18_510_1).

[\[B22c-DOCM-910144\]-N3](#)

The Contractor shall provide a preventive maintenance handbook for each equipment or family of equipment of the Converter Station (e.g. C&P system, Cooling System, Valve hall earthing switches, DC surge arresters, etc...).

The preventive maintenance handbooks shall accurately cover in detail **each maintenance task listed in the MAL/MIP**. Each maintenance task shall be classified by frequency (monthly, quarterly, yearly etc...). The aim of this part of the manual is to describe step by step by means of pictures of the actual

on site equipment how to proceed for each maintenance task mentioned in the MAL/MIP. It should include all the elements described in [\[B22c-MAIN-910172\]](#).

[\[B22c-DOCM-910148\]-N3](#)

The preventive maintenance handbooks should include the following sections as mentioned in (B22c)

- Health and safety
- The description of the sub-system or the equipment that is dealt with in this handbook;
- The geographical location of this sub-system/equipment on the layout of the installation and its use on a single line diagram (electrical, cooling, control system, ...);
- All the reference documentation (layouts, schemes);
- Reference values with the tolerance. If there is still a gap, the Contractor will indicate the actions to be performed.
- Concerning maintenance operation in the converter hall, a procedure shall be provided by the Contractor explaining the different steps to be respected in order to enter safely in the hall and to be allowed to perform maintenance on a given equipment (grounding switch, converter valve for example)
- Status of the station and equipment to prevent risks (electrical isolation or other energy isolation)
- Number of operators required (in accordance with the Employer procedures and safety rules listed in section 8.2 of (B22c) their skills (electrician, mechanic, technician, engineer,...) and their level of responsibility (Works conductor or operator), maintenance level of the task
- Tools required to perform the task, specify if spares are needed
- Duration of the task and impact on the operational scheme (outage/no outage)
- Task description
- Checks after task completion to ensure the task has been performed well
- **Operational Report:** checklist enriched with pictures (**a template will be provided by the Employer for these checklists**) allowing for the reporting of the performed preventive maintenance tasks and the observed values of sensors, etc.... During the Defect Notification Period, **this checklist is sent to the Contractor for validation of the realised maintenance tasks.**

8.17.1.4 Corrective Maintenance Handbook

[\[B22c-DOCM-910145\]-N3](#)

The Contractor shall provide an Instruction Manual for each and every single equipment of the Converter Station. These instruction manuals shall cover all the corrective maintenance aspects related to their respective equipment, and especially include a detailed procedure for equipment replacement.

Most of the times, the Original Equipment Manufacturer (OEM) already provides such information in a manual. In this case, unless specifically requested by the Employer, the Contractor does not have to provide an instruction manual in addition to the OEM manual.

[\[B22c-DOCM-910149\]-N3](#)

For the C&P system, additional documents, called “corrective maintenance handbooks” shall be provided by the Contractor.

[\[B22c-DOCM-910200\]-N3](#)

The Contractor shall provide detailed procedures (with pictures) that show and explain how to perform:

- The maintenance on the valves including all safety procedures.
 - o Earthing of the valves,
 - o Replacement of VSM d.c. capacitor,
 - o Replacement of VSM.
- Spare converter transformer exchange (with busbar reconfiguration)

These procedures shall explain step by step each task and present the safety rules to be respected.

A preliminary version of these procedures shall be provided before training sessions and used during training.

8.17.1.5 Troubleshooting Guide

[\[B22c-DOCM-910153\]-N3](#)

All the alarms or warnings of the station shall be categorised in 3 types (as mentioned in (B22c)) depending on their impact on the system's operation.

[\[B22c-DOCM-910154\]-N3](#)

The troubleshooting guide shall provide **for each alarm** (potentially showing up on the HMI):

- Description of the alarm
- The emergency degree of the alarm
- The causes: reason that leads to this alarm or warning;
- The consequences of this alarm or warning on the system operation;
- The equipment concerned by this alarm;
- Actions that must be undertaken and the associated repair actions with the reference of the documents describing these actions;
- Link with the maintenance documentation (instruction manual or corrective maintenance manual);

The troubleshooting guide will be submitted to Employer acceptance. The Employer shall have the possibility to make comments on the provided troubleshooting guide that will have to be taken into account by Contractor.

A preliminary version of the troubleshooting guide shall be provided before training sessions and used during training.

8.17.1.6 Signal mapping list

[\[B22c-DOCM-910152\]-N3](#)

The Contractor shall provide the Signals Mapping List (SML) document which provides for each signal of the C&P system:

- Transmitter (sending)
- Receptor (receiving)
- Identification codes
- Nature of the Signal

The final version of the SML shall be provided before the Taking over.

8.17.2 Asset Management

[\[CBOD-MAIN-100269\]](#)

The Contractor shall provide, populate, maintain, and update the CMMS as per requirements [B22c-CMMS-910206] of B22c, and [B22ci-MANT-930075] of (B22ci).

[\[B22c-MAIN-910053\]-N3](#)

The Contractor shall provide a Long-Term Asset Replacement Plan (LTARP). The LTARP shall provide an insight into the condition and expected degradation of all equipment forming part of the Works and, as such, the overall life and expected replacement (or refurbishment) period of assets. Disposal of equipment shall also be described. It shall provide guidelines regarding the optimum timing of refurbishment and replacement in terms of life-cycle management for all components or equipment. The plan shall be specific to the equipment supplied by the contractor for the Converter Stations in the scope of these Works. In addition to converter components (i.e. cooling systems, valve components, HV equipment, control and protection equipment), it shall also include the components defined in the civil works (B22b_i).

The end-goal of the LTARP is to draw up a long-term budget for life-extending maintenance, renovations and replacement of the converter station equipment, based on costs and conditions.

[\[B22c-DOCM-910163\]-N3](#)

The Contractor shall fill in the Employer's maintenance database according to requirement mentioned in section "Database system and graphical requirement" of (B22b_i).

[B22c-MAIN-910177]-N3

The Employer shall be granted full access (in both reading and writing) to every PLC in the Converter Station as well as to at least Pole control and Converter Control levels (as defined in (IEC62747) figure 12) of the C&P software. No access hindering (e.g. "black box") shall be tolerated for these software. Procedures on how to access, modify and compile all of these software shall be provided.

8.17.3 Spare parts

[B22c-SPAR-910055]-N3

The Contractor shall provide a **complete set of spares, including maintenance kit** (special tools and test equipment) **required for at least five years of operation** to meet the availability and reliability performances described in section 7.2. Final quantity of spares will be confirmed at call-off and may be optimised by Employer.

[B22c-SPAR-910058a]-N3

Some materials shall be stored in multiple copies (e.g. each constitutive part of the VSM (capacitor, auxiliary circuit, power electronics of VSC Valve Level), C&P electronic cards, etc...) depending on the fault rate and the estimated lifetime of each equipment.

The spare parts batch for each and every equipment for which a spare exists shall be sized according to the formula below for each Converter Station:

$$N_{spare} = floor \left(\frac{N_{same\ equipment} * T_{spare}}{MTBF} \right) + 1$$

Where:

MTBF: Mean Time Between Failure

- $N_{same\ equipment}$: quantity of the same equipment in one Converter Station.
- T_{spare} : duration of the spares (5 Service Years)
- N_{spare} : N_{spare} shall be the nearest whole number of spare for the equipment considered
- floor: floor function that takes as input a real number "x" and gives as output the greatest integer less than or equal to "x".
- For example, as a result of the above formula, the following spares shall be taken into account (assuming an acceptable MTBF for these equipment) for each Converter Station:
 - One spare single phase converter transformer
 - One spare AC phase reactor
 - One spare DC reactor per
 - One spare of each type of Auxiliary System Transformer.

8.17.4 Briefing and training

[B22c-TRAI-910119]-N3

In order to achieve high reliability and availability, it is necessary that Employer personnel acquire good know-how and understanding of the HVDC system before taking-over. Hence, in-depth training of the Employer personnel by the Contractor during the construction and commissioning phase is requested. The minimum training contents are presented below.

[B22c-TRAI-910122]-N3

At least, the training program shall deal with the following issues, including some real practices based on the documentation provided by the Contractor. The details of each training group are presented in (B22c).

Group 1 - Preliminary training: this theoretical training shall take place in the Employer offices before C&P FATs. It shall contain no more than 14 hours of lectures (excluding breaks) and may not include practical hands-on training.

[B22c-TRAI-910123]-N3

Group 2 – Maintenance training: this training shall take place on-site during the Station Subsystem Tests. It shall contain 50 hours of training and must include almost exclusively practical hands-on training on the real HVDC equipment. All preventive maintenance tasks shall be covered.

[\[B22c-TRAI-910124\]-N3](#)

Group 3 – Automation training: this training shall take place in the Employer Laboratory on the C&P Replica before the end of the Trial Period. It shall contain 35 hours of training and must include both theoretical and practical hands-on contents.

[\[B22c-TRAI-910125\]-N3](#)

Group 4 – Expert training: this training shall take place in the OEM offices after the end of C&P system FATs and before C&P cubicles are shipped to sites. It shall contain 50 hours of lectures and practical hands-on training on the C&P cubicles (or C&P Replica).

[\[B22c-VALV-910096\]-N3](#)

A valve mock-up shall be provided by the Contractor in order for the Employer personnel to develop and manage in-house skills and know-how regarding preventive and curative maintenance of the power electronics valves.

The valve mock-up shall be a true-to-life full-size (scale 1:1) replica of one valve tower or equivalent valve structure, such that working conditions (height, weight, equipment accessibility, etc.) are exactly the same as on the actual power electronics valves. All equipment related to the valves such as earthing points, water pipes, grading electrodes, cable trays, insulators, fibre optics, etc.... shall be included in the mock-up.

The valve mock-up shall be located on-site in a dedicated training room and shall be accessible at all time.

8.18 Commissioning

[\[CBOD-COMM-100132\]](#)

The Employer requirements detailed in (C14) shall be applied and supersede the requirements written in this chapter, that are just a summary. As part of Employer's requirements, the following requirements shall be applied as minimum.

[\[C14- GENE-040007/040021\]](#)

The Converter System, all subsystems and components shall be comprehensively tested by the Contractor to demonstrate that they meet the requirements of Technical Specifications. Contractor shall follow the (CIGRE_TB697).

[\[C14- GENE-040022\]](#)

From Pre-Commissioning onwards, Employer's engineering, operational and maintenance staff may be on site to witness and support the Contractor and the Contractor should make allowances to allow access to the works area as appropriate.

8.18.1 Responsibilities and Communication

[\[C14-STST-040201\]](#)

For the commissioning stages after and including Station System Tests the Employer's normal operating procedures shall be followed regarding switching, power dispatching and access to high voltage areas. Any work on high voltage Equipment will follow the National Safety legal procedures (Espace_réservé1) and Employer's operation rules.

Switching operations need to be approved by the Employer in advance.

[\[C14- GENE-040026\]](#)

For the Onshore Converter Station, as soon as it is energised from the AC grid, Employer will bear the role of:

- Outage engineer ("chargé de consignation") and Requisition engineer ("chargé de requisition") which will be responsible for isolating and safely locking the equipment in accordance with (NFC18_510_1),
- Operating engineer ("chargé d'exploitation").

Employer will therefore provide authorisation to access electrical assets and perform the operations.

[\[C14- GENE-040027\]](#)

For the Offshore Converter Station, as soon as it is energised from the AC grid, Employer will bear the role of Operating engineer ("chargé d'exploitation"). EPCI Contractor shall bear the role of Outage engineer ("chargé de consignation") and Requisition engineer ("chargé de requisition") which will be responsible for isolating and safely locking the equipment in accordance with (NFC18_510_1). Contractor shall thus be responsible to perform all the operations, manoeuvres, etc. as part of the test program as requested by the Employer Operating engineer ("chargé d'exploitation").

[\[C14- GENE-040113\]](#)

In its scheduling, the Contractor shall take into account that for the tests requiring the participation of the Employer, this participation should be not higher than 8 hours per day and consider only Business Days and business hours (7:30 am to 5:00 pm Central European Time, with a one-hour and thirty minutes break at mid-day)". Exceptional tests requiring an extended participation of the Employer will be highlighted in the schedule (dates and duration) and will require acceptance from the Employer. In particular, if a test requires the presence of the Employer for more than 8 hours a day (taking into account all necessary preparation actions and the post-test debriefing time including lockout and tagout and all necessary actions to establish safe working conditions according to national regulations), it shall be mentioned at least 6 months prior to the execution of the test

[\[C14- GENE-040088\]](#)

The Tests Schedule shall be agreed following the Inspection Test Plan analysis carried out by the Employer (in particular in regard to grid security). This agreement can evolve up to real time because of grid contingencies or constrained network situations.

The Contractor must make time allowance in his schedule for these risks and the Employer will make his best endeavour to mitigate these risks as much as possible.

[\[C14- GENE-040110\]](#)

The Employer reserves the right to make alterations to the test programme and procedures in collaboration with the Contractor.

The Contractor shall include five (5) Business Days of additional testing within their programme, in order to cover the time required by the Employer for possible additional tests of its choice.

[\[C14- GENE-040053\]](#)

From Offshore Substation First Energization Certificate, as long as the HVDC Cable Link is available, the Employer shall be responsible of supplying and bearing the cost of the energy from the Onshore AC Network for the Offshore Substation. In any other cases, the Contractor shall account for the cost or energy necessary for the Works.

[\[C14- GENE-040054\]](#)

From Onshore Station First Energization Certificate, the Employer shall be responsible of supplying and bearing the cost of the energy from the Onshore AC Network for the Onshore Converter Station, as long as energy from the tertiary (source 1) of the transformer is used.

For both construction and commissioning phases, if required energy from source 2 comes from Employer Existing AC Substation through a dedicated auxiliary transformer (see (Call_off)), the Employer will supply it at its own cost.

In any other cases, the Contractor shall account for the cost or energy necessary for the Work.

[\[C14- GENE-040056\]](#)

In case of contingency out of the Contractor's responsibility (ie in Employer's, Producer's or Cable Contractor's responsibility) where energy cannot be supplied from Onshore AC Network, Contractor shall be responsible to provide the energy and associated logistics, that will be subject to the mechanism of Initially Not Priced Subcontracted Works defined in Part 3 of Exhibit 2.

[\[C14-TRIAL-040279\]](#)

The checks and maintenance activities (scheduled maintenance) shall be carried out by the Contractor in agreement between the Employer and the Contractor on the basis of the Maintenance Inspection Plan and Maintenance Activity List proposed by the Contractor and accepted by Employer.

8.18.2 Testing of the Converter System

[\[C14- GENE-040118/19\]](#)

If some of the specified tests are not completed satisfactorily, the Contractor shall inform Employer immediately, shall eliminate all defects and shall improve the quality without additional cost for the Employer. In this case, additional tests shall be carried out.

[\[C14- GENE-040122\]](#)

If any item of Equipment supplied by the Contractor fails during any of tests or shows any signs of damage or failure during the whole course of Commissioning, it shall be repaired and defective parts shall be replaced or redesigned, including restocking of the Spare Parts which may have been used.

[\[C14- GENE-040077\]](#)

The Contractor shall submit to the Employer for each test phase:

- (a) a Tests Schedule (which shall contain a timetable including the programme and sequencing of the tests) not less than twelve (12) Months; and
- (b) An Inspection Test Plan, Commissioning Protocol, Test Schedules not less than six (6) months;
- (c) the Test Procedures not less than three (3) months.

prior to the planned date when the Works are required to be ready for each of the tests phase. The documents must be reviewed and accepted by the Employer.

[\[C14- GENE-040091\]](#)

During the entire commissioning period including pre-commissioning, sub system, system and interconnector tests, the Contractor shall provide an up-to-date Test Schedule with a seven day lookahead of the precise day to day activities that will be performed.

[\[C14- GENE-040085\]](#)

The Inspection Test Plan shall include at least the following information:

- a) List of all tests consistent with the tests required for Commissioning,
- b) Indication of the need for power supply, connection of main circuits and other information necessary for system planning, commercial and operational requirements.

[\[C14- GENE-040096\]](#)

For each test, a Test Record including the identifications of Punch List items and defects shall be provided to the Employer no later than 48 hours after the test completion for review and signature by the Employer's representative.

[\[C14-GENE-040104\]](#)

The report document shall be submitted not later than ten (10) Business Days after completion of the test stage to the Employer for review and acceptance, summarising the testing activities and demonstrating that all test criteria are passed.

8.18.3 Commissioning of the Converter System

Contractor shall follow the sequence and associated milestones of flowchart (C14_1).

[CBOD-GENE-110273]

Necessary preconditions for each subsequent stage are:

- a) The issuance of the relevant certificate from the previous stage by Employer.
- b) Provision of the relevant commissioning deliverables, spares for commissioning and [installation, operation and maintenance manuals] in relevant status depending on the stage (preliminary and red-marked up as-built documents).
- c) Provision by Contractor of relevant training to Employer for the purpose of witnessing and supporting the Contractor.
- d) Notification to the Employer with a delay indicating the commencement of the tests.

[C14-GENE-040037]

The Contractor shall give notice to the Employer within five (5) Business Days of the date on which the prerequisites are met and the relevant sub-system of the works are ready for the Station Sub-System tests.

[C14-MC-040129]

Before the Mechanical Completion start, at least the following preconditions shall be met:

- a) The FAT of the components and subsystems has been performed satisfactory,
- b) The Inspection Release Certificate (IRC) has been released by the Employer for the relevant equipment as per (QUA)
- c) In case of conditional IRC accepted by the Employer, the Contractor shall confirm a date of closure of Punch list items to not impact further installation milestones
- d) The components and subsystems have been transported, delivered and inspected without damages to their installation location within the Onshore Converter Station and the Offshore Converter Platform topside at the yard and installed,
- e) Relevant Tests Deliverables have been issued by Contractor and accepted by Employer,

[C14-GENE-040050]

For radial connections between Onshore Converter Station and OSS, no tests evidencing satisfaction of the Interconnector System tests criteria shall be carried out involving any part of the Works until:

- a) The Employer allows Contractor to energise the IAC before power withdrawal
- b) The relevant Tests Deliverables have been issued by the Contractor and accepted by the Employer,
- c) Onshore Station System Tests Certificate and Offshore Station System Tests Certificate have been issued;
- d) The Producers have connected the OWFs and is ready to inject a power capacity of 25% of the Guaranteed Active Power Link Capacity.

8.18.4 Trial Operation

[C14-GENE-040059]

The Trial Operations tests shall not commence before following conditions are fulfilled, except if stated otherwise in the Contract:

- a) Interconnector System Tests Certificate has been issued; and
- b) Telephone assistance service has been put in place at that time and is ready to be used as required in Technical Specification [B22c-TSUPP-930026] of (B22c).
- c) The Producers have connected the full extent of the OWFs and is ready to inject a power capacity of 100% of the Guaranteed Active Power Link Capacity.

[CBOD-GENE-110274]

Subject to requirements of [CBOD-GENE-110278] and [CBOD- GENE-110279], the Contractor shall successfully complete ninety (90) Days of trial operations testing consisting of:

- a) an initial trial operations burn-in period ("Trial Operations Part 0") of thirty (30) days provided that no minimum performance requirements in relation to [Converter Forced Outage, Scheduled Energy Unavailability, Converter Forced Energy Unavailability, Guaranteed Active Power Link

Capacity or reactive power capacity below that required by the Employer's Requirements and Technical Specifications] shall be applied to such Trial Operations Part 0, and followed by:

- b) subject to requirement [CBOD-GENE-110278], a further trial operations period ("Trial Operations Part 1") of forty (40) Days of operation. In case of occurrence of an event or circumstance specified in requirement [CBOD-GENE-110275], the associated time shall be discounted in accordance with requirement [CBOD-GENE-110277]; and followed by:
- c) subject to requirement [CBOD-GENE-110279], a further trial operations period ("Trial Operations Part 2") of twenty (20) Days of continuous operation.

Parts 0, 1 and 2 are together the "Trial Operations test". Part 2 (requirement [CBOD-GENE-110274.c]) must be an uninterrupted period and Part 1 (requirement [CBOD-GENE-110274.b]) may be interrupted by Part 2.

[CBOD-GENE-110275]

The events or circumstances for the purposes of the requirement [CBOD-GENE-110274.b] are:

- a) either a Converter Forced Outage, or
- b) the occurrence of a Severity 1 Fault.

[CBOD-GENE-110276]

For the purposes of the requirement [CBOD-GENE-110275.b], any period of forced or scheduled reduction in active power transmission capacity below the Guaranteed Active Power Link Capacity or any period of reduction in reactive power capacity below that required by the Employer's Requirements and Technical Specification would be counted within duration of Trial Permissible Outage, and forty-eight (48) hours of Trial Permissible Outages means the sum of:

- a) Converter Scheduled Energy Unavailability and provided that the Contractor has given notice to the Employer of such Converter Scheduled Energy Unavailability by the earlier of:
 - o twenty-four (24) hours before such Scheduled Energy Unavailability; and
 - o before 10.30am CET on the Day before such Scheduled Energy Unavailability;
- b) to the extent such notice has not been given, this shall be treated as Converter Forced Energy Unavailability.
- c) Converter Forced Energy Unavailability.

[CBOD-GENE-110277]

For the purposes of the requirement [CBOD-GENE-110274.b], "shall be discounted" shall have the meaning: the forty (40) Days of operation shall be paused at the time of the start of the event or circumstance and shall resume at the time that normal operation recommences.

[CBOD-GENE-110278]

At any time during the Trial Operations Part 1 referred to requirement [CBOD-GENE-110274.b], if any of the following is experienced then the entire forty (40) days period shall restart:

- a) more than two Converter Forced Outages occur;
- b) an aggregate of more than forty-eight (48) hours of Trial Permissible Outages occurs.

[CBOD-GENE-110279]

At any time during the Trial Operations Part 2 referred to requirement [CBOD-GENE-110274.c], if any of the following is experienced then the entire twenty (20) Days period shall restart:

- a) the occurrence of any Converter Forced Outage;
- b) any period of Converter Scheduled Energy Unavailability or Converter Forced Energy Unavailability occurs; or
- c) the occurrence of any period of forced or scheduled reduction in active power transmission capacity below the Guaranteed Active Power Link Capacity or any reduction in reactive power capacity below that required by the Employer's Requirements and Technical Specifications.
- d) The occurrence of a Severity 1 Fault.

[CBOD-GENE-110280]

All Severity 1 Faults and Severity 2 Faults occurring during Trial Operations Part 0 and Part 1 must be fully rectified by the Contractor before end of Part 1. All Severity 3 Faults (according to Employer's

definition [B22c-MAIN-910005] of (B22c)) in occurring during Trial Operations Part 0 and Part 1 must be fully rectified before Taking Over. Root cause analysis of the Fault shall be provided to the Employer not more than fifteen (15) Business Days after the occurrence of the Fault and before the end of Trial Operations Part 1 (whichever is earlier).

[\[CBOD-GENE-110281\]](#)

Subject to requirement [CBOD-GENE-110279], all Severity 1 Faults and Severity 2 Faults occurring during Trial Operations Part 2 must be fully rectified by the Contractor before Taking Over. All Severity 3 Faults occurring during Trial Operations Part 2 must be fully rectified unless the rectification of any such Severity 3 Fault may be treated as a Final Reservation. Root cause analysis of the Fault shall be provided to the Employer not more than fifteen (15) Business Days after the occurrence of the Fault and before the end of Trial Operations Part 2 (whichever is earlier).

[\[CBOD-GENE-110282\]](#)

As soon as twenty (20) Days of continuous operation applicable to successfully achieve Trial Operations Part 2 occurs and provided that the Converter Stations were operated during a cumulative period of 35 days from the start of Trial Operation tests (inclusive of the Trial Operation Part 0), then the Contractor has the right to apply for an early Taking Over Certificate and the provisions of Taking Over of the Works as per Contract. These twenty (20) days of continuous operation can take place at any time during the sixty (60) days of the period of Trial Operations Part 1 and 2.

[\[CBOD-GENE-110283\]](#)

Following conditions shall be fulfilled during the Trial Operation:

- Active power of at least 95% of Guaranteed Active Power Link Capacity has to flow continuously during 1 hour.
- Active power of at least 95% of Guaranteed Active Power Link Capacity has to flow during an aggregate of 72 hours on the Trial Operation. Only periods where active power of at least 95% of Guaranteed Active Power Link Capacity has to flow.
- Active power of at least 60% of Guaranteed Active Power Link Capacity has to flow continuously during 12 hours.

8.18.5 Acceptance

[\[C14-TRIAL-040288\]](#)

Once the Trial Operation is achieved, the Employer and the Contractor will review the following and the Employer will release the "Trial Operation Certificate" only when:

- The Contractor submits a test report confirming that the Works operated pursuant to the Trial Operation programme without any Punch List, and
- The Contractor submits to the Employer an updated set of As-Built Documentation, and
- The inspection of the punch list and the plan to fix them is jointly carried out by the Contractor and the Employer at the end of the Trial Operation and agreed by both.

9 Multiterminal readiness variant

The Employer requirements relating to multiterminal readiness variant are detailed in (CBODMT).

10 Standards, regulations and special requirements

[\[C11-STAND-010006\]-N3](#)

If no explicit standards are specified in the Technical Specifications, the following systems of regulations, standards and technical specifications shall be applied and prioritized in the following order:

- French regulations in relation to health, safety, environment, construction and design.
- European HVDC Network Code (HVDC_NetworkCode). French Grid Code (DTR_FrenchCode).
- Employer specific technical specifications and procedures in relation to health, safety, environment, construction and design (listed as applicable in the Technical Specifications).
- French standards.

- CENELEC standards
- IEC standards.
- IEEE standards.
- DNV standards.
- CIGRÉ technical brochure and recommendations.

The same prioritization applies for regulation, codes and standards specified in the Technical Specifications (unless required otherwise). Additionally, requirements made in the Technical Specifications prevail over Employer specific technical specifications (which themselves prevail over standards).

The Contractor shall confirm that the standards specified by the Employer are consistent with the above regulations, codes and standards. The Contractor may propose alternative standards, to be agreed and accepted by the Employer. If a contradiction or a conflict exist between the above listed documents, the Contractor shall inform the Employer and submit proposal to the Employer for its review and approval.

[\[C11-STAND-010007\]-N3](#)

The latest edition of standards including amendments at contract signature of each standard shall apply unless otherwise specified. The latest revision of the standards shall always apply. In regards to EN/IEC equivalent: If an IEC equivalent is under revision (FDIS positively voted) at the time of tendering the Contractor shall apply the IEC: In particular in regards to HV equipment, the Contractor shall include in the list of tests to be carried out, tests required in IEC standards which are mentioned as applicable in the Technical Specifications and which are under revision.

After contract signature, in case the Contractor requests or identifies a benefit in using an updated edition of the standards mentioned as applicable in the Technical Specifications or newly published standards (after the tender phase), it shall be approved by the Employer.

In any case if national particular conditions exists in the EN version it shall be applied.

For document with no EN equivalent such as TS, or guides, IEC document are applicable.

Application of dual logo IEEE/IEC is subject to Employer' approval and in principle not accepted.

[\[C11-STAND-010008\]-N1](#)

SI units shall be used in all documents.

11 Appendix I: Reference document and standards

[\[C11-APPD-010105\]](#)

The requirements specified in this document shall comply with the following IEC/ISO standards, additional French standards and other listed documents. Unless a specific version of a standard is mentioned, the last existing version shall apply by default. When no IEC/ISO standard exists, the Converter Stations shall comply with recognized standards and design practices.

Note that national legislation, standards and specifications may be available only in their original language and that no translation will be made available.

(Arrete_17_2001) : Arrêté du 17 mai 2001 fixant les conditions techniques auxquelles doivent satisfaire les distributions d'énergie électrique.

(Arrete_22102010_Seismic) : Arrêté du 22 octobre 2010 relatif à la classification et aux règles de construction parasismique applicables aux bâtiments de la classe dite « à risque normal ».

(B21a_ii): B.2.1.a.ii Performance Checking Sheets.

(B21a_iv) - Electro Magnetic Compatibility.

(B22a) : B.2.2.a-General Equipment Requirement.

(B22a_0) - Supplier list.

(B22a_iv) B.2.2.a.iv - Voltage Sourced Converter.

(B22a_v) : B.2.2.a.v General Control Protection and Telecommunications.

(B22a_vi) B.2.2.a.vi Onshore Converter Station Auxiliary Supply.

(B22a_vii) Gas Insulated Switchgear.

(B22a_viii) - Auxiliary Transformer.

(B22av_2) : B.2.2.a.v.2-Control System FST and FAT.

(B22av_3) : B.2.2.a.v.3-Control System Dependability.

(B22av_4) : B.2.2.a.v.4-Control System Hardware Tests.

(B22av6): B22av6 Control and Protection System Scope of Works.

(B22b_i) : Civil works and buildings.

(B22c) : B.2.2.c Maintenance in design.

(B22e) : B.2.2.e Intrastation cables.

(C14): C.1.4 Commissioning Test & Trial Operation.

(C14_1) - Commissioning flowchart .

(CBOD) HVDC Converter basis of design.

(CCG_P) : Cahier des charges general postes HTB (Structures et matériels HTB) (CCG_P).

(CIGRE_TB391) : CIGRE Report 391: Guide for measurement of radio frequency interference from HV and MV substations. 2009.

(CIGRE_TB535): Cigré Technical Brochure N0. 535, EMC within power plants and substations.

(Cigre_TB842) Dielectric testing of gas insulated systems.

(Cigre_TB852) Recommendations for testing DC extruded cable systems for power transmission at a rated voltage up to and including 800 kV.

(CISPR_11) Industrial, scientific and medical (ISM) radio-frequency equipment - Electromagnetic disturbance characteristics - Limits and methods of measurement.

(Decret_2010_1255) - French decree 2010-1255.

(Decret2016_1074) Décret no 2016-1074 du 3 aout 2016 relatif à la protection des travailleurs contre les risques dus aux champs électromagnétiques.

(Def) - CMA-TRA-RTE-90001-STR-00-Definitions.

(ERS_2_3) - Architectural.

(ERS_5_2) - ERS 5.2 Utility SCADA.

(ERS_5_4) - ERS_5_4 Telecom systems.

(ERS_5_5) - LV Cable and cable routing requirements.

(ERS_5_6) - ERS 5.6 Auxiliary Services.

(ERS_5_7) - ERS_5_7 Offshore and onshore station requirements - Cybersecurity.

(ERS_6_3) - Heating, Ventilation and Air Conditioning System.

(ERS_6_5) ERS Cooling.

(ERS_TI) - Transport and Installation.

(Exhibit_6) - Exhibit 6 - Guaranteed Performances.

(HSE) - HSE specifications.

(IEC 61000-3-6) Assessment of emission limits for the connection of distorting installations to MV, HV and EHV power systems.

(IEC 61400-21-1) Wind energy generation systems - Part 21-1: Measurement and assessment of electrical characteristics - Wind turbines.

(IEC 62271-102) - High-voltage switchgear and controlgear - Part 102 : alternating current disconnectors and earthing switches.

(IEC_60076_7) Power transformers - Part 7: Loading guide for mineral-oil immersed power transformers.

(IEC_61000_6_1) - Electromagnetic compatibility (EMC) – Part 6-1: Generic standards – Immunity for residential, commercial and light-industrial environments.

(IEC_61000_6_2) - Electromagnetic compatibility (EMC) – Part 6-1: Generic standards – Immunity for industrial environments.

(IEC_61000_6_4) - Electromagnetic compatibility (EMC) – Part 6-4 Generic standards – Emission standard for industrial environments.

(IEC_61000_6_5) - Electromagnetic compatibility (EMC) – Part 6-5 Generic standards – Immunity for power station and substation environments.

(IEC62271-203) - AC gas-insulated metal-enclosed switchgear for rated voltages above 52kV.

(ISO_IEC_17025) Exigences générales concernant la compétence des laboratoires d'étalonnages de d'essais.

(ISO_IEC_17065) Evaluation de la conformité: Exigences pour les organismes certifiant les produits, les procédés et les services.

(NC_HVDC) : ENTSO-E Network Code on High Voltage Direct Current Connections and DC connected Power Park Modules, 2015.

(NF EN 50121-2) Applications ferroviaires - Compatibilité électromagnétique - Partie 2: Emission du système ferroviaire dans son ensemble vers le monde extérieur.

(NF_EN_1991_1) - NF EN 1991-1-/NA - Annexe Nationale Eurocode 1-4 Vent.

(NF_EN_1991_1_3) - NF EN 1991-1-3/NA - Annexe Nationale Eurocode 1-3 Neige.

(NF_EN_61000_1_2) - Electromagnetic compatibility (EMC) - Part 1-2 : general - Methodology for the achievement of functional safety of electrical and electronic systems including equipment with regard to electromagnetic phenomena.

(NF_EN_62271_1) - High-voltage switchgear and controlgear - Part 1 : common specifications for alternating current switchgear and controlgear.

(SOW03) - Interface Scope of Work.

61000-3-6. Electromagnetic compatibility (EMC) – Part 3-6: Limits – Assessment of emission limits for the connection of distorting installations to MV, HV and EHV power systems.

AMAC - List of authorized chemical products of RTE.

Arrête du 9 juin 2020 relatif aux prescriptions techniques de conception et de fonctionnement pour le raccordement aux réseaux d'électricité.

Article 8.3.1 - Cahier des charges des capacités constructives (parc non synchrone de générateurs, y compris en mer).

B21a_iii Studies and models.

B21a_v - Sustainable design.

B22a_i: B.2.2.a.i-Converter Transformer.

B22a_ii: B.2.2.a.ii AC DC equipment.

B22a_iii: B.2.2.a.iii Reactors, capacitors and resistors.

B22c: Maintenance Specification.

B22ci: Converter Station O&M.

B22d - Control and Protection System Replica.

B22e_i: Intrastation cables.

B22e_ii: Intrastation cables - Engineering installation and commissioning .

B22e_iii: Intrastation cables - Execution data dossier.

B22e_iv: Intrastation cables - Technical specification 214.

BOD-01 - Offshore Converter Platform Basis of Design.

BOD-02-Interface Basis of Design.

Call off.

Call-off.

CBODMT - Multiterminal readiness variant.

CIGRE 483 - Guidelines for the Design and Construction of AC Offshore Substations for Wind Power Plants.

CIGRE TB 590 Protocol for Reporting the Operational Performance of HVDC Transmission System.
Cigre TB 754 - AC side harmonics and appropriate harmonic limits for VSC HVDC.
CIGRE TB 811 DC side harmonics and filtering in HVDC transmission systems (2020).
Cigre technical brochure 844: Feasibility study for assessment of lab losses measurement of VSC valves.
CIGRE's Technical brochure No.697, "Testing and commissioning of VSC HVDC systems", 2017.
EN 50647 Basic standard for the evaluation of workers' exposure to electric and magnetic fields from equipment and installations for the production, transmission and distribution of electricity).
ERS 2.5 Corrosion protection.
 European HVDC Network Code COMMISSION REGULATION (EU) 2016/1447 of 26 August 2016.
 French Grid Code - DTR.
 IEC 60068-3-3 : Environmental testing - Part 3-3: supporting documentation and guidance - Seismic tests methods for equipment.
IEC 60071-1 Insulation co-ordination - Part 1 : definitions, principles and rules.
IEC 60071-2 Insulation co-ordination - Part 2: Application guidelines.
 IEC 6007-12: Insulation co-ordination - Part 12: Application guidelines for LCC HVDC converter stations.
IEC 60076-22: Power transformer and reactor fittings.
IEC 60076-3: Power Transformers - Part 3: Insulation levels, dielectric tests and external clearances.
IEC 60076-5: Part 5: Ability to withstand short circuit.
IEC 60137: Insulated bushings for alternating voltages above 1000 V.
 IEC 60255-24: Measuring relays and protection equipment - Part 24: Common format for transient data exchange (COMTRADE) for power systems.
IEC 60815-4 Guide for the selection and dimensioning of high-voltage insulators for polluted conditions.
 IEC 60840: Power cables with extruded insulation and their accessories for rated voltages above 30 kV ($U_m = 36$ kV) up to 150 kV ($U_m = 170$ kV) - Test methods and requirements.
 IEC 60871-2 Ed. 1999: Shung capacitors for AC power systems having a rated voltage above 1000 V - Part 2: Endurance testing.
 IEC 60909-0 Ed. 2, 2016-01.
IEC 61000 Electromagnetic Compatibility.
IEC 61000-4-30 Power Quality Measurement Methods.
 IEC 61071: Capacitors for power electronics.
IEC 61803 Determination of power losses in high-voltage direct current (HVDC) converter stations.
 IEC 61869-1: Instrument transformers - Part 1: General requirements.
 IEC 62067: Power cables with extruded insulation and their accessories for rated voltages above 150 kV ($U_m = 170$ kV) up to 500 kV ($U_m = 550$ kV) - Test methods and requirements.
 IEC 62243: Security for industrial automation and control systems.
 IEC 62271-207: High-voltage switchgear and controlgear – Part 207: Seismic qualification for gas-insulated switchgear assemblies for .
 IEC 62271-318: High-voltage switchgear and controlgear – Part 318: DC gas-insulated metal enclosed switchgear for rated voltages including and above 100 kV.
 IEC 62501 - Voltage sourced converter (VSC) valves for high-voltage direct current (HVDC) power transmission - Electrical testing.
IEC 62747 Terminology for voltage-sourced converters (VSC) for high-voltage direct current (HVDC) systems.
IEC 62751 Determination of power losses in voltage sources converters for HVDC systems.
 IEC TS 60815-3. *Selection and dimensioning of high-voltage insulators intended for use in polluted conditions - Part 3 Polymer insulators for a.c. systems.*
IEC/IEEE 65700-19-03: Bushings for DC application.
 IEC_60076_series: Power Transformers.
 IEC_61786_2 Measurement of DC magnetic, AC magnetic and AC electric fields from 1 Hz to 100 kHz with regard to exposure of human beings - Part 2: Basic standard for measurements.
IEC60071_11. *Insulation co-ordination - Part 11: Definitions, principles and rules for HVDC system.*
 IEC60099_4: Surge arresters - Part 4: Metal oxide surge arresters without gaps for AC system.
 IEC60099_9: Surge arresters - Part 4: Metal oxide surge arresters without gaps HVDC converter stations.
 IEC60815_2: Selection and dimensioning of high-voltage insulators intended for use in polluted conditions - Part 2: Ceramic and glass insulators for a.c. systems.
 IEC60815_3: Selection and dimensioning of high-voltage insulators intended for use in polluted conditions - Part 3: Polymer insulators for a.c. systems.

IEC60815-1: Selection and dimensioning of high voltage insulators intended for use in polluted conditions - Part 1: Definitions, information and general principles.
IEC6100_2_4: Electromagnetic compatibility (EMC) - Part 2-4: Environment compatibility levels in industrial plants for low-frequency conducted disturbances.
IEC61850: Communication networks and systems in substations.
IEC61869_series: Instrument transformers.
IEC62271_100 : High-voltage switchgear and controlgear - Part 100: Alternating-current circuit breakers.
IEC62271-300: High-voltage switchgear and controlgear - Part 300: Seismic qualification of alternating current circuit-breakers.
ISO_IEC 27001: Information security, cybersecurity and privacy protection - Information security management systems requirements.
NF C18-510-1 - Opérations sur les ouvrages et installations électriques dans un environnement électrique.
NF C18-510-1 - Recueil d'instructions de sécurité électrique pour les ouvrages.
NF_EN_50180: Bushings above 1 kV up to 52 kV and from 250 A to 3.15 kA for liquid filled transformers.
NT-DI-CNER-DP-MHT-17-00339 : « Cahier des Charges Techniques : Équipements auxiliaires et appareillages Basse Tension des matériels Haute Tension ».
PRO: Project Management Specifications.
QUA 01: Quality assurance requirements.
QUA 02 : Surveillance requirements.
QUA 03 : Documentation management requirements.

12 Appendix II: List of attachments

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